International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Case Studies: Investigating DJI Mavic Air Flight Evidence

Student Name: Kafayat Omolara Animashawun
Student ID: 2025/FWSD/11317
Instructor Name: Aminu Idris
Date of Submission: 02/11/2026

EXECUTIVE SUMMARY

This assignment involved a forensic examination of digital evidence associated with a DJI Mavic Air drone, using an Android logical evidence image supplied as part of the Case Study 5 exercise. The examination was conducted on a Kali Linux workstation, with the primary objective of identifying, preserving and analysing DJI GO 4 application artefacts that could provide information about recorded flight activity.

The examination commenced with the establishment of a structured case directory, preservation of the original Android logical archive and creation of a separate working copy. SHA-256 hashing confirmed that the original and working archives were byte-for-byte identical at the time of verification. The working copy was subsequently extracted and examined, resulting in the identification of 220 files within the DJI GO 4 application directory.

The principal artefacts examined included a DJI TXT flight record, two DAT flight records, a cached MP4 video, its associated INFO metadata, six cached JPEG thumbnail images and multiple DJI GO 4 application logs. The cached video was identified as an H.264/AVC MP4 file with a resolution of 1280 × 720 pixels and a duration of approximately 4 minutes and 0.972 seconds. Its associated INFO metadata recorded a capture date of 19 June

2018 at 14:50:39 and a duration of 234.833 seconds. The difference between the two duration values was documented as an unresolved discrepancy.

The application notification log contained eight timestamped warnings relating to SD-card write speed, maximum flight distance, weak signal, high wind velocity, obstacle sensing and GPS reception. These findings provided additional context for the application activity recorded on 19 June 2018. However, the warnings were not treated as independent proof of specific physical flight events.

The two DAT records shared an identical initial binary sequence but differed in size, SHA-256 hash and subsequent content. Attempts to interpret them using Octave and SciPy were unsuccessful. The TXT flight record and selected application logs also contained binary or encoded-looking data that could not be reliably interpreted using the examination methods available.

A provisional timeline and cross-source correlation assessment were prepared. Filesystem modification timestamps were distinguished from timestamps embedded in application records and video metadata. A definitive unified UTC timeline could not be established because the timezone basis of the application event timestamps remained unconfirmed.

The raw internal microSD image and EWF forensic image were not successfully obtained and therefore could not be examined. These limitations, together with the unresolved binary structures and timestamp discrepancies, have been documented in the final evidence register and findings.

Overall, the examination identified and preserved relevant DJI GO 4 artefacts, established integrity reference values for selected evidence files, and documented findings that may support further investigation. No definitive conclusion was reached regarding the identity of the drone operator or the precise sequence of all physical flight events.

PURPOSE OF THE ASSIGNMENT

The purpose of this assignment was to apply basic digital forensic examination techniques to evidence associated with a DJI Mavic Air drone. The exercise focused on identifying and examining digital artefacts recovered from an Android logical image, particularly those associated with the DJI GO 4 application.

The examination sought to determine what information could be recovered from available flight records, cached video, metadata, images and application logs. It also aimed to demonstrate the importance of evidence preservation, cryptographic integrity verification, systematic artefact examination, timestamp analysis and cross-source correlation when investigating digital evidence.

OBJECTIVES OF THE ASSIGNMENT

The specific objectives of the examination were to:

1. Establish a structured forensic working environment and maintain a distinction between original evidence and working copies.
2. Verify the integrity of the supplied Android logical archive using SHA-256 cryptographic hashing.
3. Extract and inventory the Android logical evidence and identify relevant DJI GO 4 application artefacts.
4. Examine the available TXT and DAT flight records to determine whether interpretable flight information could be recovered.
5. Identify and analyse the cached MP4 video and its associated INFO metadata.
6. Extract and visually examine representative video frames and identify cached application images.
7. Examine DJI GO 4 application logs for timestamped warnings and other information relevant to the recorded activity.
8. Compare timestamps and metadata across available evidence sources and prepare a provisional timeline.
9. Perform cross-source correlation while distinguishing confirmed findings from unresolved observations.
10. Maintain an evidence register documenting the files examined, their integrity reference values and unavailable evidence sources.
11. Document the limitations of the examination and identify areas requiring further forensic analysis.

METHODOLOGY OF THE ASSIGNMENT

The examination followed a structured forensic workflow designed to preserve the supplied evidence and ensure that analytical activities could be traced to the relevant source files.

First, a dedicated case directory was established on the Kali Linux workstation. Separate directories were created for original evidence, working copies, analytical outputs, screenshots and report materials. The supplied Android logical archive was preserved in the originals directory, while a separate working copy was created for examination.

Second, SHA-256 cryptographic hashing was performed on the original and working archives. The matching hash values established that the two archive files were byte-for-byte identical at the time of verification. The working copy was then extracted using the unzip utility.

Third, the extracted Android logical image was examined to identify relevant DJI GO 4 application data. A recursive inventory was generated, and potentially relevant files were classified according to their locations and apparent functions. Selected flight records, video files, metadata, cached images and application logs were subjected to further examination.

Fourth, file identification, hexadecimal inspection, printable-string extraction, metadata examination and cryptographic hashing were performed on selected artefacts. Where appropriate, specialist utilities and compatible parsing libraries were used to attempt structural interpretation. Failed parsing attempts were documented, and no unsupported interpretation was assigned to binary data.

Fifth, the cached video was examined using FFprobe, and representative still frames were extracted using FFmpeg. The associated INFO metadata was reviewed for capture information, duration, resolution, synchronisation fields and GPS-related data. Cached JPEG images were also examined using file identification and image metadata utilities.

Sixth, DJI GO 4 application logs were examined through recursive searches, timestamp review and preliminary decoding attempts. Readable application notifications were extracted and compared with the video metadata and other available artefacts.

Finally, a provisional timeline, cross-source correlation assessment and forensic evidence register were prepared. Filesystem timestamps were distinguished from embedded event

timestamps, and timezone conversions were applied only where the available information supported them. Unavailable evidence sources and unresolved findings were documented as limitations.

TOOLS AND RESOURCES USED

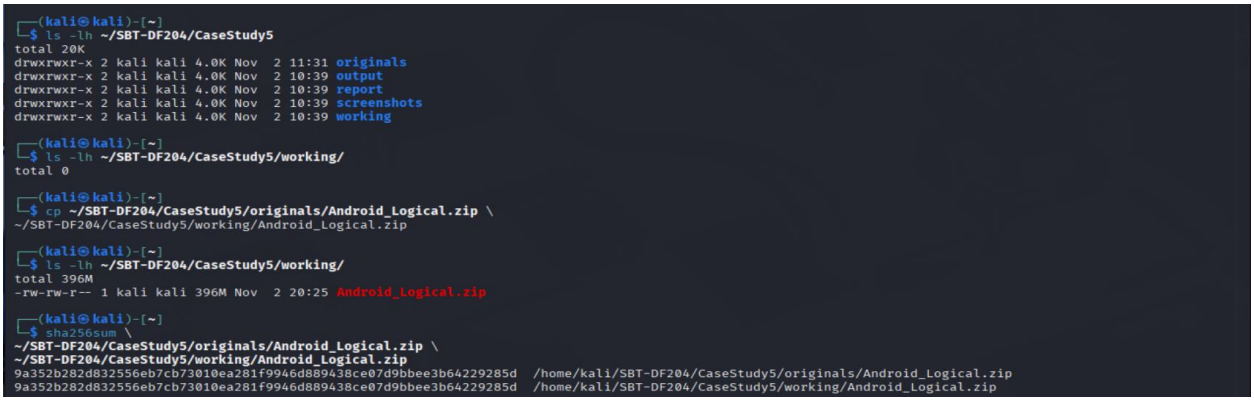
The following tools and resources were used during the examination:

S/N	Tool / Resource	Purpose
1	Kali Linux	Digital forensic examination environment.
2	Linux Terminal	Evidence handling, file inventory, hashing, metadata inspection and command-line analysis.
3	SHA-256	Verification of evidence integrity by generating and comparing cryptographic hashes.
4	file	Identification of file types and preliminary format analysis.
5	strings	Extraction of readable character sequences from binary files.
6	xxd	Hexadecimal examination of file headers and binary contents.
7	cmp	Binary comparison of the two DJI DAT flight-record files.
8	stat	Examination of file sizes, timestamps and filesystem metadata.
9	FFmpeg	Extraction of still frames from the DJI video recording.
10	FFprobe	Examination of video properties, including codec, resolution, frame rate and duration.
11	ImageMagick	Inspection and preparation of cached DJI thumbnail images.
12	Python	Supporting file analysis, timestamp calculations and evidence documentation.
13	SciPy	Attempted structural parsing of the DJI DAT files.

14	GNU Octave	Attempted examination of the DJI DAT files using MATLAB-compatible functionality.
15	Android Logical Evidence Archive	Primary evidence source containing DJI GO 4 application data and related artefacts.

Establishment of the case working directory

The forensic examination commenced with the establishment of a structured working directory for Case Study 5 on the Kali Linux examination workstation. The directory was organized into separate folders for original evidence, working copies, analytical outputs, screenshots and the final report. This structure was adopted to maintain a clear distinction between the original evidence supplied for examination and the copies used during forensic analysis. The original evidence was retained separately, while subsequent extraction and examination activities were performed within the designated working directory.



Structured case directory established for the DJI Mavic Air forensic examination.

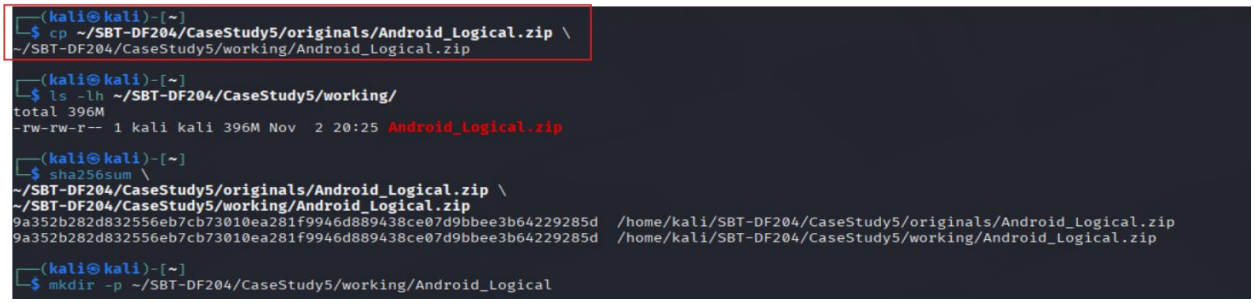
The terminal output listing:

- originals
- working
- output
- screenshots
- report

Identification and preservation of the Android logical image

The Android logical evidence archive, identified as Android_Logical.zip, was located in the Downloads directory and copied into the originals directory created for the

investigation. The archive was approximately 396 MB in size. A separate copy was subsequently created in the working directory to facilitate examination while preserving the original evidence archive. This approach ensured that the examination activities were conducted against a working copy rather than directly against the preserved original.



Cryptographic integrity verification

To establish the integrity of the evidence archive and verify that the working copy was an exact binary duplicate of the preserved original, the SHA-256 hashing algorithm was applied to both files. The resulting SHA-256 hash for the original archive was:

9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d

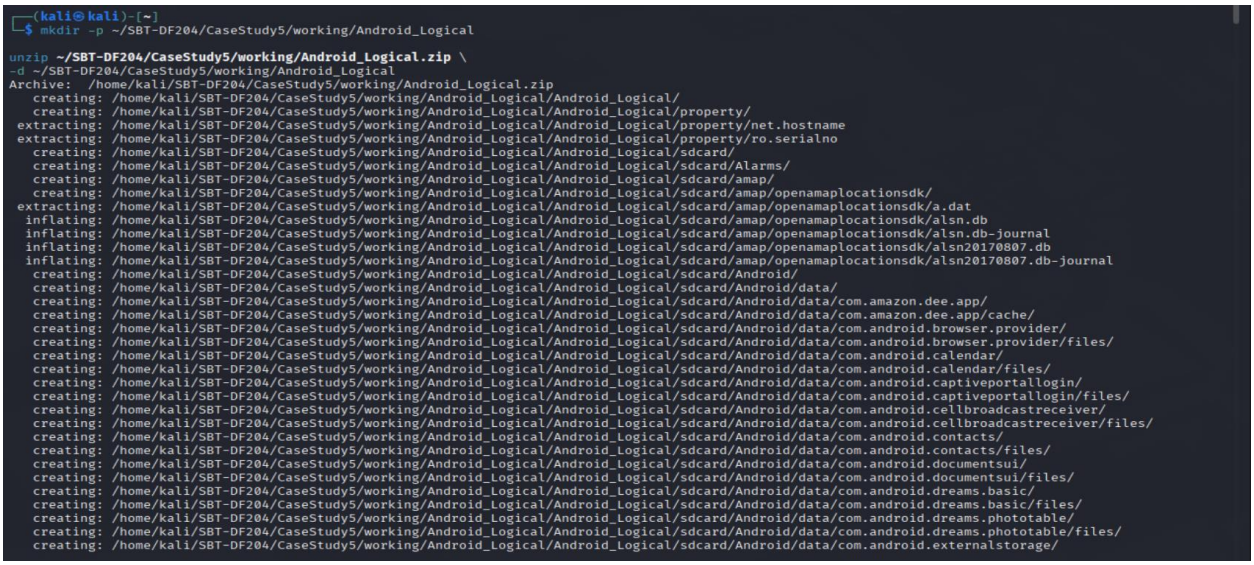


The working copy produced the same SHA-256 hash value. The matching hash values demonstrate that the two archive files were byte-for-byte identical at the time of verification. This provided an integrity checkpoint before extraction and subsequent forensic examination.

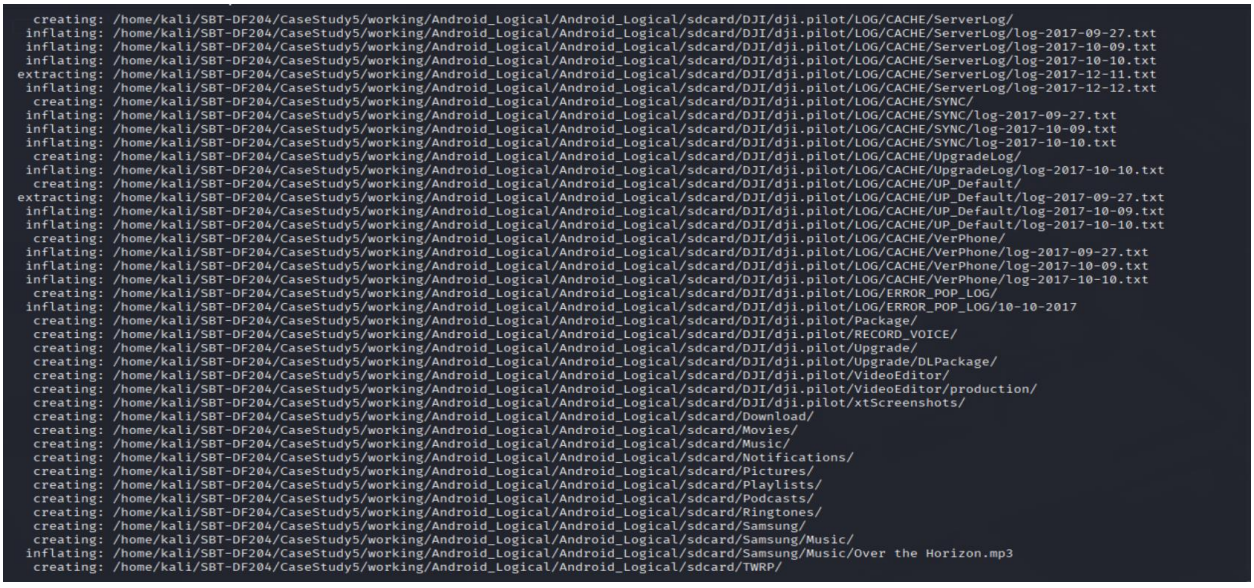
Extraction of the Android logical evidence

Following the successful integrity verification, the working copy of Android_Logical.zip was extracted into the designated working directory using the unzip utility. The extraction process produced an Android_Logical directory containing the extracted Android logical evidence. The resulting directory structure included a property directory and an sdcard directory. These directories were retained in their extracted form for subsequent

examination of device properties, application artefacts and other potentially relevant digital evidence.



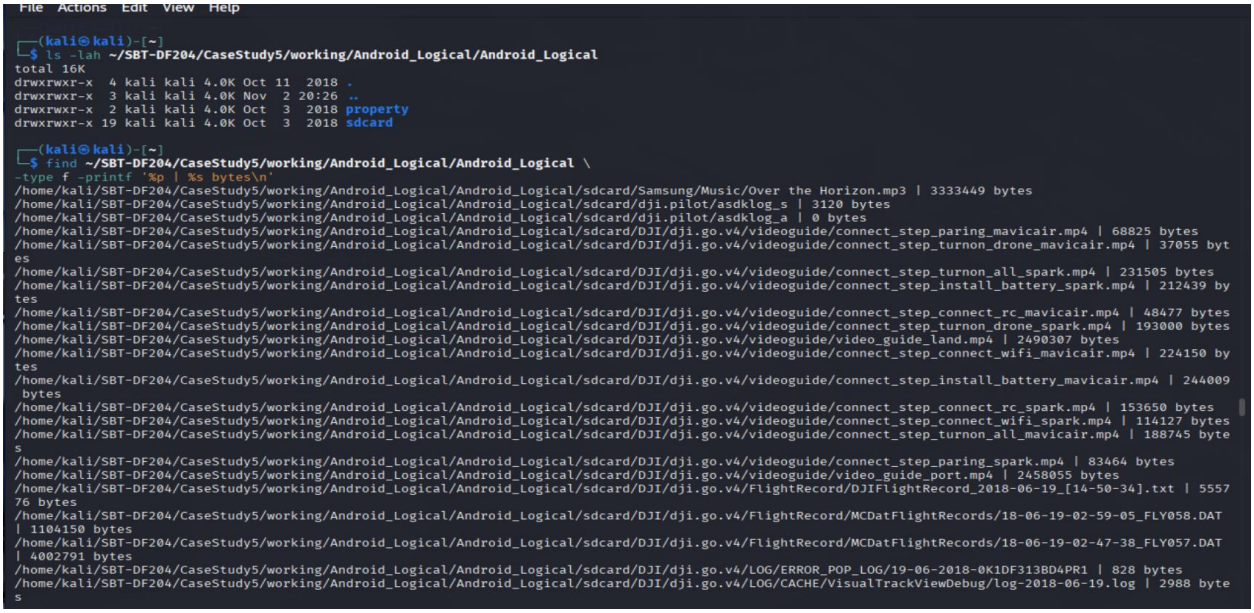
Extraction of the verified Android logical evidence archive within the designated forensic working directory.

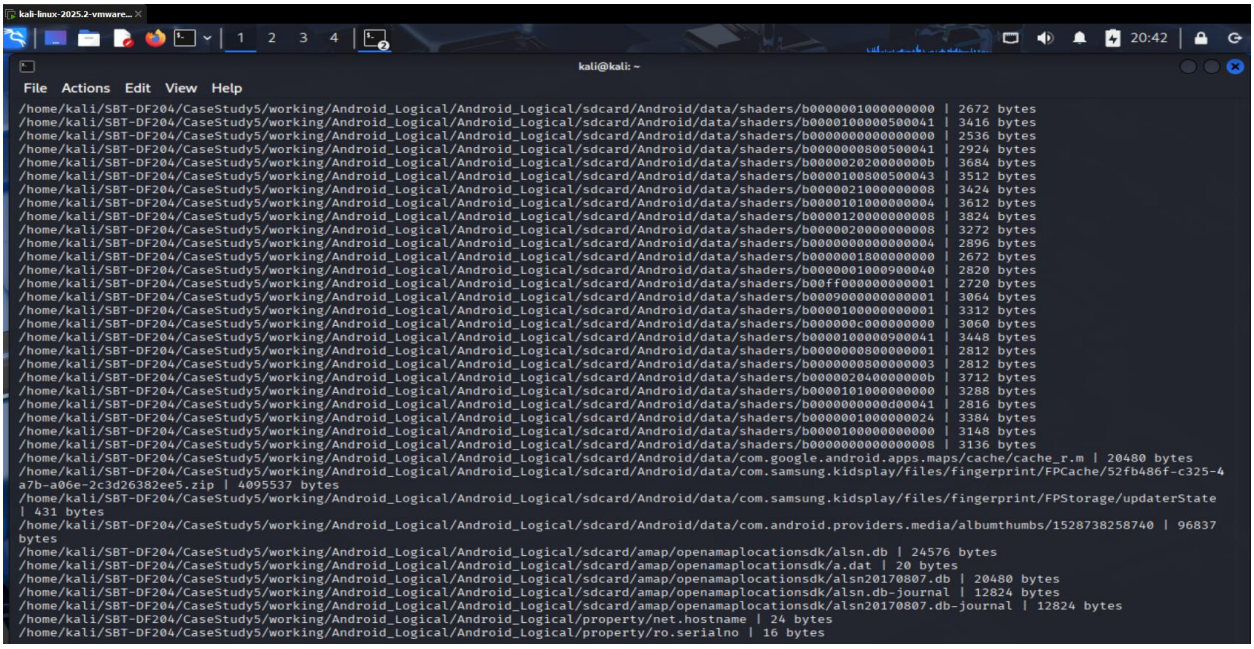


Evidence inventory and identification of relevant artefacts

Following extraction, a preliminary examination of the Android logical evidence directory was undertaken to identify files and directories associated with the DJI GO 4 application. Particular attention was directed towards the application data stored under the

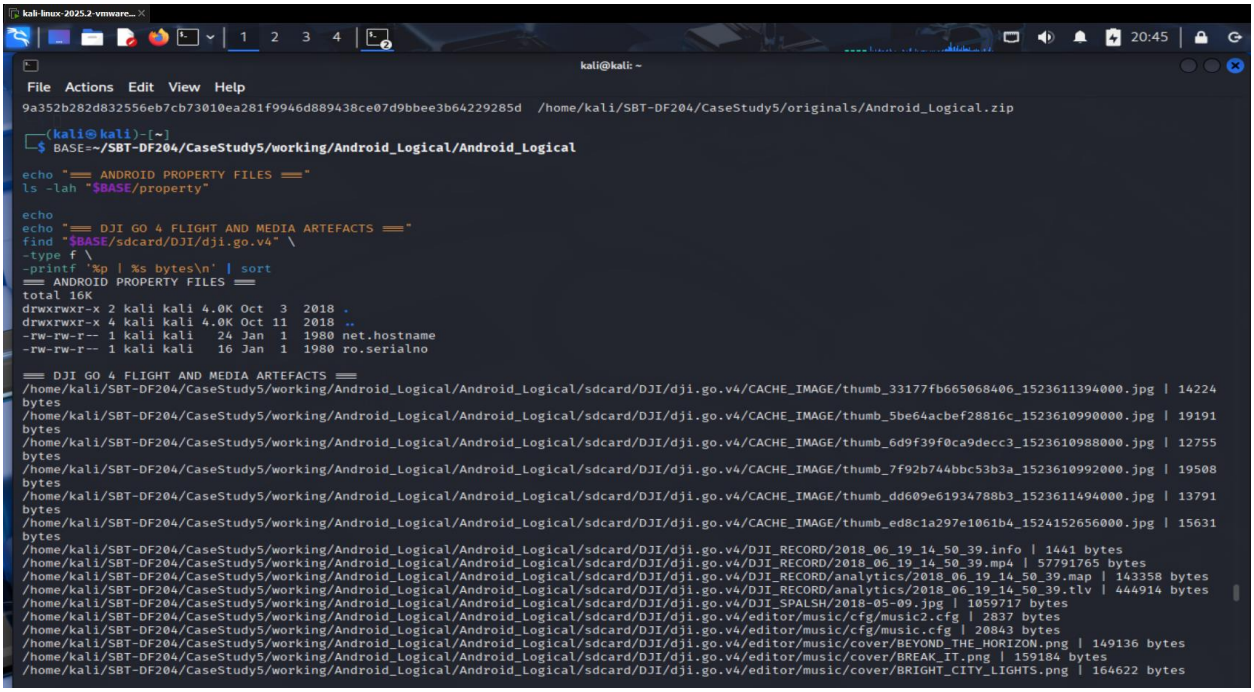
sdcard/DJI/dji.go.v4/ directory. The initial directory inspection identified several potentially relevant artefact categories, including flight record files, DAT flight records, cached media, video recordings and application analytics files. These artefacts were identified for further examination and correlation to establish their relevance to the flight activity under investigation. At this stage, their presence was documented without assuming that every identified file related to the same flight event.

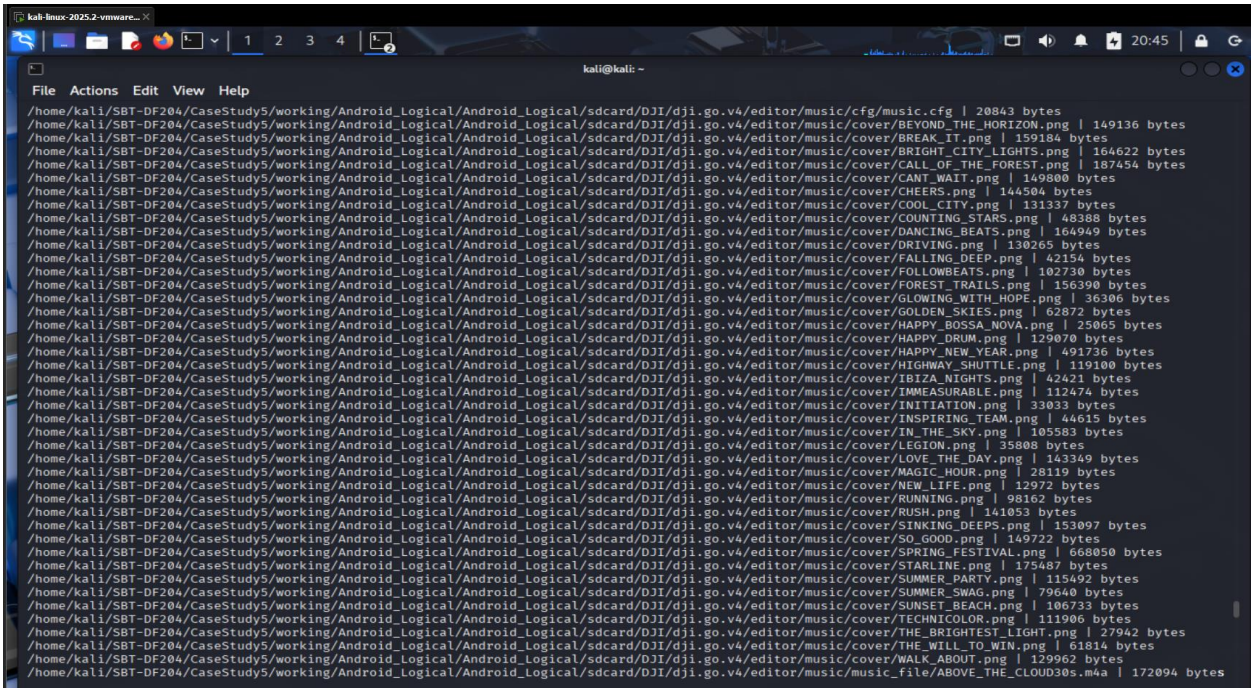


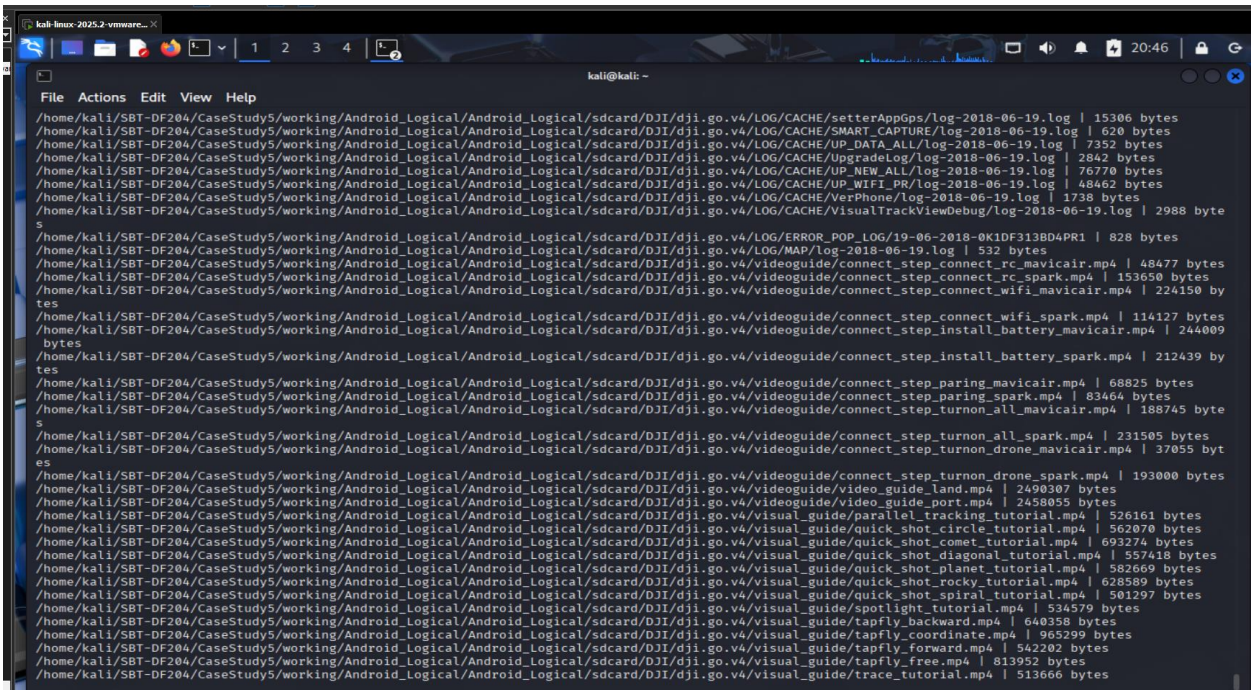


Identification and preliminary classification of DJI GO 4 artefacts

The extracted Android logical image was examined to identify application data associated with DJI GO 4. The directory structure contained a dedicated FlightRecord directory with one TXT flight record dated 19 June 2018 and two DAT flight-record files bearing the identifiers FLY057 and FLY058. The DJI_RECORD directory contained an MP4 video file, an associated INFO metadata file, and supporting analytics files in MAP and TLV formats. Six cached thumbnail images were also identified within the CACHE_IMAGE directory. In addition, several DJI GO 4 application logs were present, including logs dated 19 June 2018. These artefacts were classified as potential sources of flight, application activity, video and metadata evidence for further examination. Their relevance to the incident will be assessed by comparing their internal timestamps, metadata and recorded flight information rather than relying solely on filenames.







The most relevant artefacts are:

Artefact	Location / filename	Size
DJI flight record	FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt	555,776 bytes

DAT flight record 1	18-06-19-02-47-38_FLY057.DAT	4,002,791 bytes
DAT flight record 2	18-06-19-02-59-05_FLY058.DAT	1,104,150 bytes
Cached video	DJI_RECORD/2018_06_19_14_50_39.mp4	57,791,765 bytes
Flight metadata	DJI_RECORD/2018_06_19_14_50_39.info	1,441 bytes
Analytics map	analytics/2018_06_19_14_50_39.map	143,358 bytes
Analytics data	analytics/2018_06_19_14_50_39.tlv	444,914 bytes

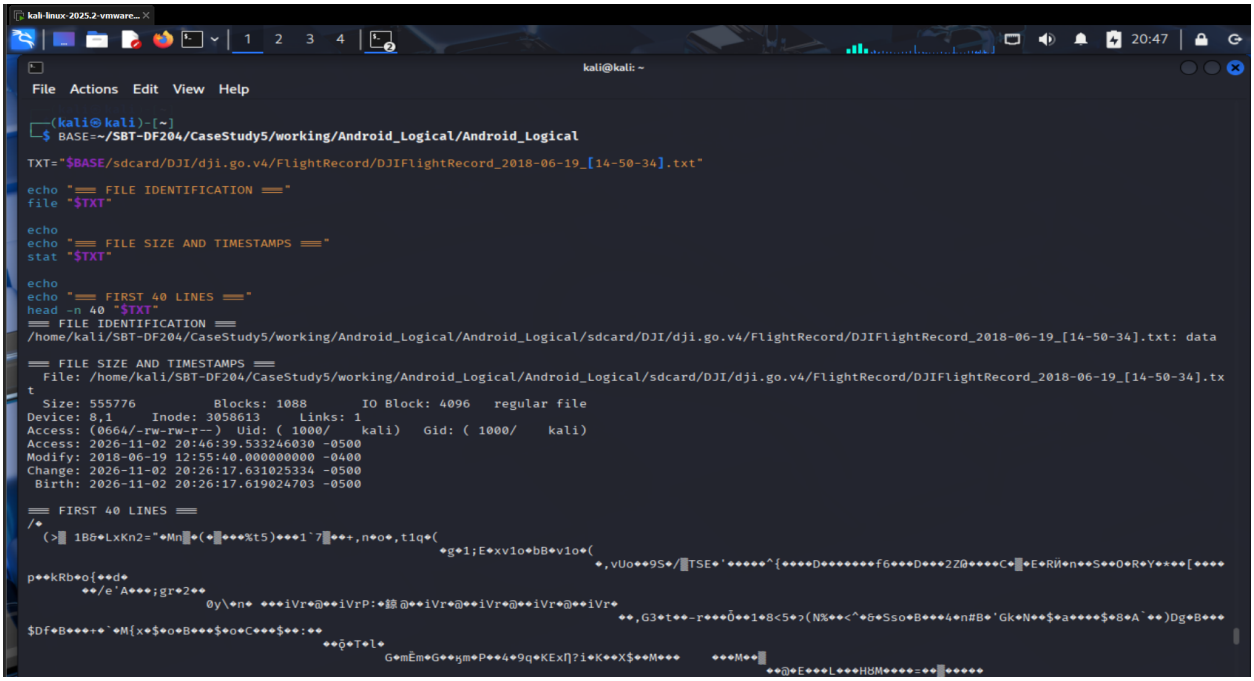
We also identified six cached thumbnail images and multiple DJI GO 4 application logs dated 19 June 2018.

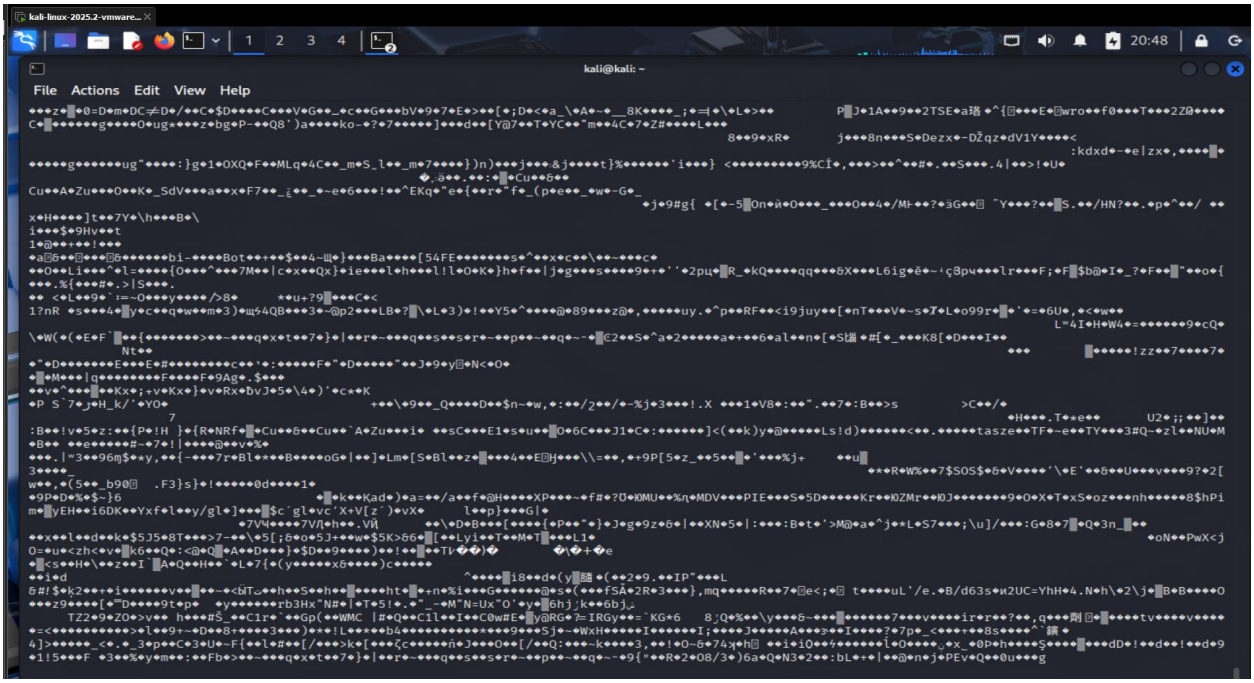
Important distinction: the filenames and dates make these artefacts relevant candidates, but they do not yet prove that every file relates to the same flight. We will establish that through metadata, timestamps and cross-source correlation.

The property directory also contains net.hostname and ro.serialno. These are device-identification artefacts. We should inspect their contents carefully and redact any sensitive identifiers in the submitted report.

Initial examination of the TXT flight record

The DJI GO 4 flight record, DJIFlightRecord_2018-06-19_[14-50-34].txt, was subjected to preliminary file identification and metadata examination. The file was reported by the Linux file utility as generic data rather than plain text. A preliminary attempt to display its opening contents using the head command produced non-readable binary characters. The file has a recorded size of 555,776 bytes and a filesystem modification timestamp of 19 June 2018 at 12:55:40, with the displayed timestamp carrying a UTC−04:00 offset. These observations indicate that the record requires format-aware decoding before its internal flight information can be examined. The original file was therefore retained unchanged, and further analysis will be conducted using appropriate forensic decoding methods.

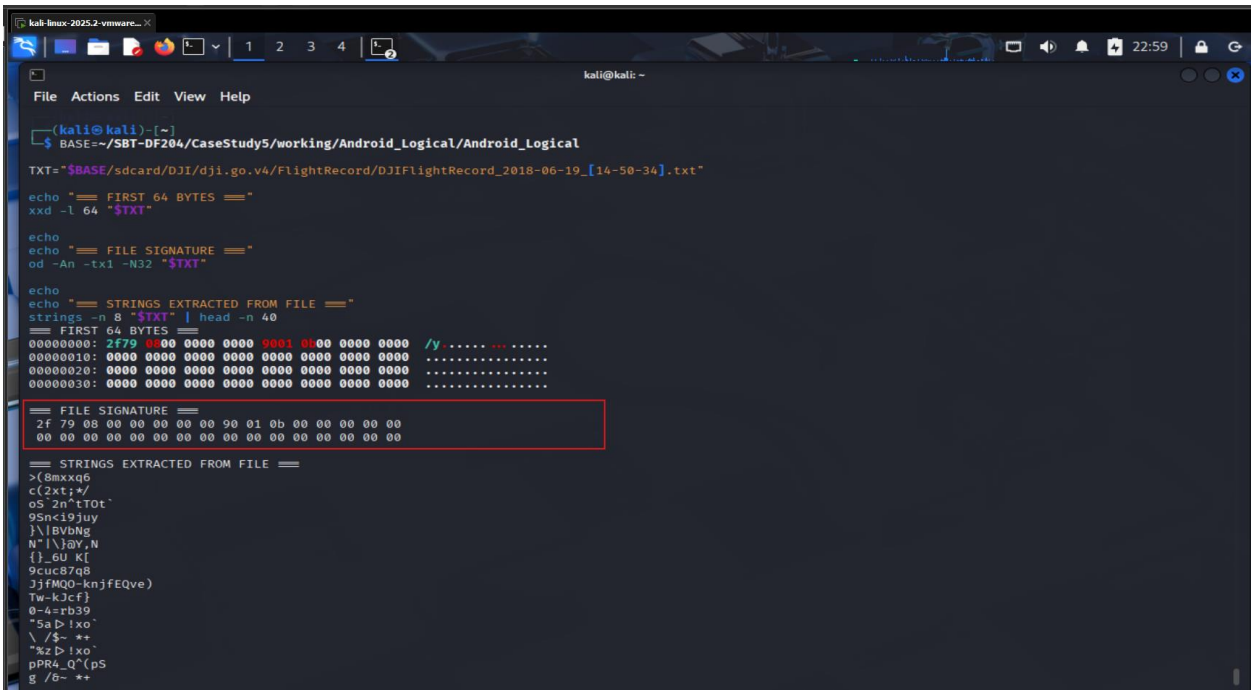


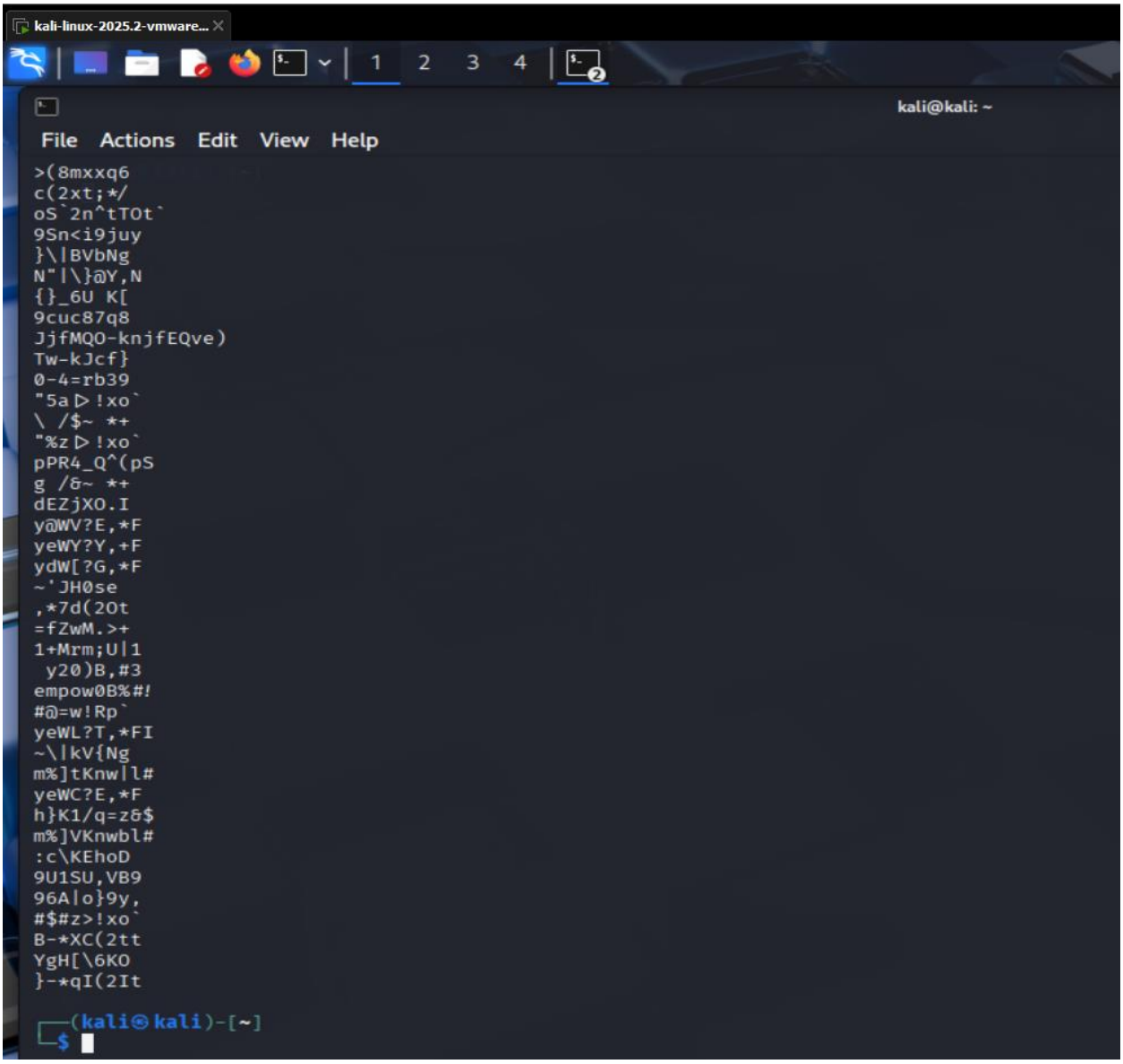


Examination of the TXT flight-record structure

The initial hexadecimal examination of the DJI GO 4 flight record revealed that the file begins with the byte sequence 2f 79 08 00 00 00 00 00 90 01 0b 00 00 00 00 00. The opening bytes did not match the commonly recognisable signatures of standard ZIP, gzip or zlib files. The strings utility extracted scattered printable character sequences, but no immediately interpretable flight information was identified. These findings indicate that the

record requires further structural examination before its contents can be interpreted. At this stage, no conclusion has been drawn as to whether the data is compressed, encrypted or stored in a proprietary binary format.

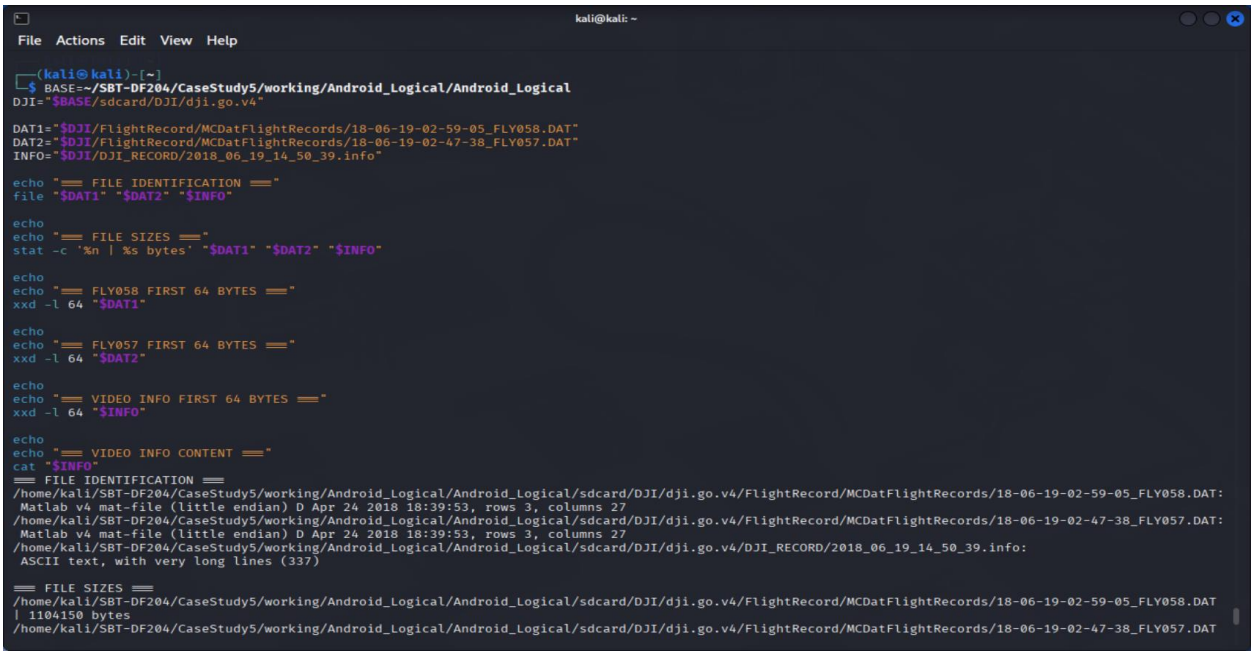




Examination of DAT flight records and video metadata

The two DAT files, FLY057.DAT and FLY058.DAT, were identified by the file utility as little-endian MATLAB version 4 MAT-files, each reporting a structure of three rows and 27 columns. Both files displayed the same build timestamp of 24 April 2018 at 18:39:53 and identical initial hexadecimal bytes, although their file sizes differed. This indicates a common file format and provides a basis for structured examination using a MATLAB-compatible parser. The associated video metadata file was identified as ASCII text and contained a capture date of 19 June 2018 at 14:50:39, a video duration field of 234,833 milliseconds, frame-rate information, resolution details and GPS position fields. These

metadata values have been recorded as potential correlation points for subsequent analysis. The timezone of the CaptureDate field has not yet been established, and the GPS coordinates will be handled as sensitive evidence.

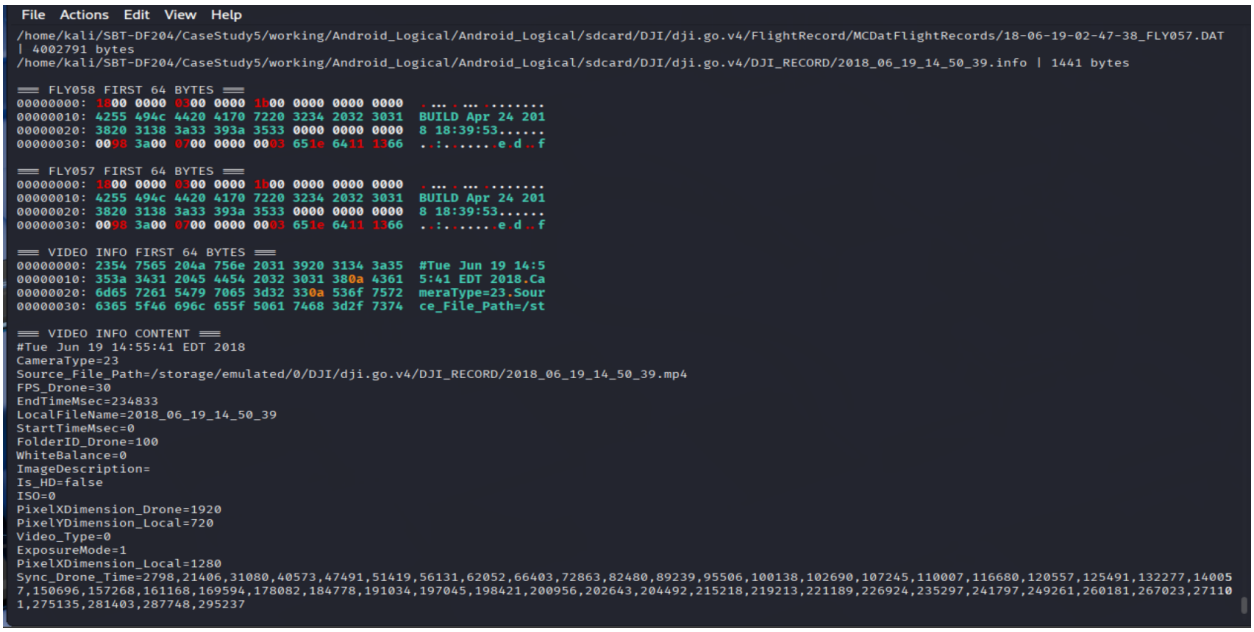


DAT flight records: Both DAT files are identified by Kali's file utility as MATLAB v4 MAT-files, with the following structure:

- Little-endian byte order.
- 3 rows.
- 27 columns.
- A recorded build timestamp of 24 April 2018 at 18:39:53.
- FLY058 size: 1,104,150 bytes.
- FLY057 size: 4,002,791 bytes.

Both files also begin with the same 64-byte sequence in your output. This strongly suggests that they share the same file structure, although it does not establish that they contain the same flight data or relate to the same flight.

This is a useful discovery: we can investigate the DAT files using a MATLAB-compatible parser rather than treating them as arbitrary binary files.

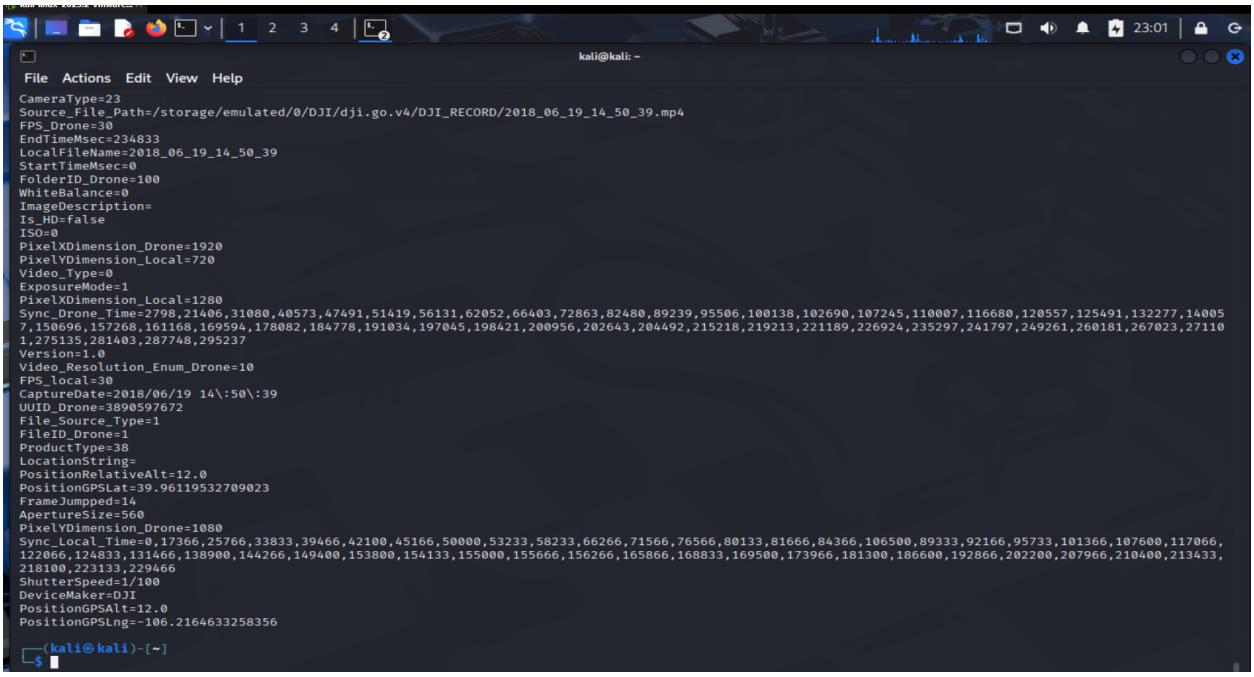


Video metadata: The .info file is readable ASCII text. It records:

- Capture date: 19 June 2018, 14:50:39.
- Metadata header timestamp: 19 June 2018, 14:55:41 EDT.
- Drone video frame rate: 30 FPS.
- Drone resolution: 1920 × 1080.
- Local resolution: 1280 × 720.
- Video duration field: 234,833 milliseconds (approximately 3 minutes 54.8 seconds).
- GPS position: latitude 39.96119532709023, longitude −106.2164633258356.
- Relative altitude field: 12.0.
- Drone position altitude field: 12.0.

The GPS coordinates are sensitive location information. Keep the exact coordinates out of any report version that will be shared publicly; use redaction or a general location description where appropriate.

Important timestamp distinction: CaptureDate and the metadata header are not the same timestamp. The header explicitly says EDT, while CaptureDate has no timezone stated in the file. We must not assume they are interchangeable or convert both to UTC until we establish the timestamp conventions.



MATLAB v4 header Inspection

The hexadecimal examination of both DAT flight records revealed identical initial 128-byte sequences. The first three little-endian 32-bit values were 24, 3 and 27, respectively, followed by a zero-valued field. An ASCII build string beginning at offset 0x10 identified the build timestamp as 24 April 2018 at 18:39:53. The remaining header bytes included non-readable binary sequences and repeated 0x55 values. Although the initial structures are identical, this observation does not establish that the two files contain identical flight data. Further examination is required to determine their internal organisation and whether the contents can be decoded.

```
┌──(kali㉿kali)-[~]
└─$ BASE=~/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical
DJI="$BASE/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords"

for FILE in \
"18-06-19-02-59-05_FLY058.DAT" \
"18-06-19-02-47-38_FLY057.DAT"
do
    echo
    echo "═══════════════════════════════════════"
    echo "$FILE"
    echo "═══════════════════════════════════════"

    xxd -g 1 -l 128 "$DJI/$FILE"
done

═══════════════════════════════════════
18-06-19-02-59-05_FLY058.DAT
═══════════════════════════════════════
00000000: 18 00 00 00 03 00 00 00 1b 00 00 00 00 00 00 00  ................
00000010: 42 55 49 4c 44 20 41 70 72 20 32 34 20 32 30 31  BUILD Apr 24 201
00000020: 38 20 31 38 3a 33 39 3a 35 33 00 00 00 00 00 00  8 18:39:53......
00000030: 00 98 3a 00 07 00 00 00 00 03 65 1e 64 11 13 66  ..:.......e.d..f
00000040: 64 66 17 11 61 05 07 64 55 55 ec 38 dc 49 bb 9e  df..a..dUU.8.I..
00000050: c6 4c 44 38 d0 d3 9b f1 85 af 24 ba 98 74 19 22  .LD8......$..t."
00000060: 2b ad 60 98 e2 c9 f6 fc 13 a4 de 55 55 55 55 55  +.`........UUUUU
00000070: 55 55 55 55 55 55 55 55 55 55 55 55 55 55 55 55  UUUUUUUUUUUUUUUU

═══════════════════════════════════════
18-06-19-02-47-38_FLY057.DAT
═══════════════════════════════════════
00000000: 18 00 00 00 03 00 00 00 1b 00 00 00 00 00 00 00  ................
00000010: 42 55 49 4c 44 20 41 70 72 20 32 34 20 32 30 31  BUILD Apr 24 201
00000020: 38 20 31 38 3a 33 39 3a 35 33 00 00 00 00 00 00  8 18:39:53......
00000030: 00 98 3a 00 07 00 00 00 00 03 65 1e 64 11 13 66  ..:.......e.d..f
00000040: 64 66 17 11 61 05 07 64 55 55 ec 38 dc 49 bb 9e  df..a..dUU.8.I..
00000050: c6 4c 44 38 d0 d3 9b f1 85 af 24 ba 98 74 19 22  .LD8......$..t."
00000060: 2b ad 60 98 e2 c9 f6 fc 13 a4 de 55 55 55 55 55  +.`........UUUUU
00000070: 55 55 55 55 55 55 55 55 55 55 55 55 55 55 55 55  UUUUUUUUUUUUUUUU
```

```
┌──(kali㉿kali)-[~]
└─$ python3 - <<'PY'
import struct
import os

base = os.path.expanduser(
    "~/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical/"
    "sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords"
)

for name in [
    "18-06-19-02-59-05_FLY058.DAT",
    "18-06-19-02-47-38_FLY057.DAT"
]:
    path = os.path.join(base, name)

    with open(path, "rb") as f:
        header = f.read(20)

    print("\nFile:", name)
    print("Header bytes:", header.hex(" "))

    values = struct.unpack("<5I", header)
    print("Five little-endian 32-bit integers:", values)
PY

File: 18-06-19-02-59-05_FLY058.DAT
Header bytes: 18 00 00 00 03 00 00 00 1b 00 00 00 00 00 00 00 42 55 49 4c
Five little-endian 32-bit integers: (24, 3, 27, 0, 1279874370)

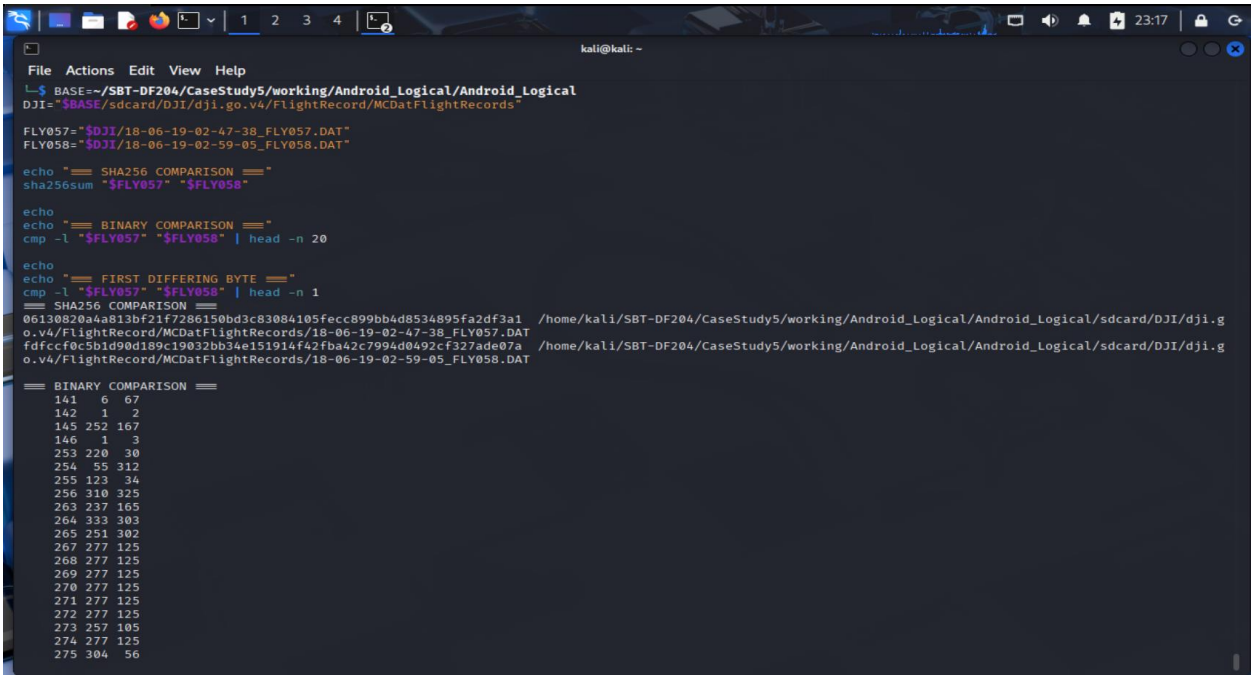
File: 18-06-19-02-47-38_FLY057.DAT
Header bytes: 18 00 00 00 03 00 00 00 1b 00 00 00 00 00 00 00 42 55 49 4c
Five little-endian 32-bit integers: (24, 3, 27, 0, 1279874370)
```

The first 20 bytes produce these five little-endian 32-bit integers:

Offset	Value	Observation
0x00	24	Header field
0x04	3	Header field
0x08	27	Header field
0x0C	0	Header field
0x10	1279874370	Begins with ASCII BUIL

Binary comparison of FLY057 and FLY058

A SHA-256 comparison was performed on the two DAT records. FLY057 produced the hash 06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1, while FLY058 produced fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade07a. A byte-level comparison using cmp identified the first difference at byte position 141 (offset 140 when zero-indexed). The files therefore share an identical initial sequence but are not binary-identical. Their different sizes and cryptographic hashes further establish that they are distinct files. The significance of their common header and subsequent differences will be investigated during further structural analysis.





The results confirm that FLY057 and FLY058 are two different binary files, despite sharing identical initial 128-byte sequences.

Findings

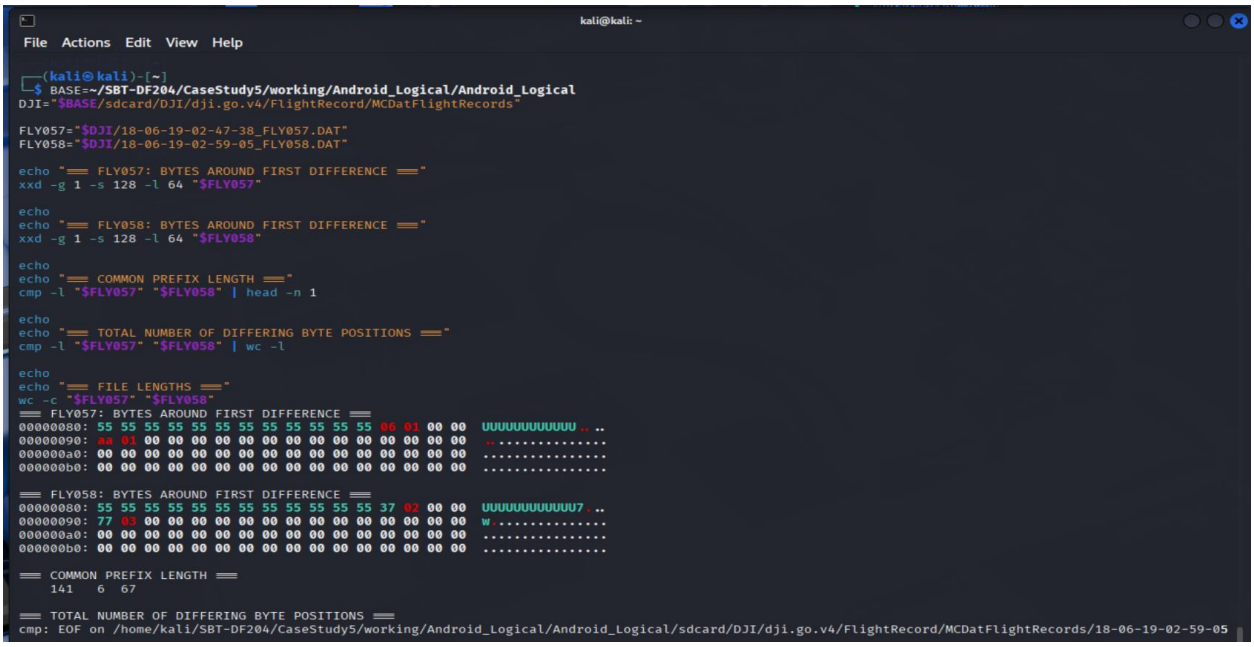
Test	FLY057	FLY058
File size	4,002,791 bytes	1,104,150 bytes
SHA-256	06130820…fa2df3a1	fdfccf0c…327ade07a
Initial 128 bytes	Identical	Identical
First differing byte	Byte 141	Byte 141

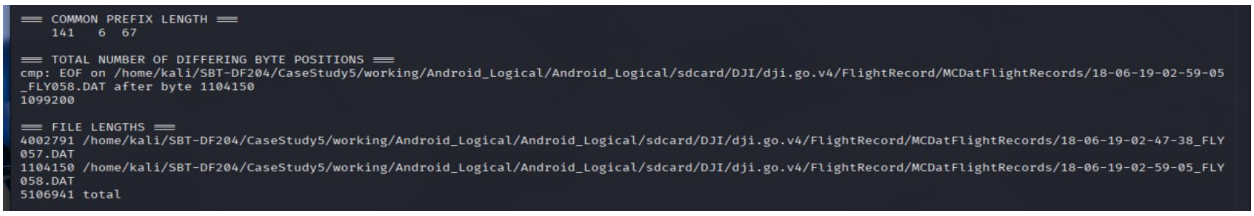
The cmp output reports byte positions starting at 1. Therefore, the first difference occurs at offset 140 (zero-based). This is significant because the files share a common header and initial content, but their binary contents diverge shortly afterwards. Their different SHA-256 hashes independently confirm that they are not byte-for-byte identical.

We should not infer that they represent the same flight, different flights, or different stages of one flight solely from this comparison.

Examination of the binary differences between FLY057 and FLY058

The latest output reveals more than just different file hashes. It shows that the records begin to diverge at byte 141, immediately after a common sequence of header and padding bytes.





Key observations

1. The files share an initial 140-byte prefix. Both files contain the same bytes from offset 0x00 through 0x8B (decimal bytes 1–140).
2. The first differences occur at offsets 0x8C and 0x90.

Offset	FLY057	FLY058
0x8C	06 01 00 00	37 02 00 00
0x90	aa 01 00 00	77 03 00 00

Interpreted as little-endian 32-bit unsigned integers, these byte sequences are:

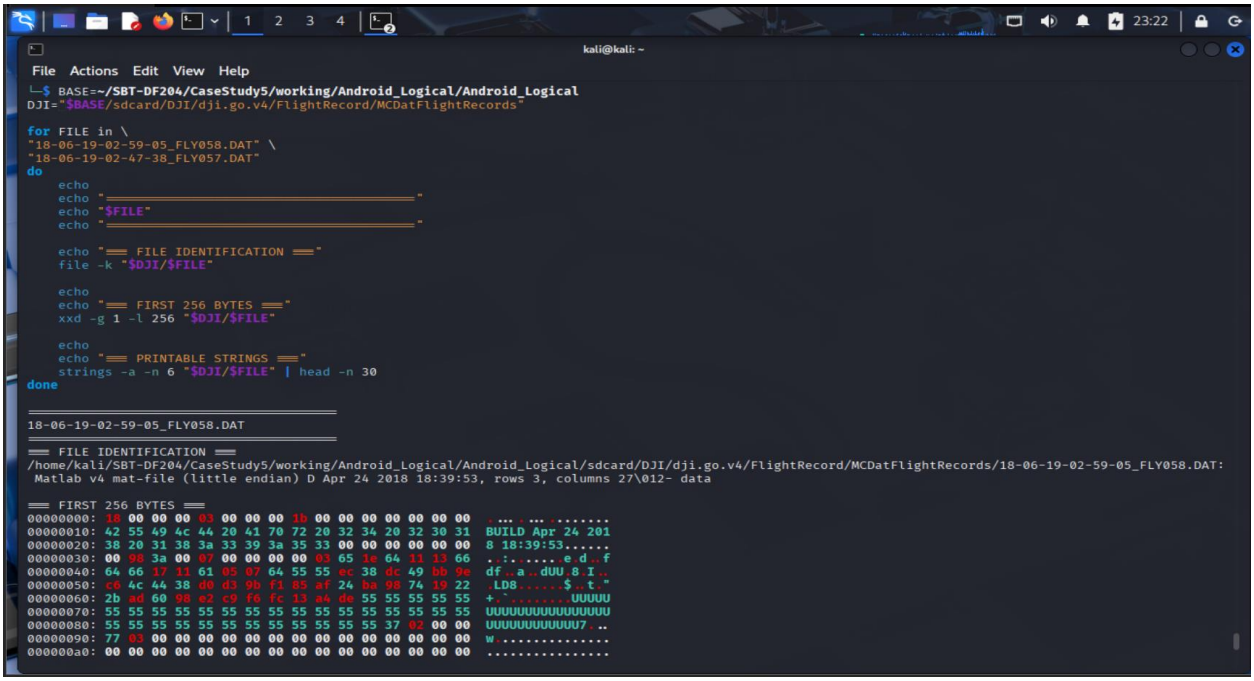
- FLY057: 262 and 426.
- FLY058: 567 and 887.

These are simply numerical interpretations of the bytes. We do not yet know what the fields represent.

3. The comparison reports 1,099,200 differing byte positions: The shorter file, FLY058, ends at 1,104,150 bytes, while FLY057 continues to 4,002,791 bytes. The cmp output therefore reports an EOF condition. Its count represents differing positions within the overlapping portion; it does not count the additional trailing bytes in FLY057.

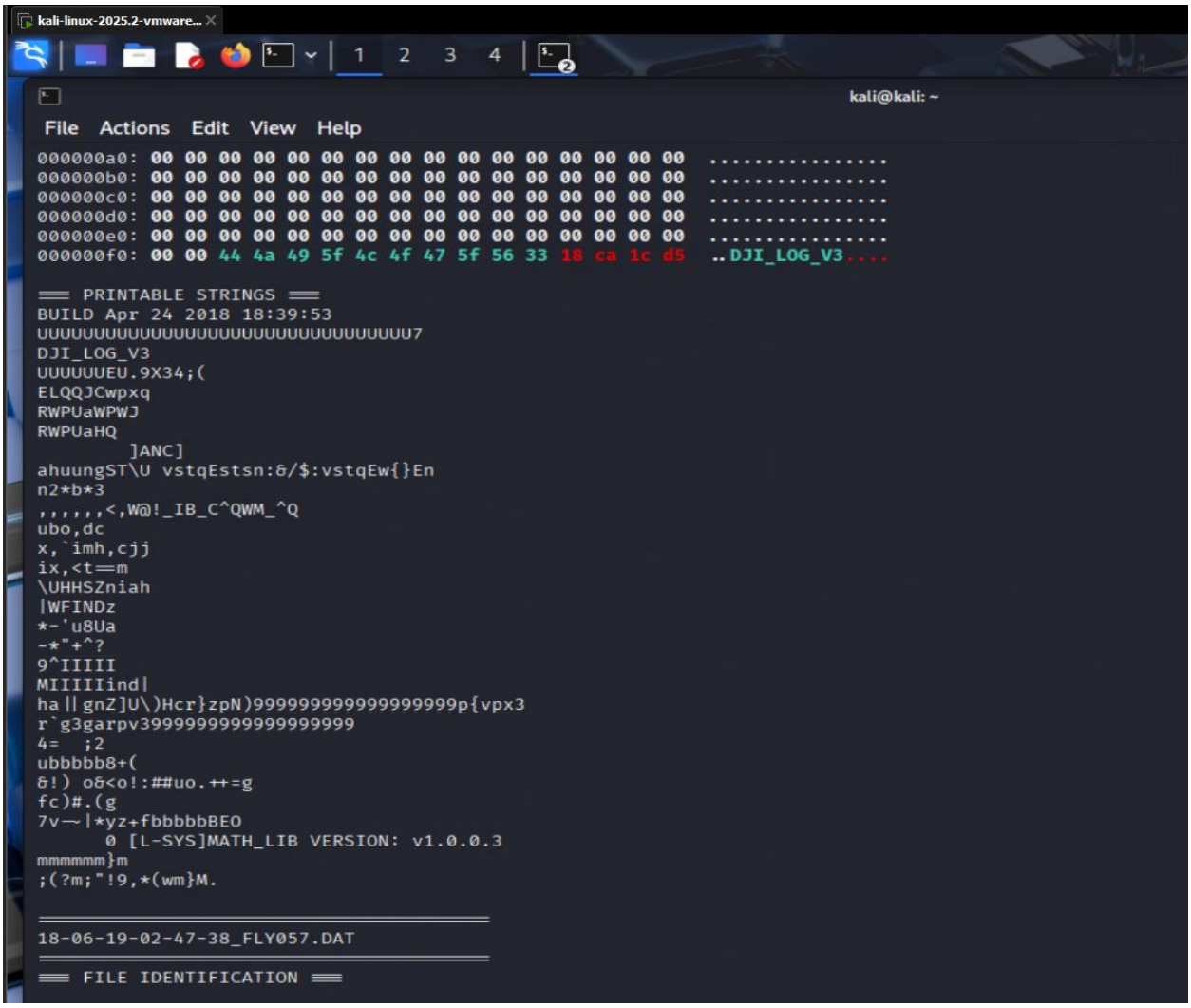
Preliminary examination of DJI DAT flight records

The two DJI DAT flight records, 18-06-19-02-59-05_FLY058.DAT and 18-06-19-02-47-38_FLY057.DAT, were examined using the Linux file utility, hexadecimal inspection with xxd, and printable-character extraction with strings. The file utility identified both records as MATLAB version 4 MAT-files, reporting a little-endian format, three rows, 27 columns and a common build timestamp of 24 April 2018 at 18:39:53. However, subsequent attempts to load the records using Octave and SciPy were unsuccessful. The MATLAB file-format identification is therefore treated as unverified rather than a confirmed classification.

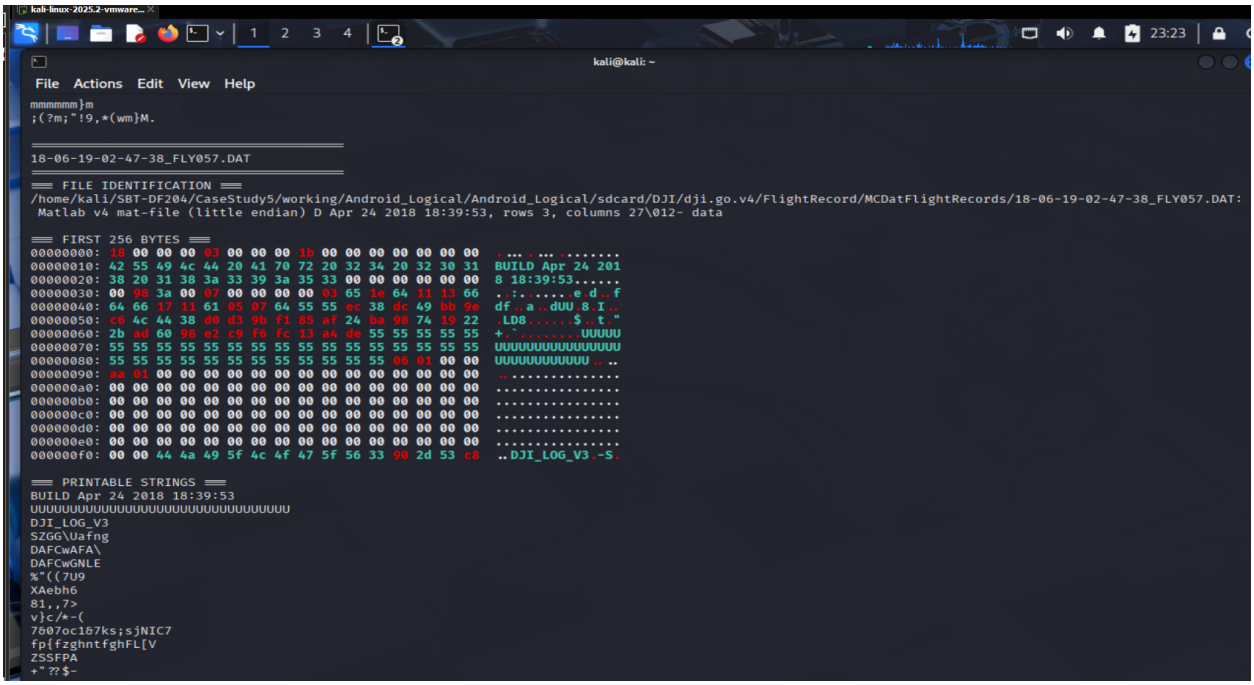


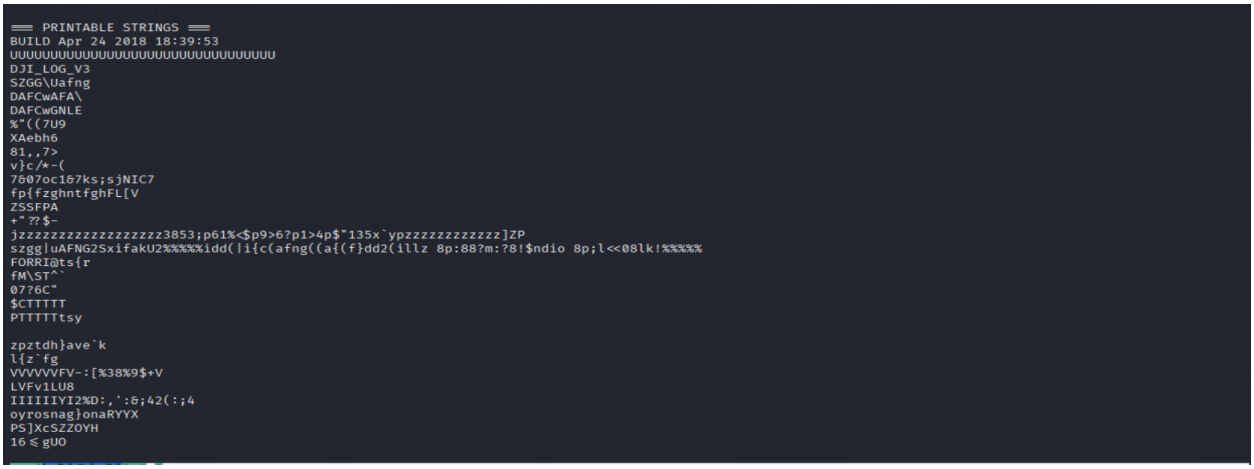
Hexadecimal examination revealed that both files share an identical initial 140-byte sequence. The printable-character extraction also identified the strings BUILD Apr 24 2018 18:39:53 and DJI_LOG_V3 in both records. These strings provide useful indicators for further format investigation, although they do not, by themselves, establish the complete file structure or the meaning of its internal data. Beyond the common initial

sequence, the records contain different binary values and printable-character patterns. The extracted strings were largely non-interpretable, and no flight parameters could be reliably established from this preliminary examination.



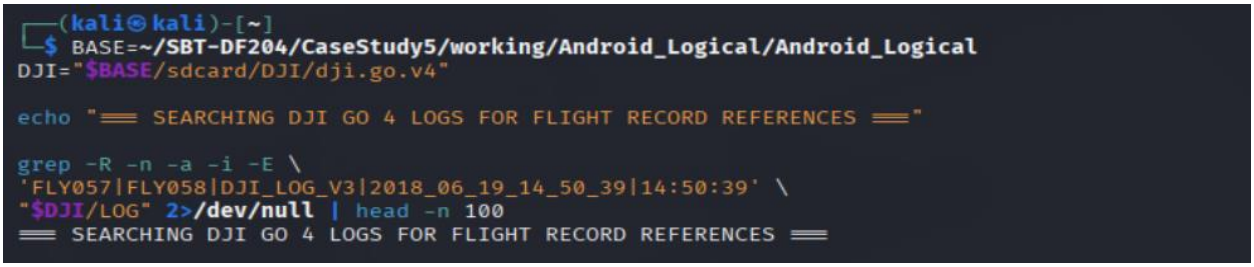
The findings indicate that the two files are distinct DJI flight-record artefacts requiring further format-aware analysis. No assumption has been made about whether their binary contents are compressed, encrypted or encoded using a proprietary DJI structure. Their relationship to the incident will be assessed through further examination and correlation with the available flight-record, application-log and video metadata evidence.





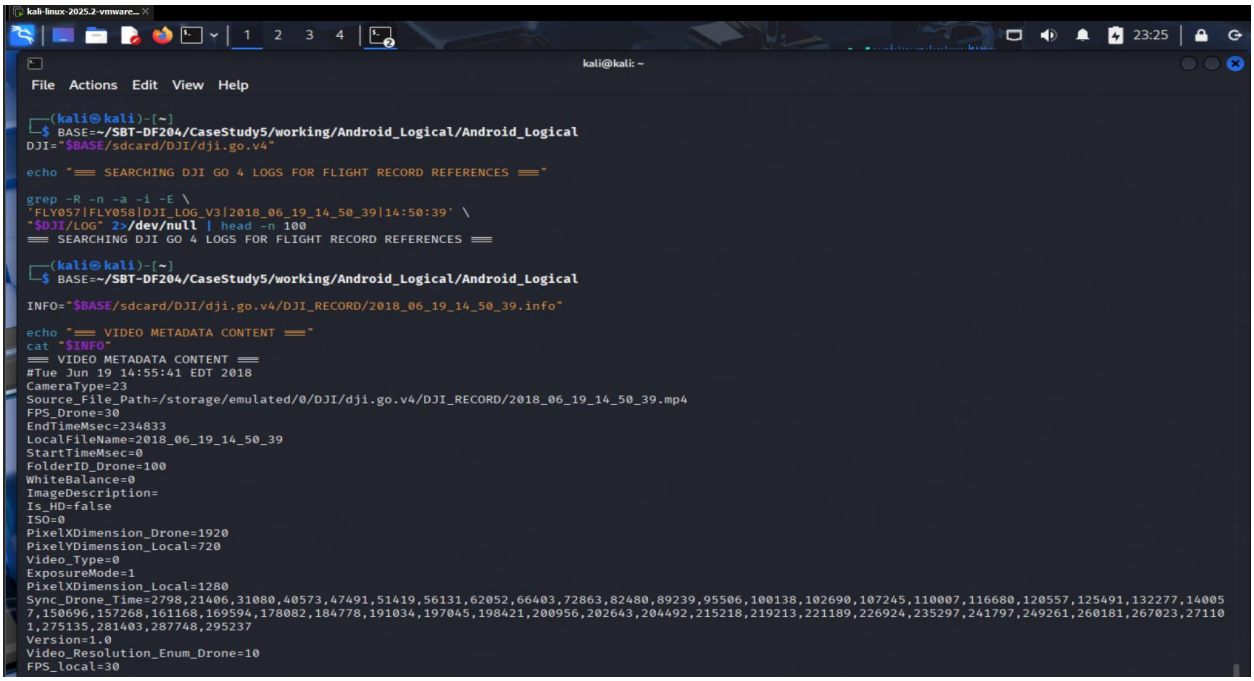
Examination of DJI GO 4 application logs

The DJI GO 4 application log directory was searched recursively for references to the two identified DAT flight records (FLY057 and FLY058), the internal identifier DJI_LOG_V3, and the associated cached video filename and capture time. The search returned no matching entries. Consequently, no direct textual association between the DAT records and the cached video could be established from the application logs examined at this stage. This negative search result has been documented as part of the examination. It does not establish that the artefacts are unrelated, as relevant information may be stored in binary records, alternative log files or under different identifiers.

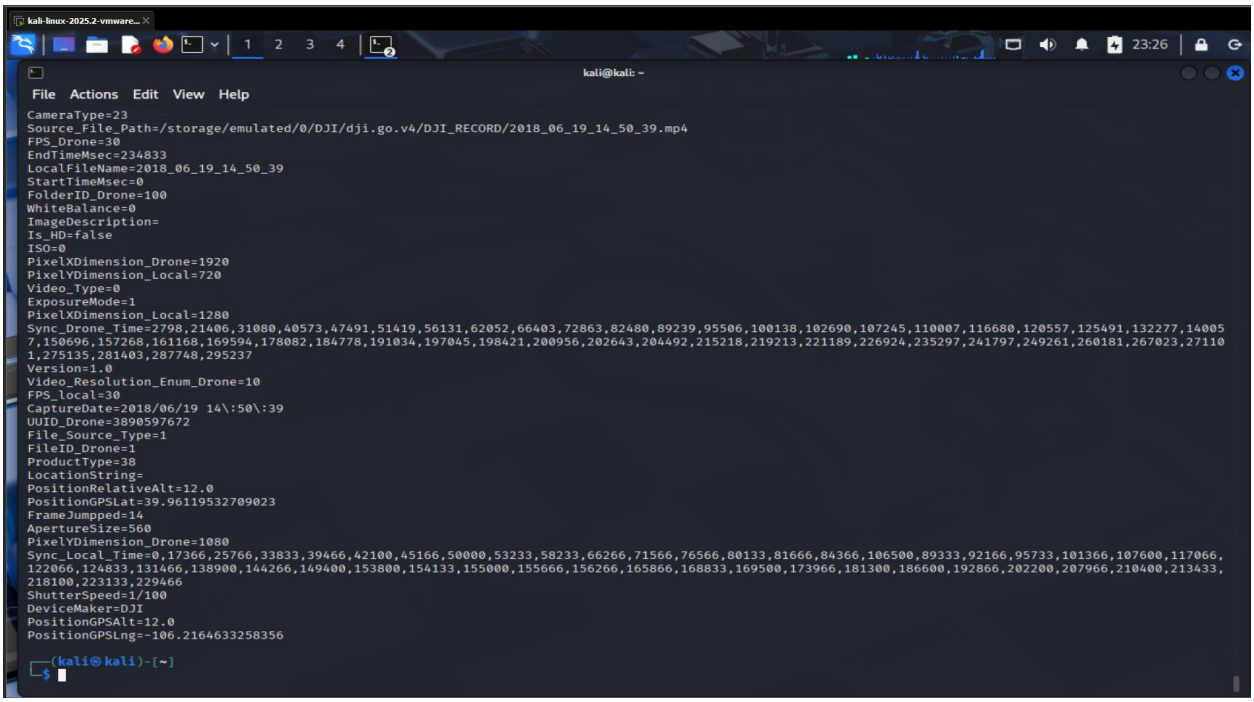


Examination of cached video metadata

The metadata file, 2018_06_19_14_50_39.info, associated with the cached MP4 video was examined to identify its recorded capture details, technical properties and available synchronisation information. The file was found to contain a capture date of 19 June 2018 at 14:50:39, a video start time of 0 milliseconds and an end time of 234,833 milliseconds, indicating a recorded duration of approximately 3 minutes and 54.833 seconds. The metadata also records a drone frame rate of 30 frames per second and drone video dimensions of 1920 × 1080 pixels, alongside local video dimensions of 1280 × 720 pixels. Additional fields include the device manufacturer (DJI), camera type, product type, video file identifier, drone UUID and synchronisation timing arrays (Sync_Drone_Time and Sync_Local_Time). These synchronisation arrays may assist in examining the relationship between drone-recorded time and locally recorded video time. The metadata header records Tuesday, 19 June 2018 at 14:55:41 EDT, while the CaptureDate field records 14:50:39 without an explicit timezone designation. These timestamps have therefore been preserved as recorded and will not be converted or treated as equivalent until their timezone basis is established.



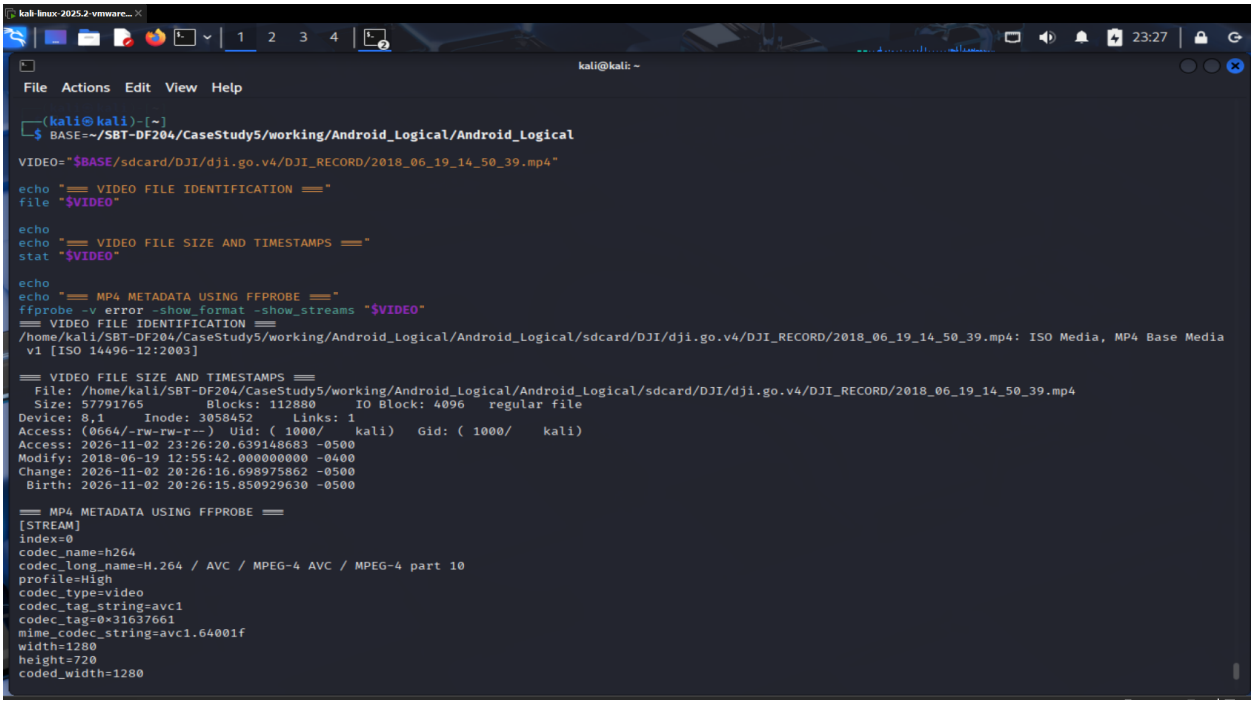
The metadata also contains GPS latitude, longitude and altitude fields. These have been identified as potentially relevant location evidence and will be handled as sensitive information in the report, with precise coordinates redacted from the general findings. The metadata provides useful correlation points, but its fields alone do not establish the identity of the drone operator or prove the circumstances of the flight.



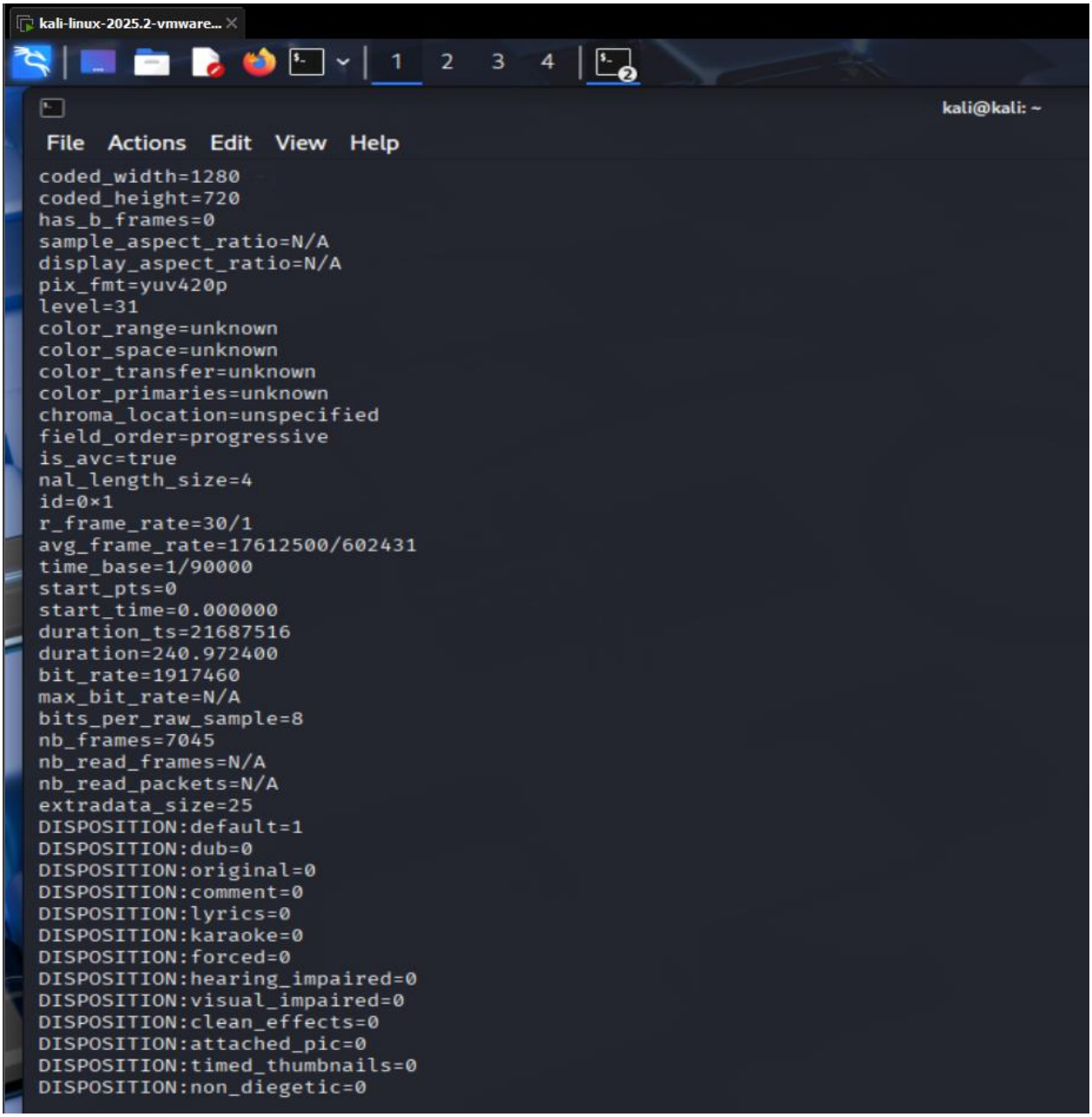
Field	Recorded value	Forensic significance
CaptureDate	2018/06/19 14:50:39	Recorded video capture date/time
StartTimeMsec	0	Beginning of the video timeline
EndTimeMsec	234833	End of the video timeline
FPS_Drone	30	Drone video frame rate
PixelXDimension_Drone	1920	Drone video width
PixelYDimension_Drone	1080	Drone video height
Sync_Drone_Time	Multiple values	Drone-side synchronisation points
Sync_Local_Time	Multiple values	Local-side synchronisation points
DeviceMaker	DJI	Recorded device manufacturer

Examination of the cached MP4 video

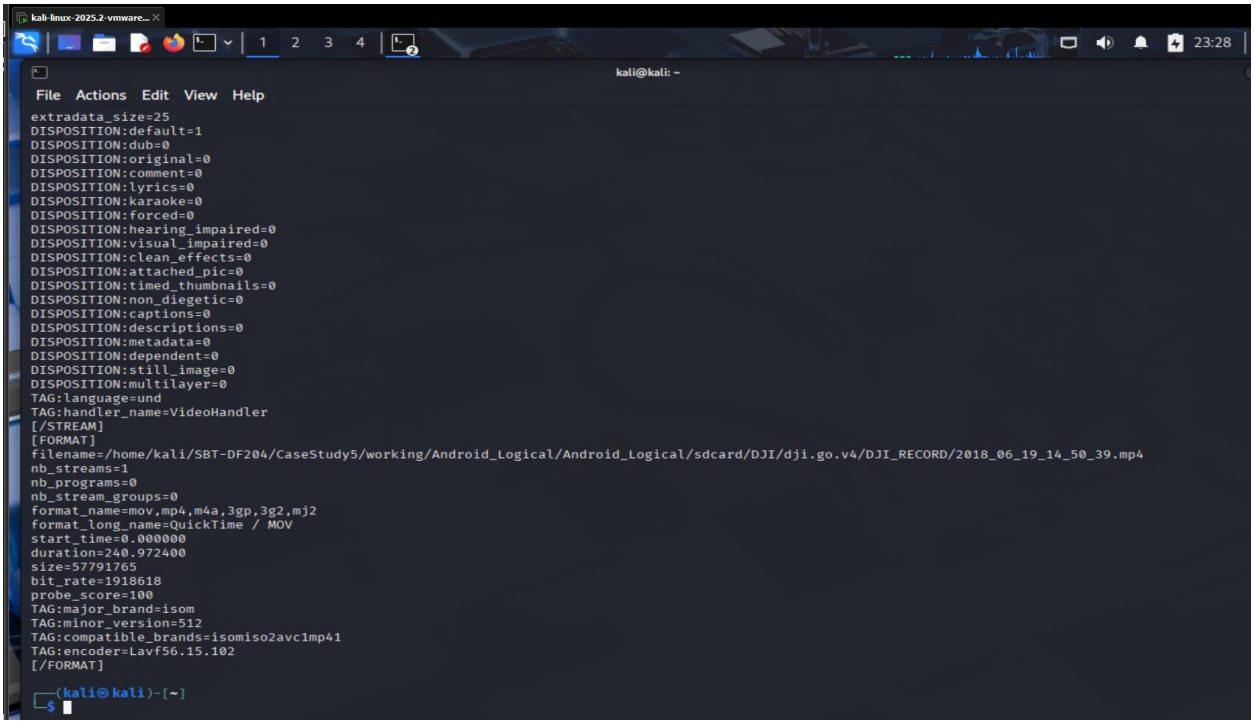
The cached video file, 2018_06_19_14_50_39.mp4, was examined using the Linux file utility, stat and FFprobe. The file utility identified it as an ISO Base Media/MP4 file. Its recorded size was 57,791,765 bytes. The filesystem metadata showed a modification timestamp of 19 June 2018 at 12:55:42, with a displayed UTC−04:00 offset. This timestamp represents filesystem metadata and should not, without further corroboration, be treated as the actual recording start time.



FFprobe identified one video stream encoded using H.264/AVC, with a resolution of 1280 × 720 pixels and a nominal frame rate of 30 frames per second. The video stream contained 7,045 frames and had a reported duration of 240.972400 seconds (approximately 4 minutes and 0.972 seconds). The MP4 format metadata also reported the same duration and file size.



A comparison with the associated INFO file revealed a difference in the reported video duration. The INFO file records an EndTimeMsec value of 234,833 milliseconds, equivalent to approximately 234.833 seconds, whereas FFprobe reports 240.972400 seconds for the MP4. The difference is approximately 6.139 seconds. The INFO file also records drone video dimensions of 1920 × 1080 and local video dimensions of 1280 × 720, which is consistent with the 1280 × 720 dimensions identified in the MP4 itself for the local video stream. These differences and correspondences have been recorded for further investigation rather than treated as evidence of alteration.



The difference between the two duration values requires attention. The INFO file's EndTimeMsec value is 234.833 seconds, while the MP4 container reports 240.972400 seconds. The difference is approximately 6.1394 seconds.

Cryptographic hashing of cached video artefacts

SHA-256 hashing was performed on the cached MP4 video and its associated INFO metadata file to establish cryptographic integrity reference values for subsequent examination. The MP4 video produced the SHA-256 hash 2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00, while the INFO metadata file produced a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a. These hash values were recorded in the examination notes to support future integrity verification and identification of the exact files examined. Hashing does not establish the authenticity of the original recording or the accuracy of its metadata; it provides a means of detecting subsequent changes to the files.

```
┌──(kali㉿kali)-[~]
└─$ BASE=~/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical

VIDEO="$BASE/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4"
INFO="$BASE/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info"

echo "== SHA-256 HASH: MP4 VIDEO =="
sha256sum "$VIDEO"

echo
echo "== SHA-256 HASH: INFO METADATA =="
sha256sum "$INFO"
== SHA-256 HASH: MP4 VIDEO ==
2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00  /home/kali/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4

== SHA-256 HASH: INFO METADATA ==
a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a  /home/kali/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info

┌──(kali㉿kali)-[~]
└─$
```

Extraction of representative video frames

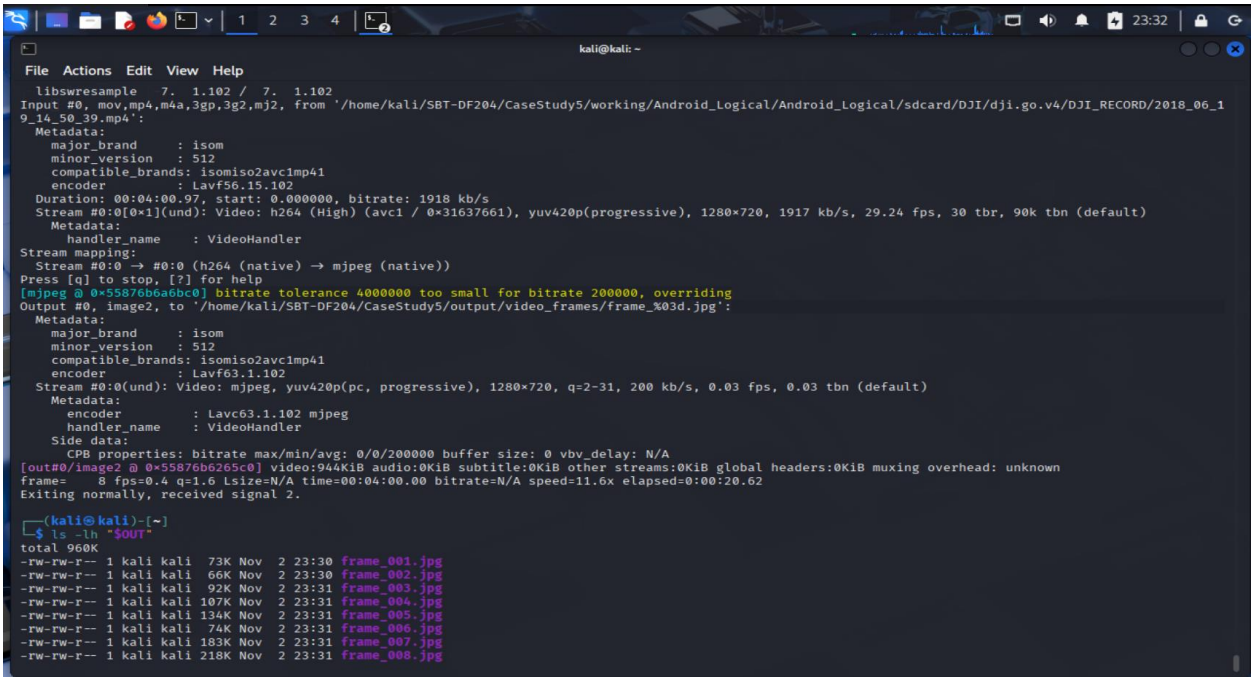
Eight representative still frames were extracted from the cached DJI MP4 video using FFmpeg, with a sampling interval of approximately 30 seconds. The extracted JPEG images were saved in the designated output/video_frames directory for visual examination. The extraction process produced eight image files, with individual file sizes ranging from approximately 66 KB to 218 KB. This sampling approach provides an overview of the video content at selected intervals while avoiding the need to extract and review every frame. The extracted frames will be examined for visible environmental features, landmarks, objects and changes in viewpoint that may assist with the investigation.

```
kali-linux-2025.2-vmware...
1 2 3 4                                                                    23:32
kali@kali: ~
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ BASE=~/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical

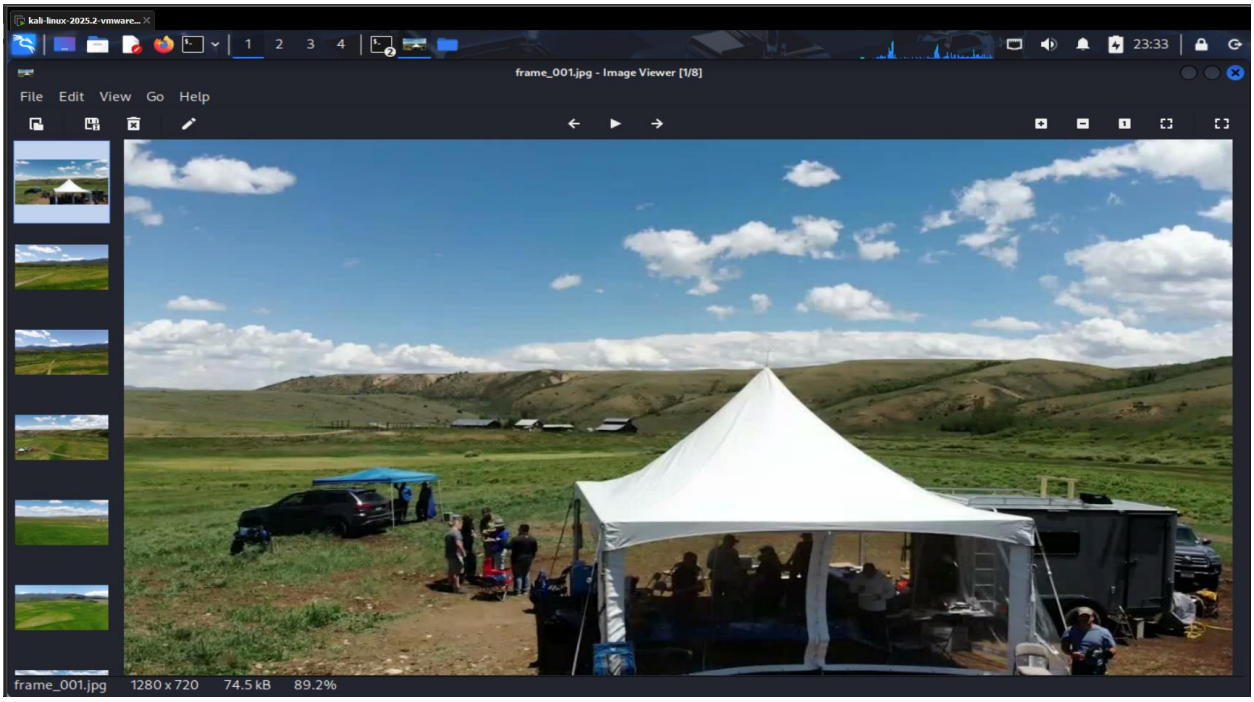
VIDEO="$BASE/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4"
OUT="$HOME/SBT-DF204/CaseStudy5/output/video_frames"

mkdir -p "$OUT"

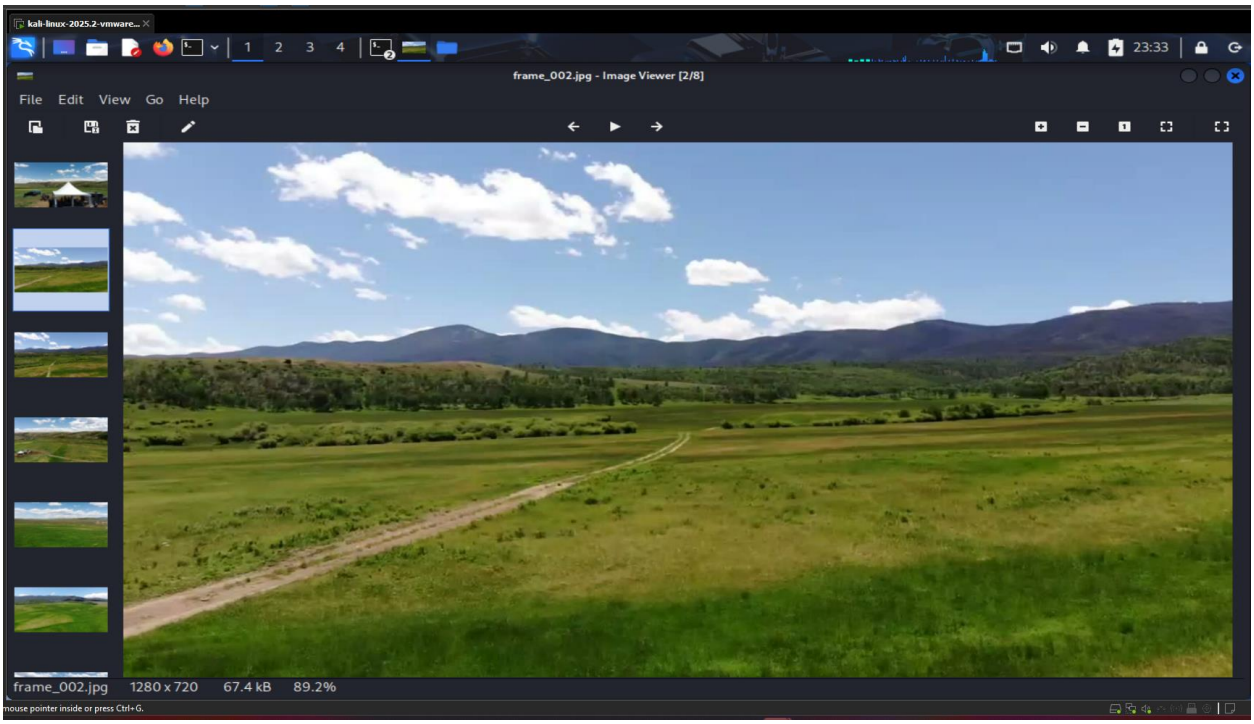
ffmpeg -i "$VIDEO" \
-vf "fps=1/30" \
"$OUT/frame_%03d.jpg"
ffmpeg version 9.0.2-3 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 16 (Debian 16.2.0-3)
  configuration: --prefix=/usr --extra-version=3 --toolchain=hardened --libdir=/usr/lib/x86_64-linux-gnu --incdir=/usr/include/x86_64-linux-gnu --arch=amd64 --enable-gpl --disable-stripping --disable-pocketsphinx --disable-libcaca --disable-libmfx --enable-gnutls --enable-libaom --enable-libass --enable-libbs2b --enable-libcdio --enable-libcodec2 --enable-libdav1d --enable-libflite --enable-libfontconfig --enable-libfreetype --enable-libfribidi --enable-libgme --enable-libgsm --enable-libharfbuzz --enable-libmp3lame --enable-libmysofa --enable-libopenjpeg --enable-libopenmpt --enable-libopus --enable-librubberband --enable-libshine --enable-libsnappy --enable-libsoxr --enable-libspeex --enable-libtheora --enable-libtwolame --enable-libvidstab --enable-libvorbis --enable-libvpx --enable-libwebp --enable-libx265 --enable-libxml2 --enable-libxvid --enable-libzimg --enable-openal --enable-opencl --enable-opengl --disable-sndio --enable-libvpl --enable-libdc1394 --enable-libdrm --enable-libiec61883 --enable-chromaprint --enable-frei0r --enable-ladspa --enable-libbluray --enable-libdvdnav --enable-libdvdread --enable-libjack --enable-libjxl --enable-libpulse --enable-librabbitmq --enable-librist --enable-libsrt --enable-libssh --enable-libsvtav1 --enable-libx264 --enable-libzmq --enable-libzvbi --enable-lv2 --enable-sdl2 --enable-libplacebo --enable-librav1e --enable-librsvg --enable-shared
  libavutil      61.  1.102 / 61.  1.102
  libavcodec     63.  1.102 / 63.  1.102
  libavformat    63.  1.102 / 63.  1.102
  libavdevice    63.  1.102 / 63.  1.102
  libavfilter    12.  1.102 / 12.  1.102
  libswscale     10.  1.102 / 10.  1.102
  libswresample   7.  1.102 /  7.  1.102
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from '/home/kali/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4':
  Metadata:
    major_brand     : isom
    minor_version   : 512
    compatible_brands: isomiso2avc1mp41
    encoder         : Lavf56.15.102
  Duration: 00:04:00.97, start: 0.000000, bitrate: 1918 kb/s
  Stream #0:0[0×1](und): Video: h264 (High) (avc1 / 0×31637661), yuv420p(progressive), 1280×720, 1917 kb/s, 29.24 fps, 30 tbr, 90k tbn (default)
    Metadata:
      handler_name    : VideoHandler
Stream mapping:
  Stream #0:0 → #0:0 (h264 (native) → mjpeg (native))
```



The first extracted video frame (frame_001.jpg) depicts an open grassy landscape with rolling hills in the background. A large white marquee tent, a smaller blue canopy, and several vehicles are visible in the foreground, with multiple individuals gathered around the structures. The scene appears to have been recorded during daylight under partly cloudy weather conditions. These visual observations have been documented as preliminary scene characteristics for comparison with other extracted frames and the available video metadata. No conclusion regarding the precise location or identity of individuals has been drawn from this frame alone.



The second extracted frame (frame_002.jpg) presents a broader view of the surrounding landscape. A dirt track is visible running diagonally across the grassy foreground towards the middle distance. Beyond the open grassland, a line of trees and vegetation separates the foreground from an elevated range of hills or mountains. The scene is illuminated by daylight under partly cloudy conditions. Compared with the first frame, the camera viewpoint has shifted away from the temporary structures and vehicles, providing a wider view of the surrounding terrain. These features have been documented for comparison with subsequent frames and the location metadata recovered from the associated video file.

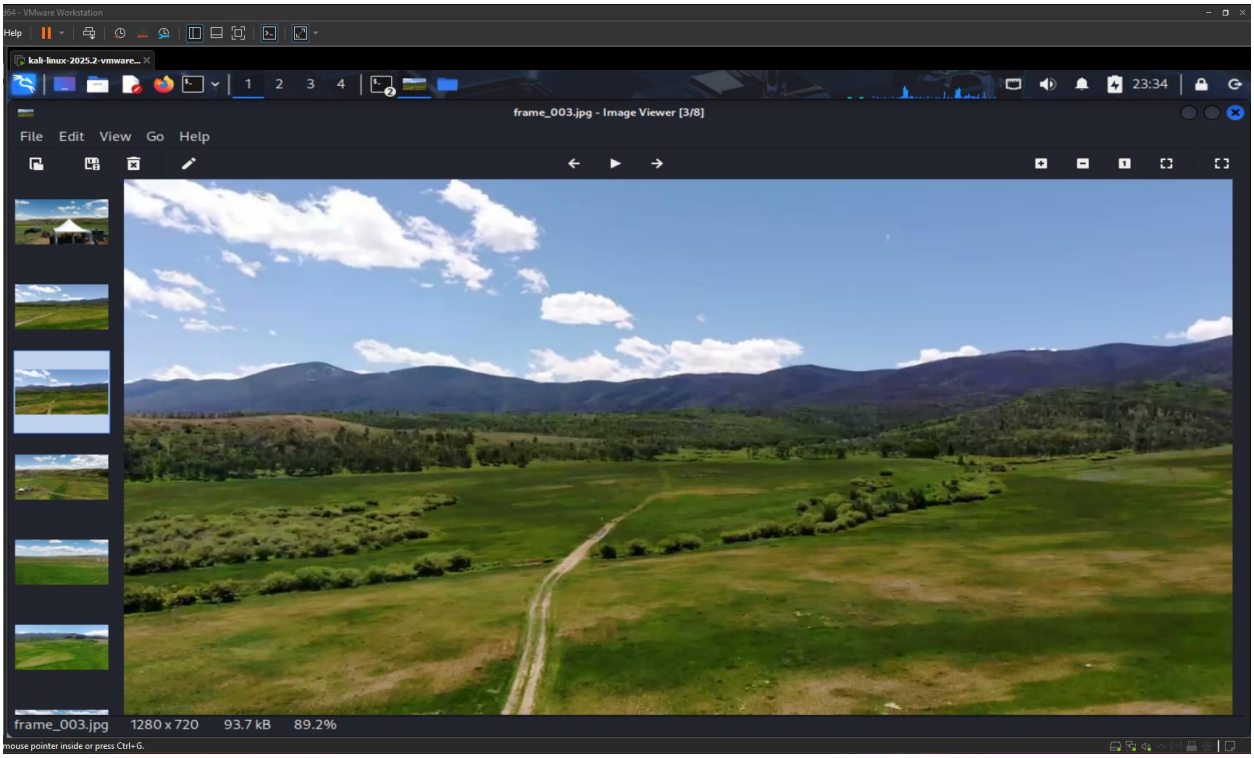


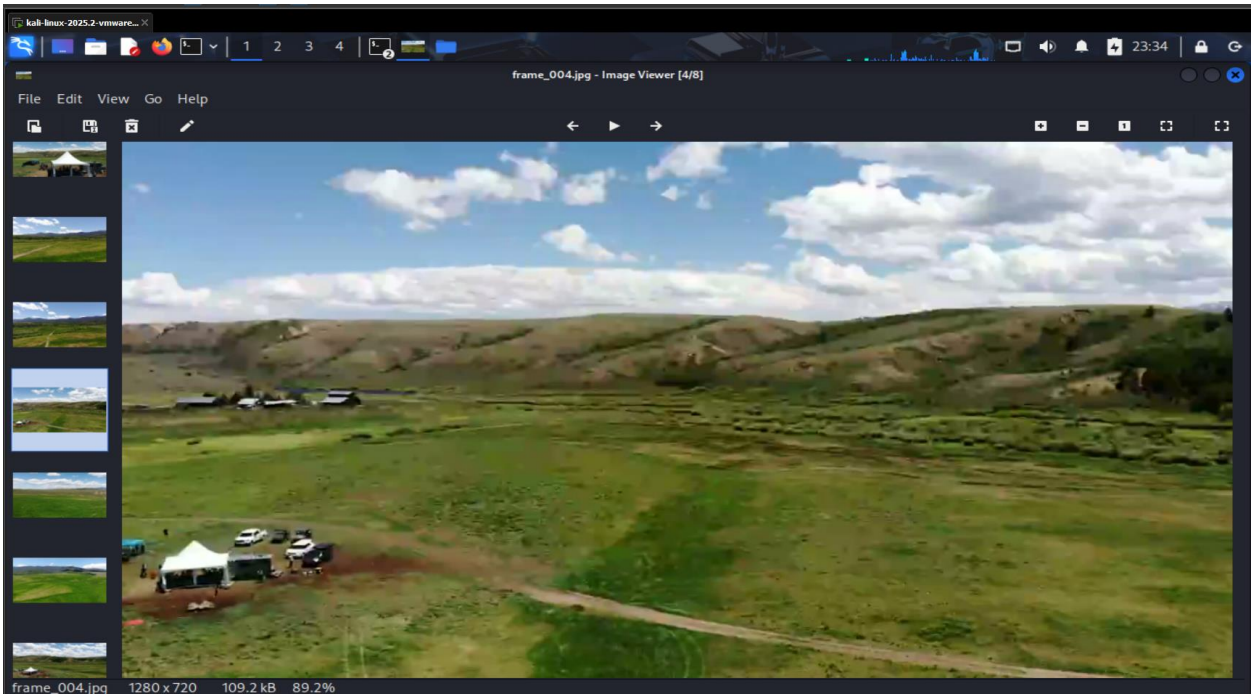
Visual examination

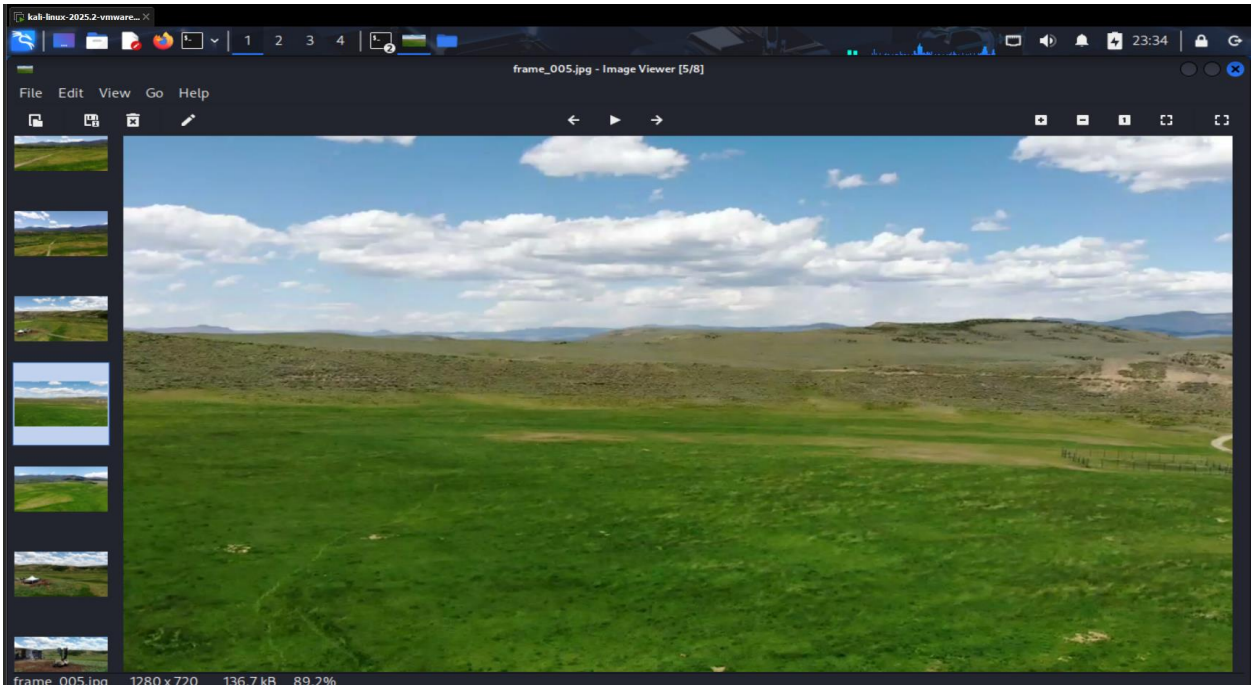
The second extracted frame (frame_002.jpg) shows a wide, open grassland landscape. A dirt track runs diagonally from the lower-left foreground towards the middle distance. Dense vegetation and a line of trees separate the grassland from a range of elevated terrain in the background. The sky is partly cloudy, with substantial visibility across the landscape.

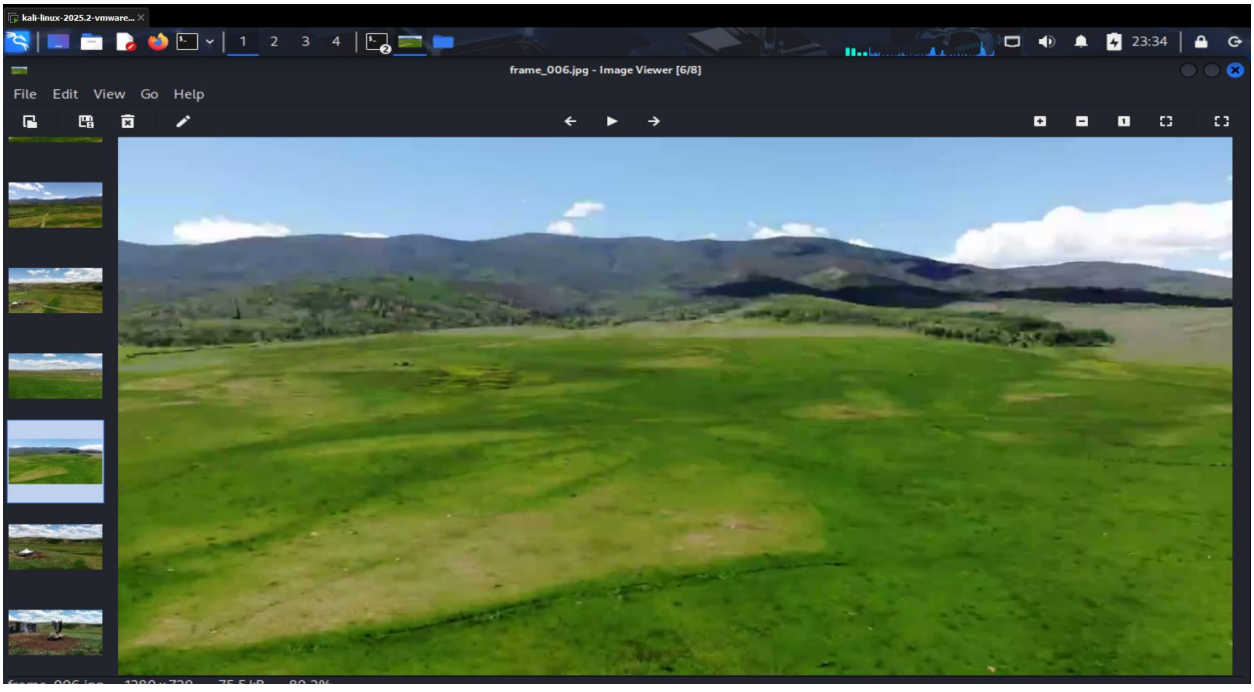
Notable comparison with Frame 1:

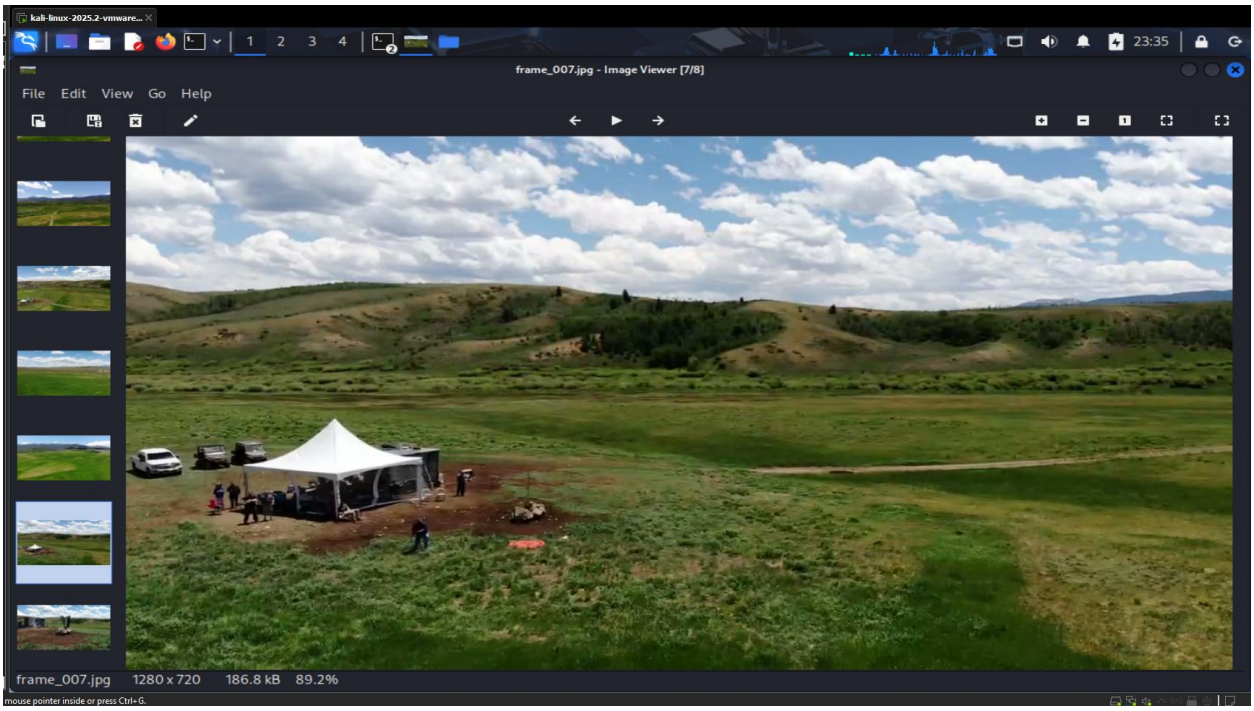
- The camera viewpoint has shifted away from the tent, vehicles and gathering.
- The dirt track and surrounding grassland are now prominent.
- A distant mountain or hill range is visible.
- The open terrain provides additional environmental features for possible location correlation.

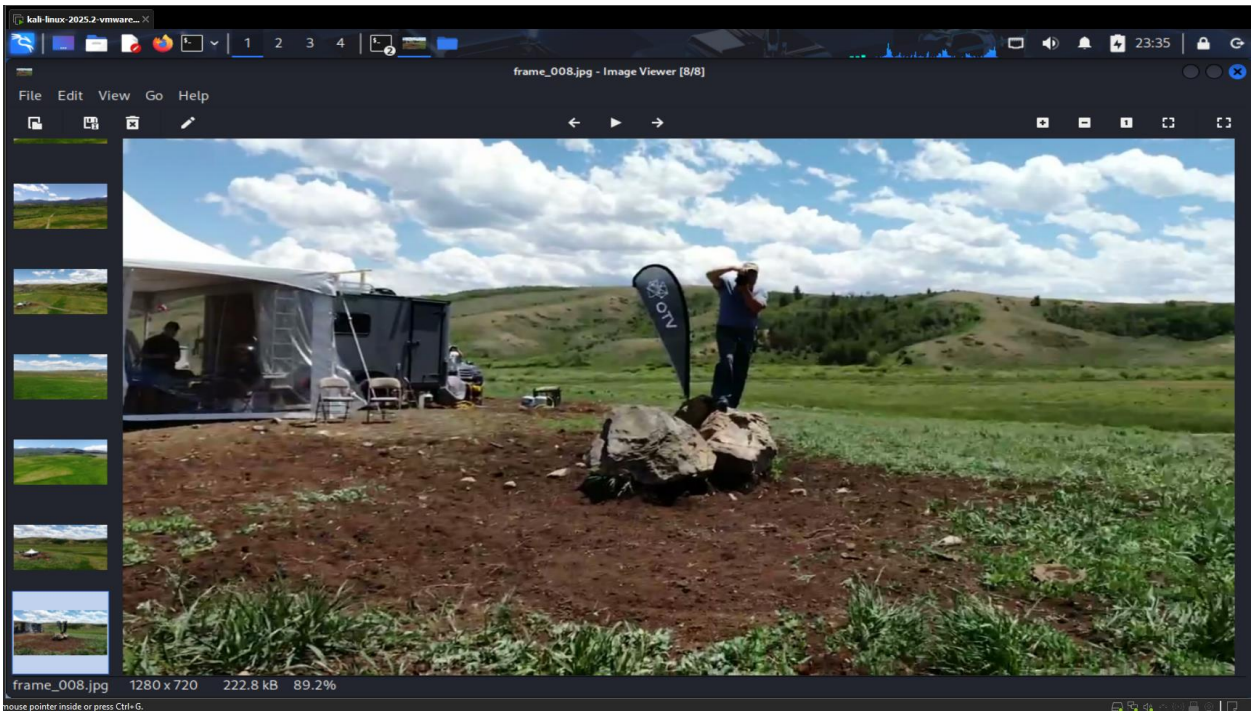

kali-linux-2025.2-vmware...
23:34
frame_003.jpg - Image Viewer [3/8]
File Edit View Go Help
frame_003.jpg 1280 x 720 93.7 kB 89.2%


kali-linux-2025.2-vmware...
23:34
frame_004.jpg - Image Viewer [4/8]
File Edit View Go Help
frame_004.jpg 1280 x 720 109.2 kB 89.2%


kali-linux-2025.2-vmware...
1 2 3 4
23:34
frame_005.jpg - Image Viewer [5/8]
File Edit View Go Help
frame_005.jpg 1280 x 720 136.7 kB 89.2%


kali-linux-2025.2-vmware...
1 2 3 4
23:34
frame_006.jpg - Image Viewer [6/8]
File Edit View Go Help

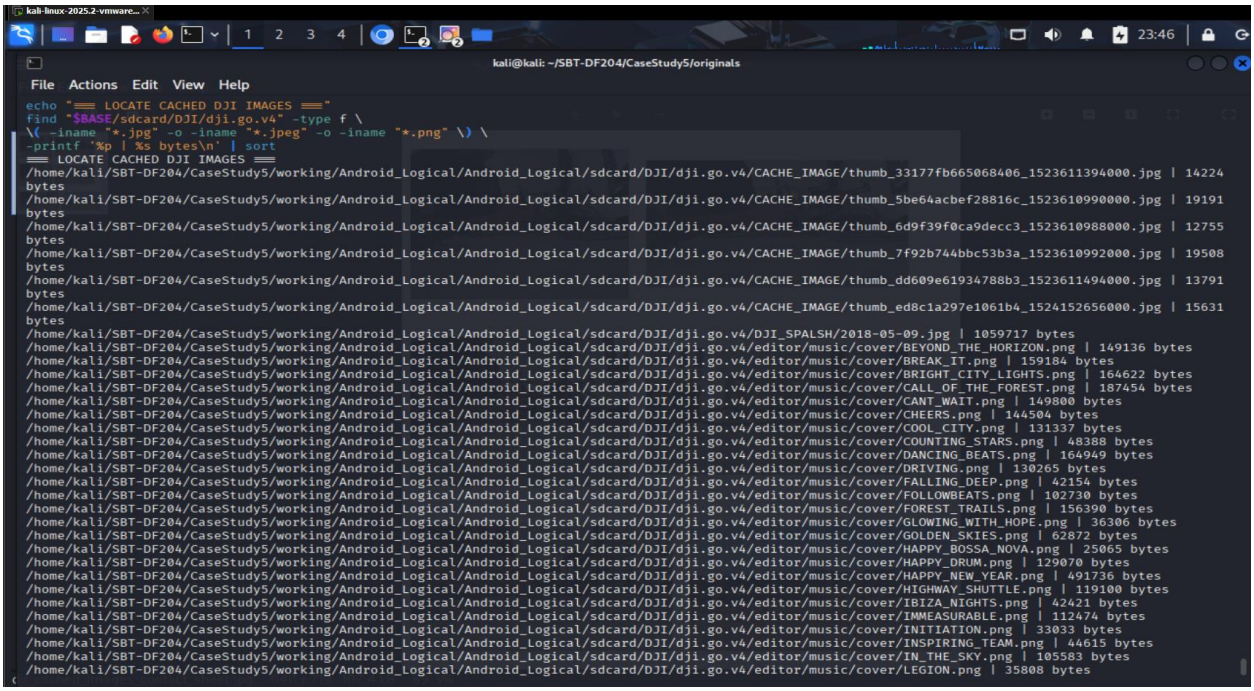




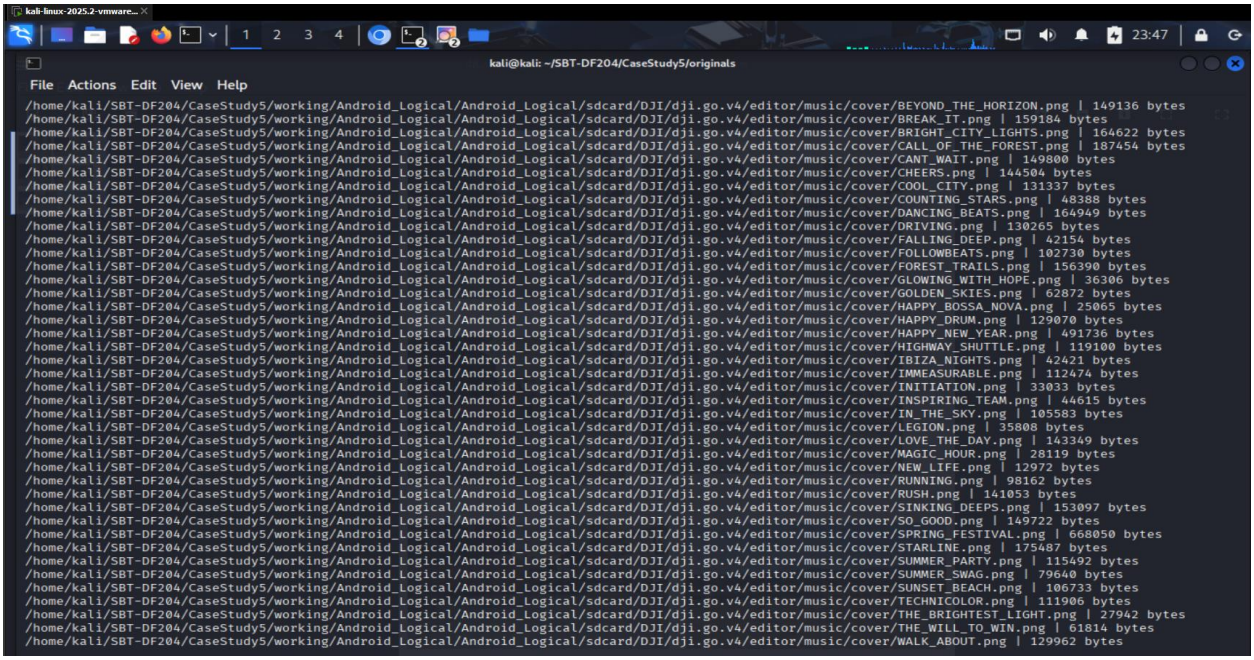
xamination of DJI GO 4 Cached Images

The DJI GO 4 application directory was searched recursively for JPEG and PNG image files to identify visual artefacts potentially relevant to the investigation. The search identified six JPEG thumbnail files within the CACHE_IMAGE directory. The thumbnails

were named using unique hexadecimal identifiers followed by numeric timestamp components. Their file sizes ranged from 12,755 bytes to 19,508 bytes.



The search also identified additional image files in other application directories, including a DJI splash-screen image and several music-cover graphics associated with the application's video-editing functionality. These were distinguished from the six cached thumbnails and excluded from the initial incident-related image review because their directory locations and apparent application purposes differ.

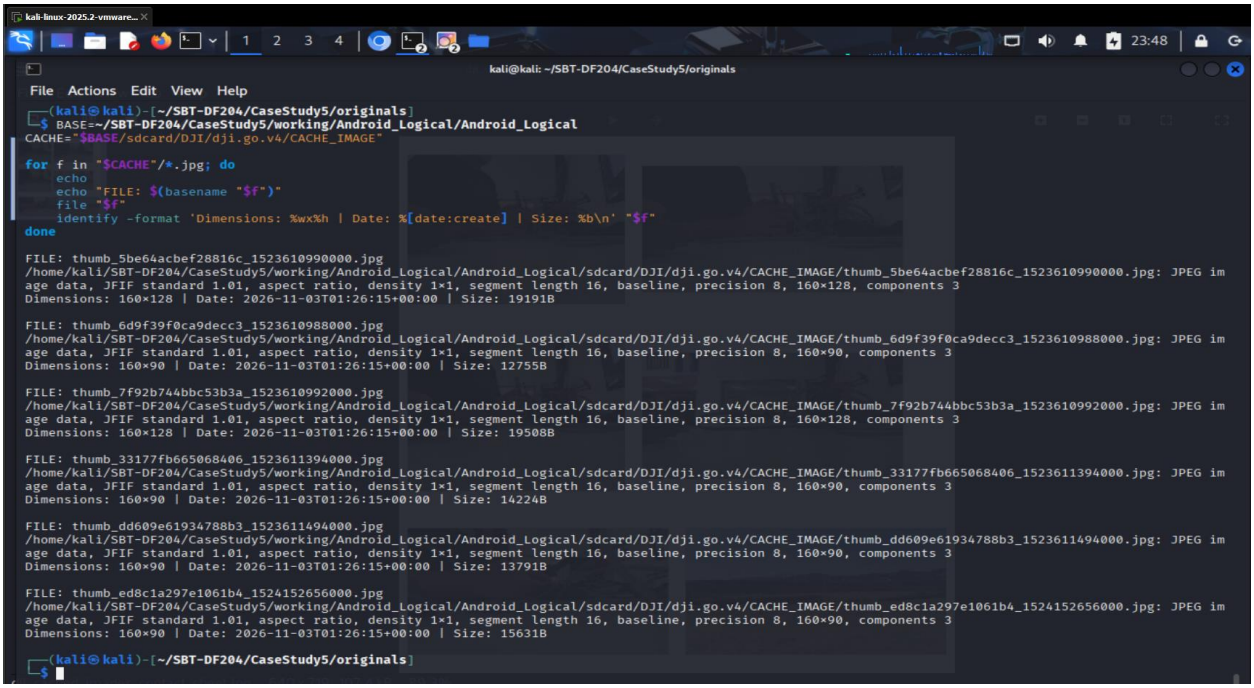


The six cached thumbnails have been retained for further examination, including image metadata inspection and visual comparison with the video frames extracted from the associated MP4 file. Their filename timestamps will also be investigated to determine their temporal relevance. At this stage, no direct association between these thumbnails and the flight recorded on 19 June 2018 has been established.

Technical Examination of Cached DJI Images

The six JPEG files recovered from the DJI GO 4 CACHE_IMAGE directory were examined using the Linux file utility and ImageMagick's identify command. All six files were recognised as valid JPEG images using the JFIF 1.01 standard. Their dimensions comprised four images measuring 160 × 90 pixels and two measuring 160 × 128 pixels. Individual file sizes ranged from 12,755 to 19,508 bytes.

The examination also identified creation-date metadata displayed as 3 November 2026 for all six images. As this date may reflect the extraction or processing environment rather than the original capture event, it has not been treated as the photographs' capture date. The numeric timestamp components embedded in the filenames have been preserved for further temporal analysis.

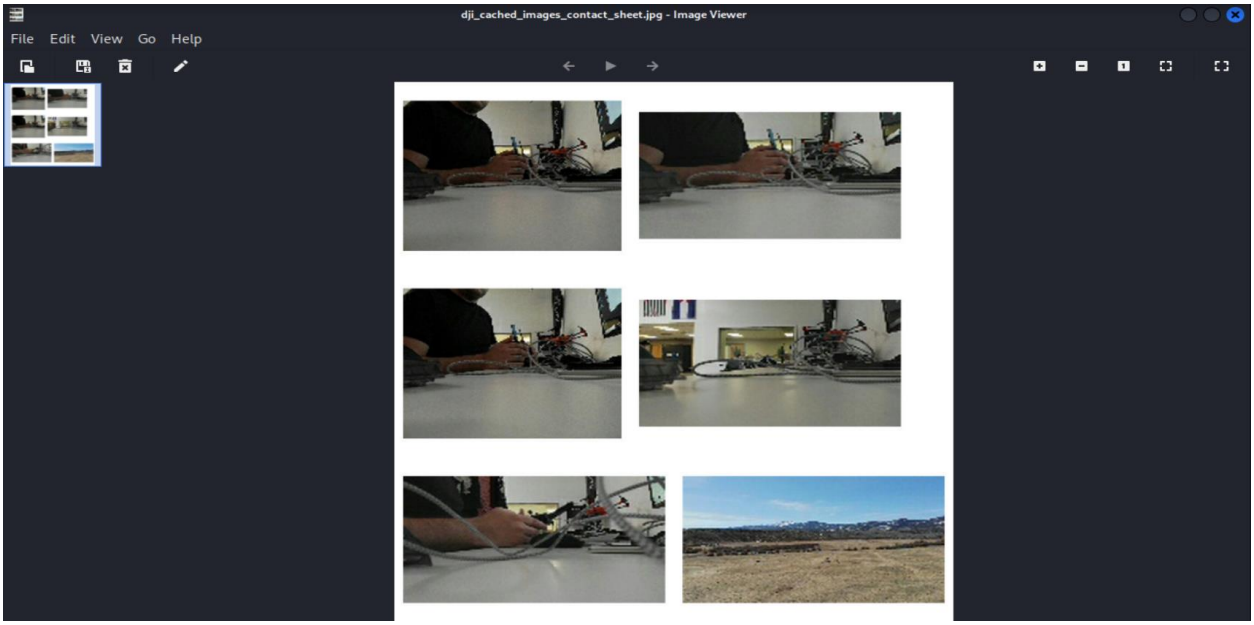


A contact sheet containing the six cached images was generated to facilitate comparative visual examination. The images will be reviewed for visual content and potential relationships with the flight video and other recovered DJI artefacts.

Cached image metadata findings

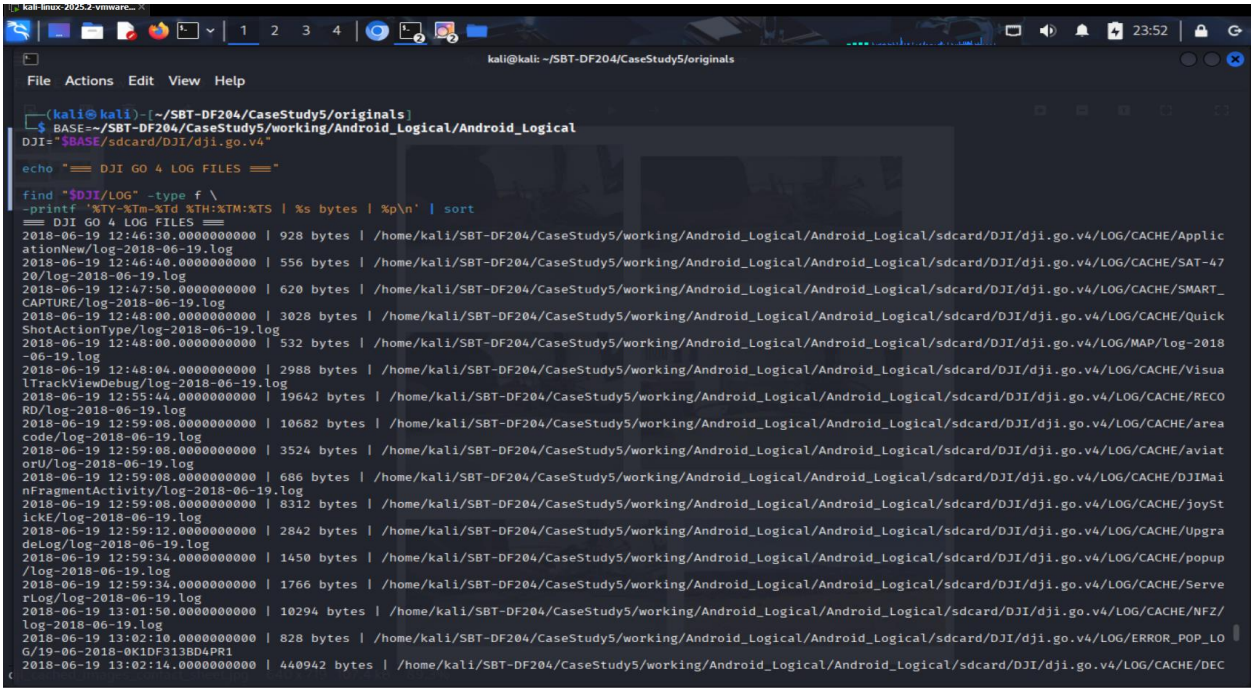
All six files were identified as valid JPEG images.

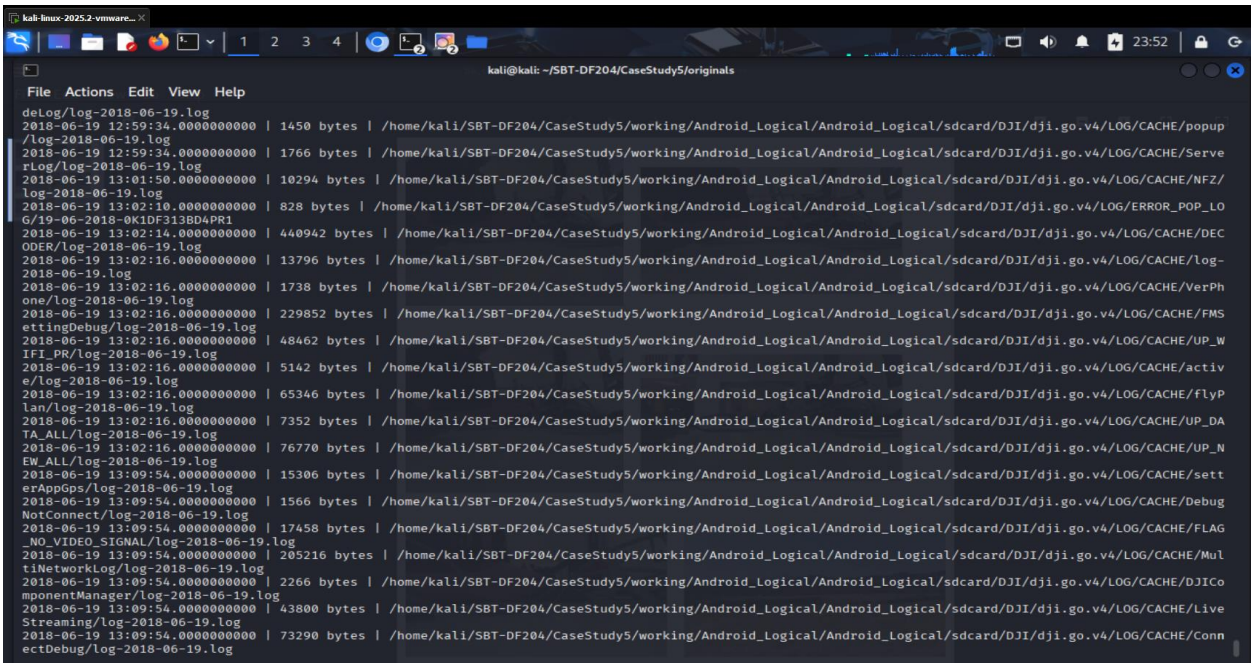
Thumbnail	Dimensions	Size
thumb_5be64acbef28816c	160 × 128	19,191 bytes
thumb_6d9f39f0ca9decc3	160 × 90	12,755 bytes
thumb_7f92b744bbc53b3a	160 × 128	19,508 bytes
thumb_33177fb665068406	160 × 90	14,224 bytes
thumb_dd609e61934788b3	160 × 90	13,791 bytes
thumb_ed8c1a297e1061b4	160 × 90	15,631 bytes



DJI GO 4 application log inventory

The DJI GO 4 application log directory was inventoried recursively to identify available application logs and their associated filesystem metadata. The examination identified multiple log files dated 19 June 2018, including records associated with recording, active application activity, flight planning, connection debugging, video decoding and other application components. The RECORD, active, flyPlan and ConnectDebug logs were identified as potentially relevant to reconstructing application activity surrounding the cached video. The larger DECODER log was also retained for further examination. The file modification timestamps were recorded as filesystem metadata and have not, at this stage, been interpreted as definitive timestamps for flight or recording events.

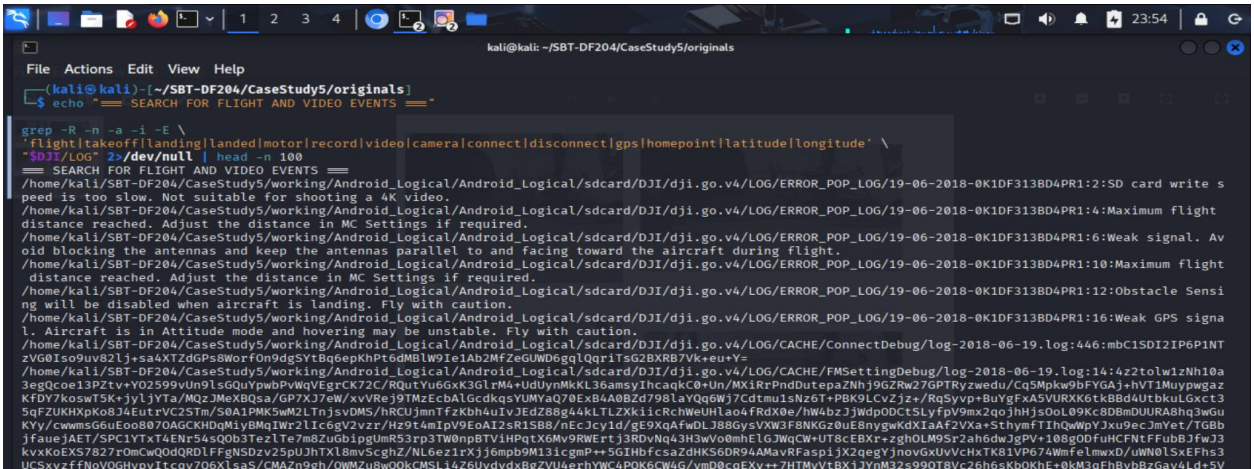


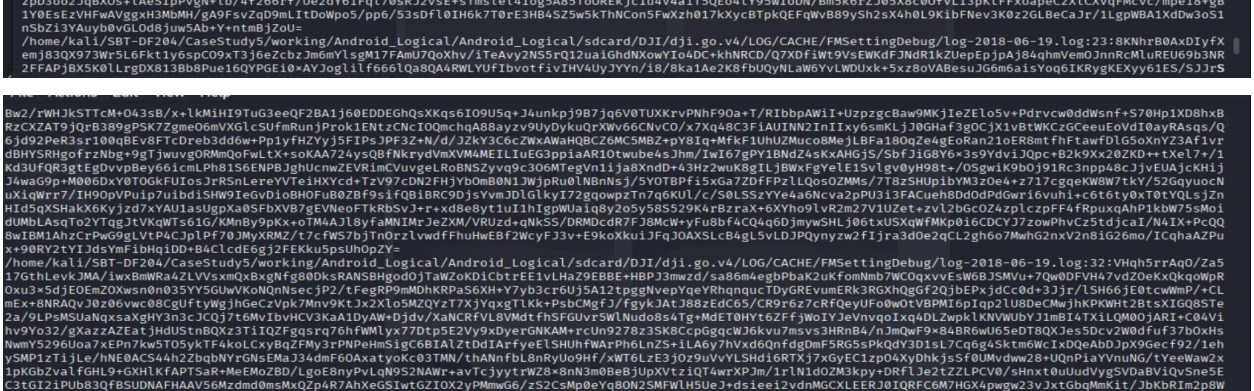


DJI GO 4 application log inventory and initial review

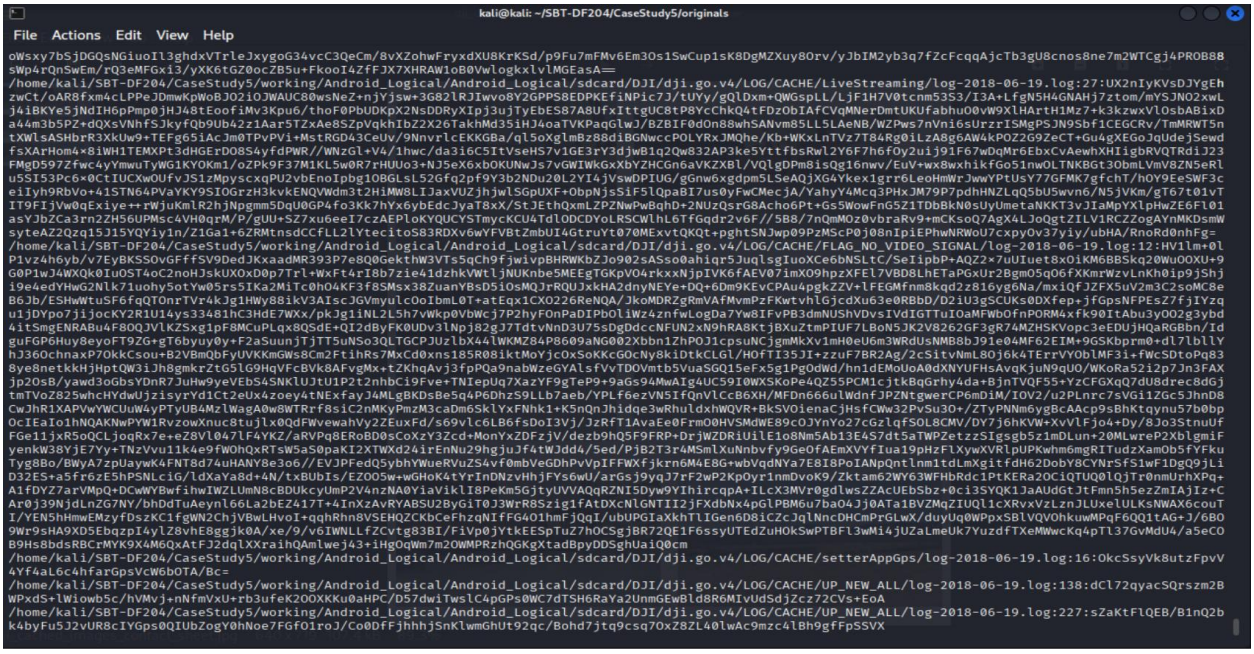
The DJI GO 4 application log directory was examined to identify available logs and determine whether they contained information relevant to the recorded flight and cached video. The inventory identified multiple log files dated 19 June 2018, including application, recording, flight-plan, decoder, GPS-setting, connection-debug and other component logs. An initial keyword search identified an error/notification log containing messages

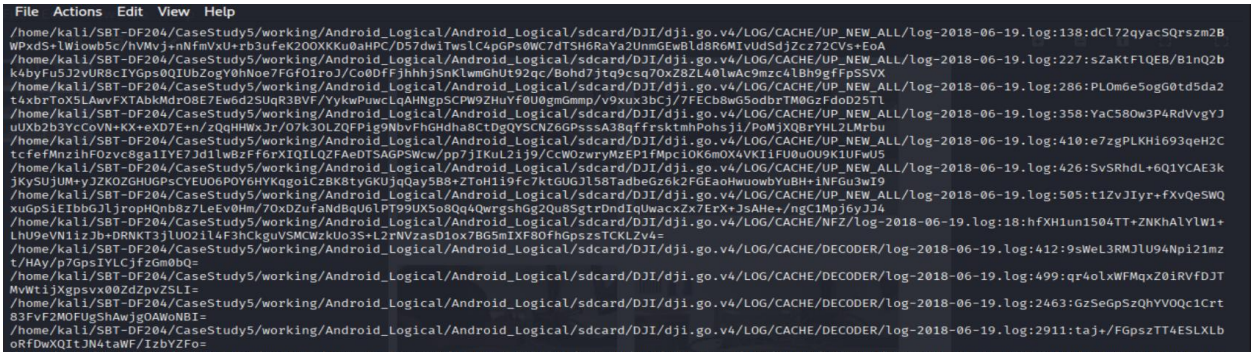
concerning slow SD-card write speed, maximum flight distance, weak signal, obstacle-sensing behaviour during landing and weak GPS reception. These messages were recorded as application-generated warnings presented for further examination. They provide contextual evidence of conditions or warnings presented by the application, but do not independently establish that a particular flight event occurred. The data also returned long encoded-looking strings in some debug logs; these were not treated as interpretable evidence.





```
wF+hG5iP3Hphwi3DMiXWQ+OsJZ3+6KrG0U0XpWMC5zp0h6vF/Y6t5jL/rxnj8W/Iyqp0h3oGg+lJywg75pTR5TN56ucBvKxlEX5GT2nkLTxqX2AlFFpHaKaKEGIcoD4rTqEOMMKGdBIZRy8Xprvh3/dUcj
IYXvvKrjeNImAmtiUIo9M+Q4ZGWbHHu3x+dqdDcZ8wyFvNHMBpx8O/62BTXse0as/Ao9HW/xokDMhsTxeRUegAnsqW+fCzQoVIMebhX7FoFfxfv4+aBIA4nV4HGhPk1V2OcRrMbePAq6xr3f0bghn1SERJ
1KB+6DqF0DSfJp0a1lV1j4QEyctkYcYdsP8pyS+C/rwrD9bhUEiwADhS0SCfVZ/mOSx6iOqnUKSabcFYX4UstDhLfgLnNpYw8W2BY/FMqbkTcPTFdb0P7NYVgSzlae1S0fvw/H+vWlcUWjGSyjgY070g1b
h8b6vKTY2Gyg7DVivVXgwNUtPhZT920Z3PXSfIw3N58unB8wZ6ewnen8UiU0gFtuTuh1YfSCO+AsSD05mR1AA6V73zwq6qSdn7di897K2dPhC9lH+OiBgDhVmIsEFRQx0TzA9e9l191rxLipdJoVArnMll
qXi/rkz8EjdU2k3jtD4wAw7xR9c+v7L9btxEkIwBywd1YoRFINcPYqMqiTgSoUM0161q/ntoVsnwyvh/Zb55XyRz3VQRCGCLw2KrFXCqPMFF775xeONS89no8IbdxHRwx76eu1ZGnJxm/VNh2h
sHT86A/pxu6s1fMHjsBoDXMiW63ghLMqd7SefImngPsCG4r0jKIZe4bdB4ydJlOh6t6sAFm5D+1TVSrmxBlZThYtV3rQnZLMu7LdRUluH3J2cQzGMExE3AO2FNMRh3lkaMBVj9Opu6MUgMl6nSp/AD3BI
+XL+aqixQeQ4xBHEFCbhY+aE4LgZJSalXvLlqipo/x9L+pl03U/dTk7s1TmVX6T2a0yQvo2r7wMb55H2zTYWg+eKf10kP3+nWAOeYd6ZzoHGsc+NA80EWxK8D83u6HRSt6QxhsnbqGSyZXR6913aCwDSqf
noRjx8gP/fm5TAM9PAa33X7SSQvM3rWzrbVUTDl8szR/GP+XVnaGij1K7x9dw/6+I991SGZZGH/DxBtRyt52qCpKFdNcegmknlFqlenOiQKS/J6LeEbmpMieOhKLwrtxoxWmQJE1NhgZG4Zh7WZl/NCi6c
HXhpR7zlvxaK6EdJAfUmbwN0GPm1LJUtC8QfVby8Gt1LmvrgiPKIujbUnv26gFHhqIrS/XfBDcq/BvxOfXa6YKGlliWScxhbbCwZjBbt9dCwTBhKSZbk3YOloEX+ABIUA2MYpHGCNzvcMTcaGsqzeSLLbR
ZTvNQnYskfM2xr41si j2Qaa/7H6WysVqVVk34FlTwyPG5UV46sfOV/W0s8Mb0H/tqL/CYmhImi53NixDbaUN3EYkuFUs1+qkmuwA1LUEsXNMt8MGHj4xNbI5SRk/dgKQBO68eWRpZp3/ySmuTB8biDCJNr
fIC0sB2jjXR9qr2GNZrpEgQ8nKhoSQ2o2NFhhD/xb4uSDJVOXVmy4yjBCcLcmzKmUj5rvdZTfB/8ASy/FVa+g48nBxN5khbSsDfl7yHMhGKX1BZTuQTsZIMKNrW5oZAV/cqwlXao+dylOTL9UABWT1rOuJ
GuaVmo0FK3rcdM7iUzkrWso8aWYcw2eAmmELCWRLQDFWgSryz26bjxzRhFjRddXBK0fqj4cofNio7MNZ/SgpedhnqSG+HJeSogKp/in4hfCoOkeEpLGwPwPdiZrVnIyMVvYgtMIBw4JZUEMcyKVzdzHSx1
uSOYUjinw82SeIWvexI3bWOgsZ5xBtmA795kFUbJiTBdrTPPzBJ965u848AN1aitBT0jG/uTZhWnQHL9B9ceGw/drXGNcu5EyoMW4vi5Sn/URr4xzAOEOZQRlSusc0OjVIOh9V4P9qklBfAASCijiU+wmx
F5RwLO3CIun1ZUdFVEfthZ4/Ydd/3ceD7mfPfCLYVrGVjaIRM=
/home/kali/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/FMSettingDebug/log-2018-06-19.log:81:TqGISIZThkIaN8
eCwObQqcnss1QKh+YANmxdc5prZzuLHEK9G2kOMbYEXUZSUJzThvbT/fUbGo3tGhjGUtbQcjavvyAPYow0sZDciBqnLMaKnegsGTFHiCXmZd81k/xF6+TESSJjLaQdHi0cCEPebUhJ9xx1V33A3i9roTI
/u4NRaIFwTjPEgAYfqf+iSWiH2cl8dlNdm9e7+yIJI28Pf/NBMsSnGdO2udE2qPAJa9dMXXmGge0nuhKmHJoJOg0ySpXla/ZG/siZGzlOaBmQwRlrWMpEDXip9p9AbR/BBILkYKSnqev+OpGkQczvD+Cg1
MRHEw0qih4eUNhnssF8ZgI3KEsid9UnStB4nQM4CXhCnIITfTO6eSsvWDUOv/b8vmKzhWMVUt2tBtAuxl95sV+tNYHgidgFYWakFr1obAe3HcMyRRIvY2GvGdHSs+s0/cpZtrWyB3tsDZaRQh7SjDxSnjcF
I4YkcZDIt+R7EVFMq8oLaYguKH8h1aC+mObq6n9MLyO5DUV0K91T0JKfuMAxZmV7ILB05KKzh/qNdeC7l8+yE1zi6zOTxHAUVvQupuBwZj/ZPSljA0zToJIQ0SIPg6B/1ae6l7LsWN8eoHh/Nc4i6g9zb
1KkWtLSQ6bf1+pljddV4/i2utogHuXUNDKUicvdbk/legmGC3V/Acj+FpkRg2DBzHCzzxIL1mowKFGoH2ROU+M9L4+pQavPmiGchIZc7KBwFrgXFPjHcufvRgrJwep4l5cgC4yxPh6Wuz4C3P2Q5sOn1V
ghes/+NZhR0Xuf58s4Y3XtcDb1wSfGS1mEUL19zE0nUIHWTinouvW+B8Ibr8HBNs9g678RDCT0YXMfUt+JCvelr+J5TOYf91dKq12462GF0xyNasFG388ADENqdfSw4+4Kua2itELcS/ROX4kmhbxhV4O8
pwkCbnGaYz9y/3PtBEvwIE4aAmGYXR8+mkOXLsLdf6uoZbTDo9+mcw867NO9TEmcjy0/pci8014S1LpKXYDzaq9VadOBmyVkMP13znyfTJ+WPMO18/VvQ2uH+pOoZxmOKOsd5pf5k3opLC6EXvt3Q6Yv
uZfviD5T9Kt3SzA8k9Mgl1Qrpb0vi7lvt8GzOsZlhkP401RtzgubRri6Ov3dCY79XzyYYH7+sChfQxQzmCtcYDN30KIDnYyCd6BtkRKwLy85zR5xjJ8hvf4zJHL2q4U8t5qIvO4TNO3IwLM5ohgD7pXddR
Ffmc8nBujyyfGsSYfKtKns0KOGp1mRZ8fUmNNlAu/W9R04kjJ76Qvn/L0sebeId8n9k+sSlK/TY9p+tDPdr1hmXmyYrkgUw02n7R8YgCGfge+Wb8RHvKmxwtDpmnpiX/F10pTKTDmOwqpvoP2qs3yLfQF
jkp+hSFimcvyoNwe5XKZe4gB+N+ZyodaDVxLxFWLmgbhc04mB7d6zyYv/mp5CHh/R3y8Qt0b17jc80rYKAnkOsaJ8MGFYClsb/PZGbIi56vGVqODRniKoayZjQ36hpxen6gvs4y0remyKaSL1S1uBv9U0A
54ouMFRM+7ijqzMvguVgD/NaiN/T+csjMmYbloUPG82emlfFvbmaNM+L1Jw9wY9h7sT77mk32lkI5nBq1rBlnAR4nSsjYMn/InroH5T0gNKja69A4/SlClaBKiLvZRe3p9+zxKaPKaF4oFy1o007b13BlI
14sqxxI28jG3ZEwW4Q+MVK4KxTRbbu2di7C259b9W5Zqleszd0G6TkRVuDqRbgfKXYZtfvi8cAkzJI/n51q47R1oMHABXDeUjgTy6G7LBRNQJjQehcoU2DhZ0PAKRRmYpOoMuhXvod/CNu6ueS7Zf1K688
YrgIMvFg2iqo+TPtjPoPbxNIaxuVqOYwcXerXN/c+Ow7vWoEHyNQQxXEzdAVCUJsH5ZtijlaWtjeLRb/HXHqHfmQYuxlErdHvpWYRQFVZPbGt1h6ITLuT9xGiztgAgnqRnOsFxgMUp0B08v+cAyZlEm2+g
pjfhNsUqeB33liUCxvXflnnTC4nZZIHc1bkH8Jj7sRG6auUuIZQoaC3auKi+hwZlpG7S2az+h65ybgpS2J87YwuME3gX6SM/ZspNkpX5uauf6goHhQFVoWwR8f/Iwh6VKOsd725WSnw5pv83obSA1Pach9
tAYyz0/etXQryOKwVDTt0Un1Nuu2DTzSMhIj1FEUkieLwGxWw=
/home/kali/SBT-DF204/CaseStudy5/working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/FMSettingDebug/log-2018-06-19.log:93:nS4Y0akkhcesXk
g/uDhqyGCL0j57wePCaovIsr18V6/dKsZfHdK/rbk2ad89HEgndjKXCnVdRK3ilgnNGuMAXAYSVOnd/sLC2MD7NO5tOvWATyMgkzAUxry7Kcvdvio/lzTiAw253xMFDBgydzNAN0VRurDahFgpshZGa/tC
RaAE9bko7iCofOclE+KBQxf1X7pBmTH75i5oXKMmvTrdT/mjo56RJOHrCCmy2JFTxwMzawNLeJHyeRCbVXPXVjLmMo9Q9RcxVyYMiX/pof/Tx4frUwd3JBFaoCpnMtH/t/fyJcLKvCBbFBLOv98sgTVpVu
D9IoZowJLhqhTk5HO7zSzRgCZ3Aluc4rNm/AAAe7rb7eB4vR+HXhvSSNO8Ng8GX2QuSbzROTWigrG5PfHelCZ6AA1RaVV2eJYti0rNaSAj9FL20WgS2jx3pJN5uKF31SUEQ6tsVtllL55tzCBZLLb+L9x5
AVRwmIz3JIO3aTwIQTHDRlENcnLO812nxwr3XLNR0nWA7IYYXJl1yhRv3pIeHoyE68MmQS5Q5WP0MfnQlmuaofJGh+EoTmom6JOOQnFsI/dpL0FIK54fTHI17SHlyHALA1OVh71Xp9GXthyX06BBA73il1
zGDnXRzk91HGWugIBpuXGV8hILS/8fm5UKHYtHNY+NP+Y1zznjw7bl0U
```





The DJI GO 4 logs contain an error/notification log dated 19 June 2018 with messages relating to flight distance, signal strength, landing and GPS conditions.

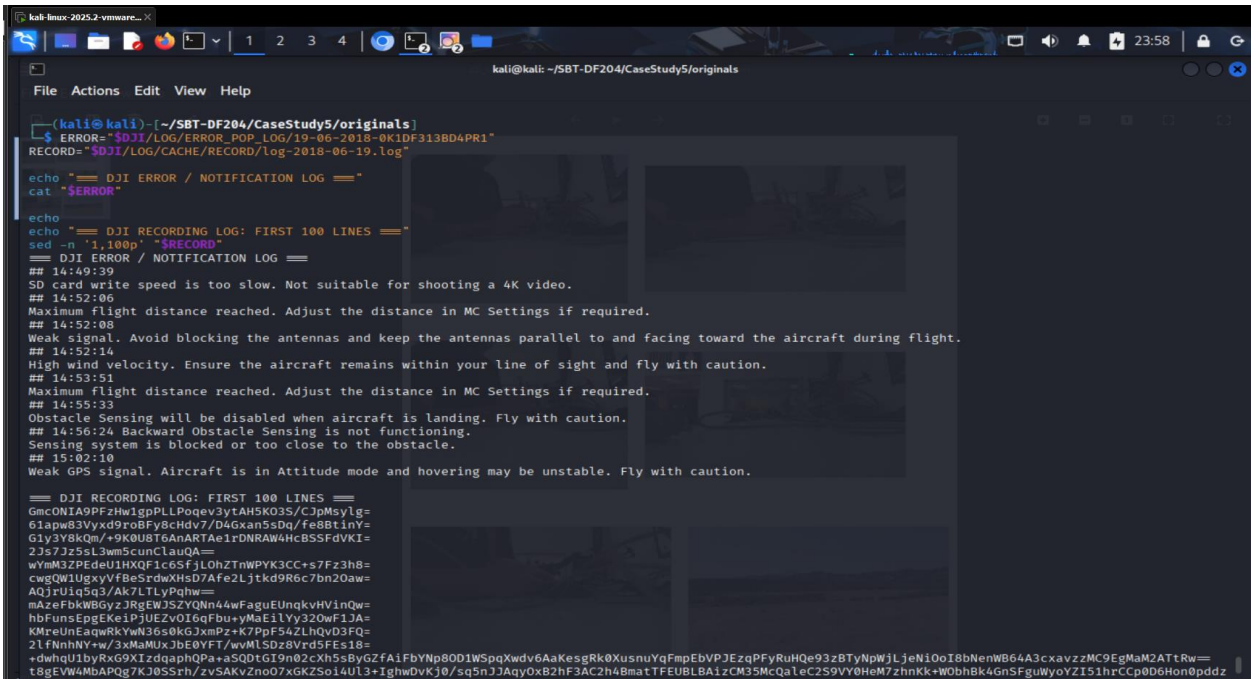
The key findings from the output are:

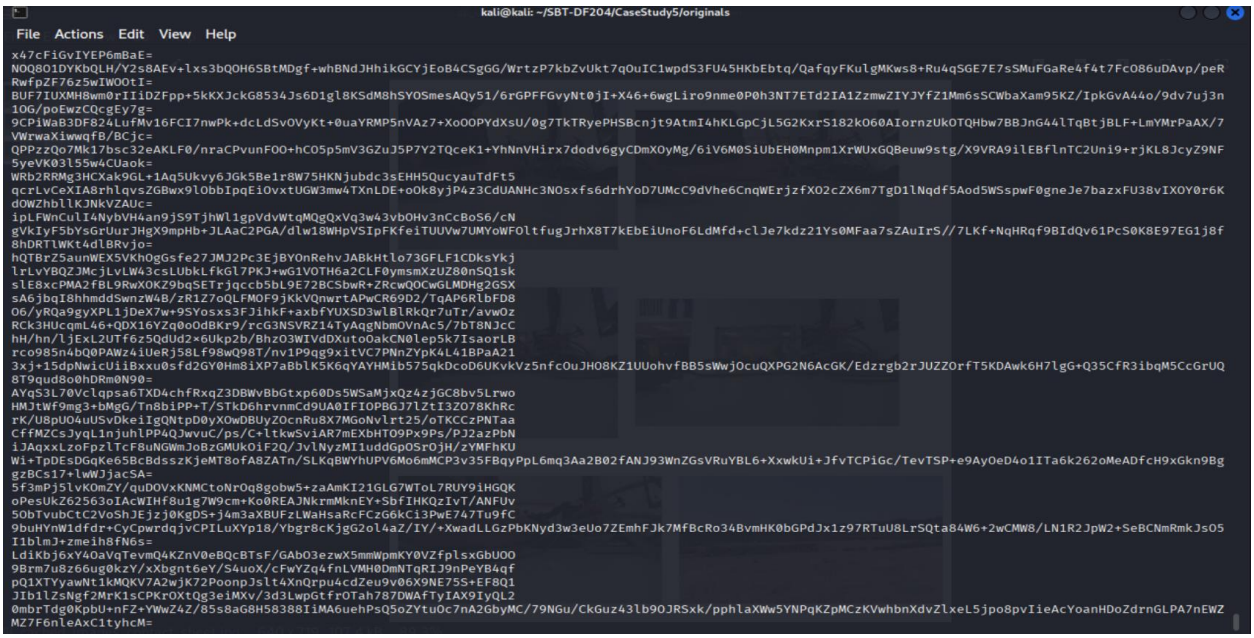
- SD-card warning: “SD card write speed is too slow. Not suitable for shooting a 4K video.”
- Flight-distance warning: “Maximum flight distance reached. Adjust the distance in MC Settings if required.” This appears twice.
- Signal warning: “Weak signal. Avoid blocking the antennas…”

- Landing-related warning: “Obstacle Sensing will be disabled when aircraft is landing.”
- GPS warning: “Weak GPS signal. Aircraft is in Attitude mode and hovering may be unstable.”

Examination of timestamped DJI GO 4 error and notification records

The DJI GO 4 error and notification log, identified as 19-06-2018-0K1DF313BD4PR1, was examined to recover timestamped application warnings relevant to the recorded activity. Eight entries were identified, covering the period from 14:49:39 to 15:02:10 on 19 June 2018. The earliest entry, at 14:49:39, reported that the SD-card write speed was too slow for 4K video recording. Subsequent entries recorded maximum flight-distance warnings at 14:52:06 and 14:53:51, a weak-signal warning at 14:52:08, and a high-wind warning at 14:52:14. At 14:55:33, the application displayed a warning that obstacle sensing would be disabled when the aircraft was landing. A further entry at 14:56:24 reported that backward obstacle sensing was not functioning, while the final entry, at 15:02:10, reported weak GPS reception and indicated that the aircraft was in Attitude mode. These entries provide timestamped evidence of warnings generated by the DJI application during the recorded period. They have been retained for correlation with the cached video, INFO metadata and other flight records. The warnings are not, in isolation, proof that the corresponding aircraft events occurred. Their timestamps have also been preserved as recorded pending confirmation of the applicable timezone.


kali-linux-2025.2-vmware...
23:58
kali@kali: ~/SBT-DF204/CaseStudy5/originals
File Actions Edit View Help
(kali㉿kali)-[~/SBT-DF204/CaseStudy5/originals]
$ ERROR="$DJI/LOG/ERROR_POP_LOG/19-06-2018-0K1DF313BD4PR1"
RECORD="$DJI/LOG/CACHE/RECORD/log-2018-06-19.log"
echo "=== DJI ERROR / NOTIFICATION LOG ==="
cat "$ERROR"
echo
echo "=== DJI RECORDING LOG: FIRST 100 LINES ==="
sed -n '1,100p' "$RECORD"
=== DJI ERROR / NOTIFICATION LOG ===
14:49:39
SD card write speed is too slow. Not suitable for shooting a 4K video.
14:52:06
Maximum flight distance reached. Adjust the distance in MC Settings if required.
14:52:08
Weak signal. Avoid blocking the antennas and keep the antennas parallel to and facing toward the aircraft during flight.
14:52:14
High wind velocity. Ensure the aircraft remains within your line of sight and fly with caution.
14:53:51
Maximum flight distance reached. Adjust the distance in MC Settings if required.
14:55:33
Obstacle Sensing will be disabled when aircraft is landing. Fly with caution.
14:56:24 Backward Obstacle Sensing is not functioning.
Sensing system is blocked or too close to the obstacle.
15:02:10
Weak GPS signal. Aircraft is in Attitude mode and hovering may be unstable. Fly with caution.
=== DJI RECORDING LOG: FIRST 100 LINES ===
GmcONIA9PFzHw1gpPLLPoqev3ytAH5KO3S/CJpMsylg=
61apw83Vyxd9roBFy8cHdv7/D4Gxan5sDq/fe8BtinY=
G1y3Y8kQm/+9K0U8T6AnARTAe1rDNRAW4HcBSSFdVKI=
2Js7Jz5sL3wm5cunClauQA==
wYmM3ZPEdeU1HXQF1c6SfjLOhZTnWPYK3CC+s7Fz3h8=
cwgQW1UgxyVfBeSrdwXHsD7Afe2Ljtkd9R6c7bn2Oaw=
AQjrUiq5q3/Ak7LTLyPqhw==
mAzeFbkWBGyzJRgEWJSZYQNn44wFaguEUnqkvHVinQw=
hbFunsEpgEKeiPjUEZvOI6qFbu+yMaEilYy32OwF1JA=
KMreUnEaqwRkYwN36s0kGJxmPz+K7PpF54ZLhQvD3FQ=
2lfNnhNY+w/3xMaMUxJbE0YFT/wvMlSDz8Vrd5FEs18=
+dwhqU1byRxG9XIzdqaphQPa+aSQDtGI9n02cXh5sByGZfAiFbYNp8OD1WSpqXwdv6AaKesgRk0XusnuYqFmpEbVPJEzqPFyRuHQe93zBTyNpWjLjeNiOoI8bNenWB64A3cxavzzMC9EgMaM2ATtRw==
t8gEVW4MbAPQg7KJ0SSrh/zvSAKvZnoO7xGKZSoi4Ul3+IghwDvKj0/sq5nJJAqyOxB2hF3AC2h4BmatTFEUBLBAizCM35McQaleC2S9VY0HeM7zhnKk+WObhBk4GnSFguWyoYZI51hrCCp0D6Hon0pddz


kali@kali: ~/SBT-DF204/CaseStudy5/originals
File Actions Edit View Help
x47cFiGvIYEP6mBaE=
NOQ8O1DYKbQLH/Y2s8AEv+lxs3bQOH6SBtMDgf+whBNdJHhikGCYjEoB4CSgGG/WrtzP7kbZvUkt7qOuIC1wpdS3FU45HKbEbtq/QafqyFKulgMKws8+Ru4qSGE7E7sSMuFGaRe4f4t7FcO86uDAvp/peR
RwfpZF76z5wIWOOtI=
BUF7IUXMH8wm0rIIiDZFpp+5kKXJckG8534Js6D1gl8KSdM8hSYOSmesAQy51/6rGPFFGvyNt0jI+X46+6wgLiro9nme0P0h3NT7ETd2IA1ZzmwZIYJYfZ1Mm6sSCWbaXam95KZ/IpkGvA44o/9dv7uj3n
10G/poEwzCQcgEy7g=
9CPiWaB3DF824LufMv16FCI7nwPk+dcLdSvOVyKt+0uaYRMP5nVAz7+XoOOPYdXsU/0g7TkTRyePHSBcnjt9AtmI4hKLGpCjL5G2KxrS182kO60AIornzUkOTQHbw7BBJnG44lTqBtjBLF+LmYMrPaAX/7
VWrwaXiwwqfB/BCjc=
QPPzzQo7Mk17bsc32eAKLF0/nraCPvunFOO+hCO5p5mV3GZuJ5P7Y2TQceK1+YhNnVHirx7dodv6gyCDmXOyMg/6iV6M0SiUbEH0Mnpm1XrWUxGQBeuw9stg/X9VRA9ilEBflnTC2Uni9+rjKL8JcyZ9NF
5yeVK03l55w4CUaok=
WRb2RRMg3HCXak9GL+1Aq5Ukvy6JGk5Be1r8W75HKNjubdc3sEHH5QucyauTdFt5
qcrLvCeXIA8rhlqvsZGBwx9lObbIpqEiOvxtUGW3mw4TXnLDE+oOk8yjP4z3CdUANHc3NOsxfs6drhYoD7UMcC9dVhe6CnqWErjzfXO2cZX6m7TgD1lNqdf5Aod5WSspwF0gneJe7bazxFU38vIXOY0r6K
dOWZhbllKJNkVZAUc=
ipLFWnCulI4NybVH4an9jS9TjhWl1gpVdvWtqMQgQxVq3w43vbOHv3nCcBoS6/cN
gVkIyF5bYsGrUurJHgX9mpHb+JLAaC2PGA/dlw18WHpVSIpFKfeiTUUVw7UMYoWFOltfugJrhX8T7kEbEiUnoF6LdMfd+clJe7kdz21Ys0MFaa7sZAuIrS//7LKf+NqHRqf9BIdQv61PcS0K8E97EG1j8f
8hDRTlWKt4dlBRvjo=
hQTBrZ5aunWEX5VKhOgGsfe27JMJ2Pc3EjBYOnRehvJABkHtlo73GFLF1CDksYkj
lrLvYBQZJMcjLvLW43csLUbkLfkGl7PKJ+wG1VOTH6a2CLF0ymsmXzUZ80nSQ1sk
slE8xcPMA2fBL9RwXOKZ9bqSETrjqccb5bL9E72BCSbwR+ZRcwQOCwGLMDHg2GSX
sA6jbqI8hhmddSwnzW4B/zR1Z7oQLFMOF9jKkVQnwrtAPwCR69D2/TqAP6RlbFD8
O6/yRQa9gyXPL1jDeX7w+9SYosxs3FJihkF+axbfYUXSD3wlBlRkQr7uTr/avwOz
RCk3HUcqmL46+QDX16YZq0oOd8Kr9/rcG3NSVRZ14TyAqgNbmOVnAc5/7bT8NJcC
hH/hn/ljExL2UTf6z5QdUd2×6Ukp2b/BhzO3WIVdDXutoOakCN0lep5k7IsaorLB
rco985n4bQ0PAWz4iUeRj58Lf98wQ98T/nv1P9qg9xitVC7PNnZYpK4L41BPaA21
3xj+15dpNwicUii8xxu0sfd2GY0Hm8iXP7aBblK5K6qYAYHMib575qkDcoD6UKvkVz5nfcOuJHO8KZ1UUohvfBB5sWwjOcuQXPG2N6AcGK/Edzrgb2rJUZZOrfT5KDAwk6H7lgG+Q35CfR3ibqM5CcGrUQ
8T9qud8o0hDRm0N90=
AYqS3L70VclqpsaGTXD4chfRxqZ3DBWvBbGtxp60Ds5WSaMjxQz4zjGC8bv5Lrwo
HMJtWf9mg3+bMgG/Tn8biPP+T/STkD6hrvnmCd9UA0IFIOPBGJ7lZtI3ZO78KhRc
rK/U8pUO4uUSvDkeiIgQNtpD0yXOwDBUyZOcnRu8X7MGoNvlrt25/oTKCCzPNTaa
CffMZCsJyqL1njuhlPP4QJwvuC/ps/C+ltkwSviAR7mEXbHTO9Px9Ps/PJ2azPbN
iJAqxxLzoFpzlTcF8uNGWmJoBzGMUkOiF2Q/JvlNyzMI1uddGpOSrOjH/zYMFhKU
Wi+TpDEsDGqKe65BcBdsszKjeMT8ofA8ZATn/SLKqBWYhUPV6Mo6mMCP3v35FBqyPpL6mq3Aa2B02fANJ93WnZGsVRuYBL6+XxwkUi+JfvTCPiGc/TevTSP+e9AyOeD4o1ITa6k262oMeADfcH9xGkn9Bg
gzBCs17+lwWJjacSA=
5f3mPj5lvKOmZY/quDOVxKNMCtoNrOq8gobw5+zaAmKI21GLG7WToL7RUY9iHGQK
oPesUkZ62563oIAcWIHf8u1g7W9cm+Ko0REAJNkrmMknEY+SbfIHKQzIvT/ANFUv
5ObTvubCtC2VoShJEjzj0KgDS+j4m3aXBUFzLWaHsaRcFCzG6kCi3PwE747Tu9fC
9buHYnW1dfdr+CyCpwrdqjvCPILuXYp18/Ybgr8cKjgG2ol4aZ/IY/+XwadLLGzPbKNyd3w3eUo7ZEmhFJk7MfBcRo34BvmHK0bGPdJx1z97RTuU8LrSQta84W6+2wCMW8/LN1R2JpW2+SeBCNmRmkJsO5
I1blmJ+zmeih8fN6s=
LdiKbj6xY4OaVqTevmQ4KZnV0eBQcBTsF/GAbO3ezwX5mmWpmKY0VZfplsxGbUOO
9Brm7u8z66ug0kzY/xXbgnt6eY/S4uoX/cFwYZq4fnLVMH0DmNTqRIJ9nPeYB4qf
pQ1XTYyawNt1kMQKV7A2wjK72PoonpJslt4XnQrpu4cdZeu9v06X9NE75S+EF8Q1
JIb1lZsNgf2MrK1sCPkrOXtQg3eiMXv/3d3LwpGtfrOTah787DWAfTyIAX9IyQL2
0mbrTdg0KpbU+nFZ+YWwZ4Z/85s8aG8H58388IiMA6uehPsQ5oZYtuOc7nA2GbyMC/79NGu/CkGuz43lb9OJRSxk/pphlaXWw5YNPqKZpMCzKVwhbnXdvZlxeL5jpo8pvIieAcYoanHDoZdrnGLPA7nEWZ
MZ7F6nleAxC1tyhcM=


eNpEWLJ5y/gvyhW9ACGU6odnzjIfnrdUSkfJptk2bh5ZMb+GXgMEFDrRvvBDAsyI
p5N1iegWaUbPwAFqeBYUdYRe94dToOnt6dtNb/w0aXUEVKRl2rWaiEbuC/seENHDy60P/aumvHtDghGpGVW2XcEz0bDP41rq39W/Y1NIW3EwI/gh1cKFGDaaVnnyYnFZ1CpgTuYVBPg3Lu1tm9Q2Xm0pF9
K6CMM2iZczUvJJT+o=
ZK9Hnm0yzqOoZ/JBqP4D302iZHxbfXISYYKS9eW5bxhir4JLkkDVD51mExBMfJSS
QKU1xtv4tFG8uvRT8S4k76NGamgns4WuYQm9TMOfSfzMbHMvfTh+WEhUcW8W5FQP
MeI7l23bqQmt9tQYpKeZfWwzXIbHgkLzoG7dE7GsJgQBJKSyG+MntGsFa9+ZY7qr
q+Q3yR4aURWxcyvJJc0ooA/EpUTAme3Zwv2K64cpBCGVS4at3ltyenfTnxCz7QMBTOro/TTxGU3GcixGZPp6WMLI6MdSOIM6aZBp1B1+ferjwj32zH6n+U+zoxkVoZNui4Hqp7Essz6O3gE4qat1amPSZ4
JdEnLGNyO7rLfpHbk=
GhLyyxSwSyCAm9cp81kZWT3Wz8FWo0IBLZu3tnWrgx5Cw1dzyaScmvXEDHv/Kb14
Jwko0RlSbzGPzXYQ1wJcuw3J8zCyuR1HhJHooVDL79GU9MGH+tQMlMXH3uFQiW9M
G5F6/18UGmzAx/2NxGBvpRTe8QTVc6PytD4irDAITX25C1uoh8fb3/z9FhpDoVVByeUHNnQnsjHV4Yl3UMM3TjJnaQgvcecBxQ/5LNY5CWpNNKa0tByc7xPv15wvJx8nUxG9m4tcx+RUP2+APFOsBXccEz
blxne0AftyYfIbOSg=
ZOO2eNcmULF0Zpun9aRn7BPsiwVd9NTTnjLj4LSFWbaRnvL1StGFzaZXiScvBjzR
D78FlgzOU9eUTLds3SwkYqMmyHRTc/Gj1a7qXcoYJyAnwGZK3TAzrCFfPmoJHQaN
nwOBuhxXqyxD2oM3rtmg4enbfwvkcOgMvs8DUSIg87yhXrODyxDUjpN3FT5ttDFv
0kQy9O/970bvCOwHfoL2Uvf6/ytFTJbpsNbXweYprS4idVB2fQ8yxITvuN+8VogSwKik3palrAhctGdLei/Fm8rtCZSKS4EkqVw8TaUmjRY+NudfItE0mFX+qQv8+dtZVV3SBwU1S5PaXehp0gzHDlqZYW
Vj7qJE1e4yDYtet+8=
YseIc6/ER1kqVonwnqhHXc+AFEMx6GpdwgBqwV7GFOv32yBLrVluxvWofr3r98EZ

Timestamped warning findings

Recorded time	Application warning
14:49:39	SD-card write speed too slow for 4K recording
14:52:06	Maximum flight distance reached
14:52:08	Weak signal
14:52:14	High wind velocity
14:53:51	Maximum flight distance reached again
14:55:33	Obstacle sensing disabled during landing
14:56:24	Backward obstacle sensing not functioning
15:02:10	Weak GPS signal; aircraft in Attitude mode

A few observations are particularly important:

1. The SD-card warning occurred one minute before the INFO metadata's recorded capture time of 14:50:39.
2. The maximum-distance, weak-signal and high-wind warnings occurred during the period surrounding the video recording.
3. The obstacle-sensing warning explicitly refers to aircraft landing. This is relevant evidence, but it is an application warning—not independent confirmation that landing actually took place at 14:55:33.
4. The weak-GPS warning at 15:02:10 occurred later in the notification sequence and may relate to subsequent aircraft activity.

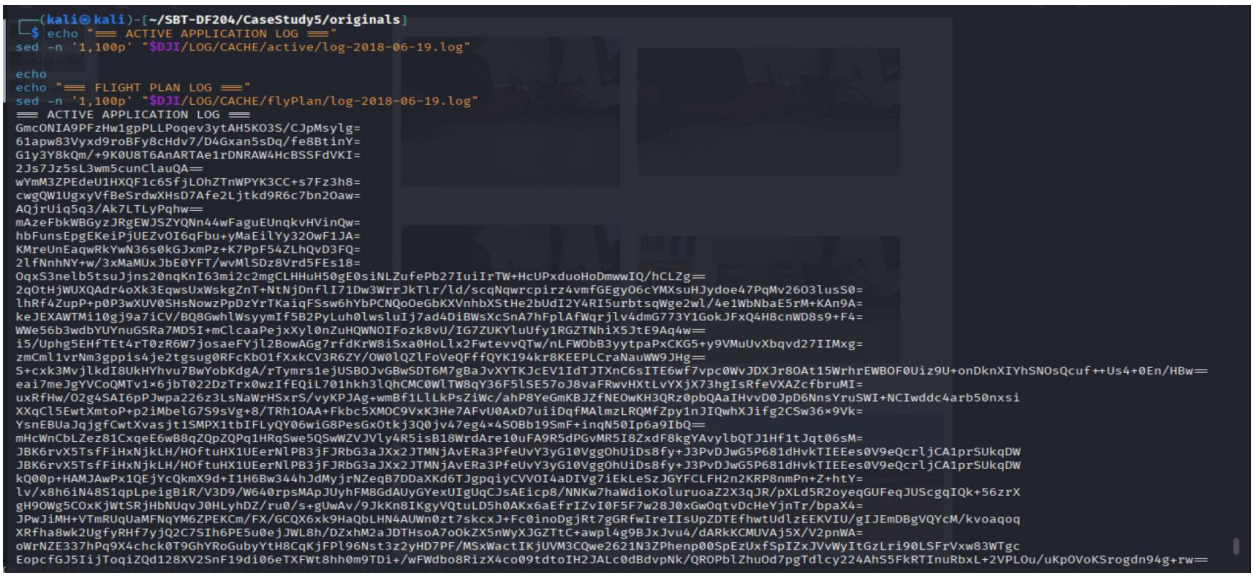
The MP4 duration is approximately 4 minutes and 0.972 seconds. If its CaptureDate and the notification timestamps use the same clock and timezone, its nominal end would be around 14:54:40. That would place the 14:55:33 landing-related warning approximately 53 seconds later.

Examination of DJI GO 4 application logs

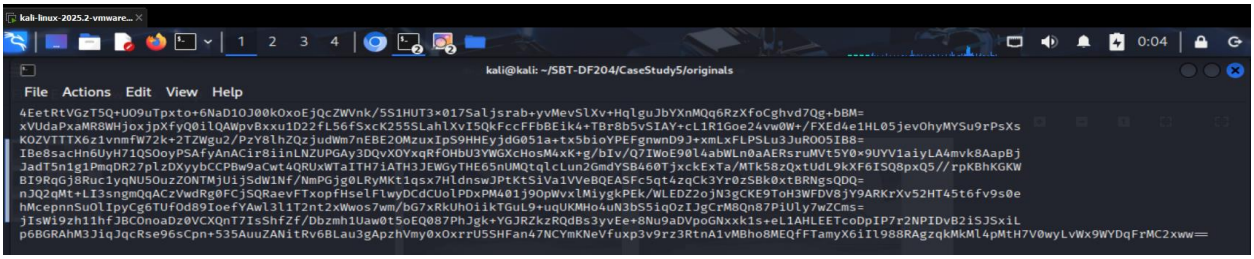
1. Active application log

The DJI GO 4 active application log, located at DJI/LOG/CACHE/active/log-2018-06-19.log, was examined to identify application activity and events associated with the flight period of 19 June 2018. The file contained multiple lines of alphanumeric strings with

characters commonly associated with Base64 encoding, including uppercase and lowercase letters, numbers, and padding characters. However, the displayed content did not contain directly readable timestamps, event descriptions, or identifiable flight-status messages.



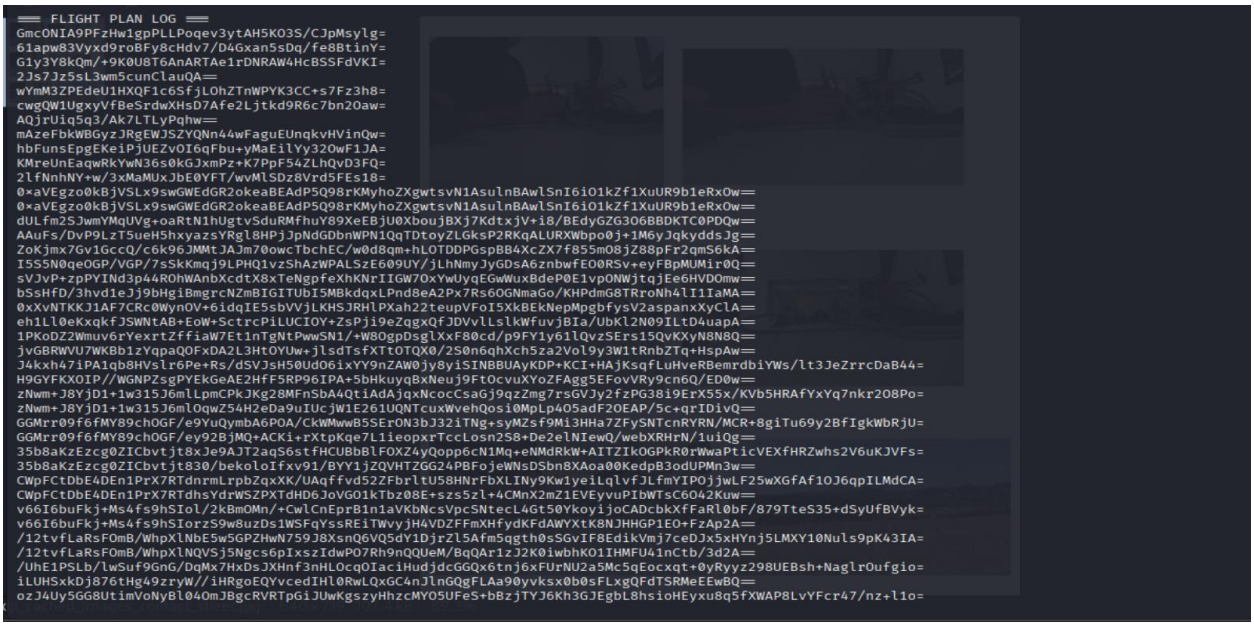
The file was therefore not interpreted as a conventional plaintext application log. Although the character patterns suggest that the entries may represent encoded or otherwise transformed data, the encoding scheme and underlying content could not be established through the initial examination. No attempt was made to assign operational meaning to the entries without a validated decoding method. The original file was preserved for potential further examination.

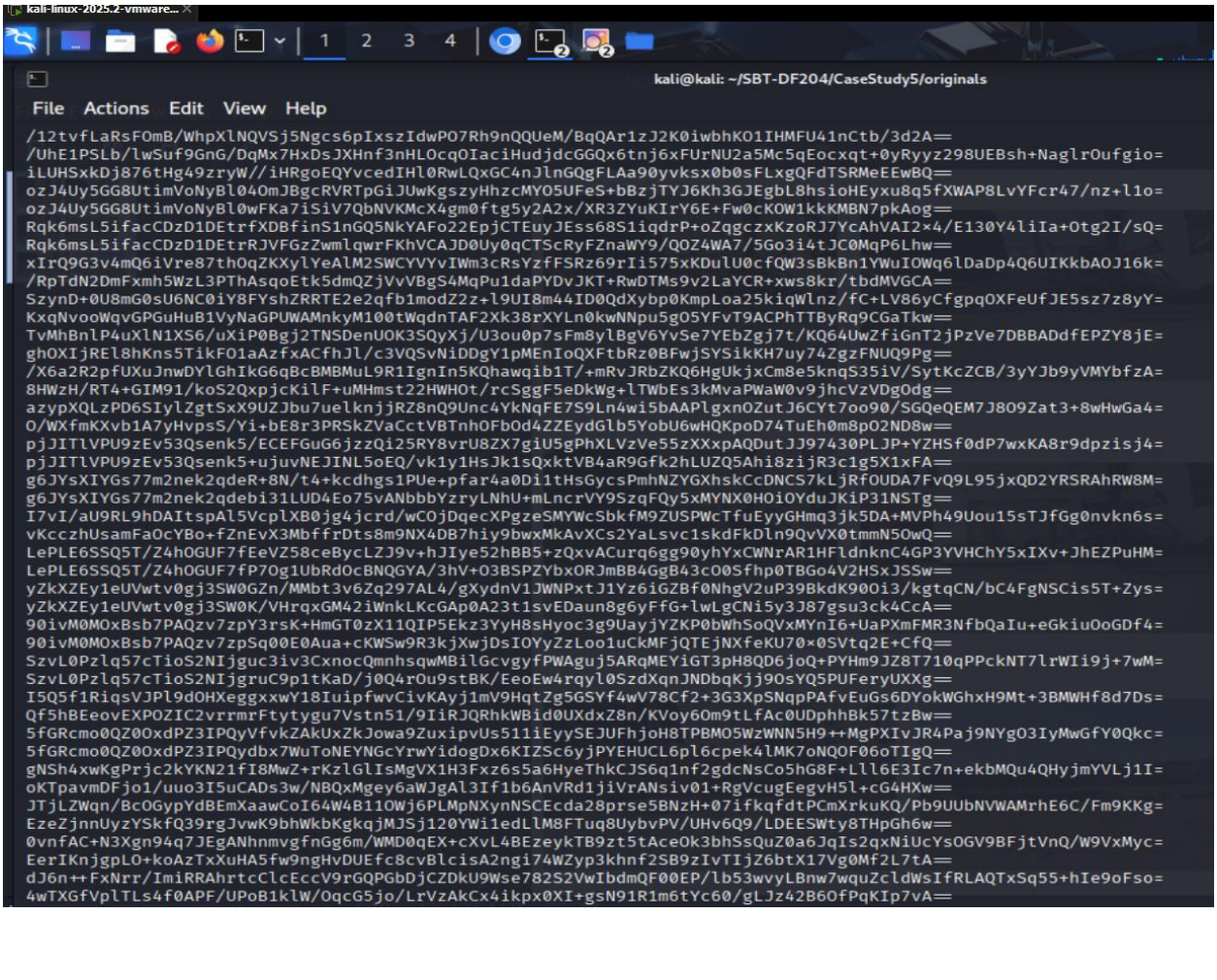


2. Flight-plan log

The DJI GO 4 flight-plan log, located at DJI/LOG/CACHE/flyPlan/log-2018-06-19.log, was also examined for information relating to flight planning, aircraft movement, and associated application activity. Similar to the active application log, the file consisted predominantly of long alphanumeric strings containing characters consistent with

Base64-formatted data. The displayed output did not expose directly interpretable flight-plan details, coordinates, timestamps, or descriptive event records.

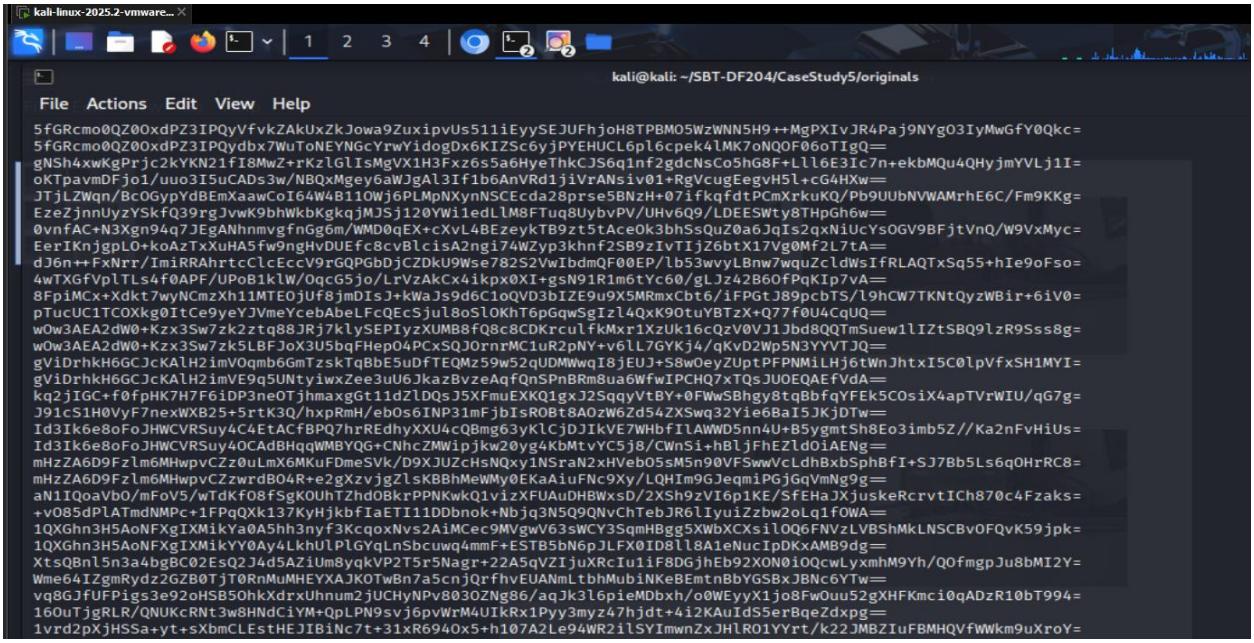




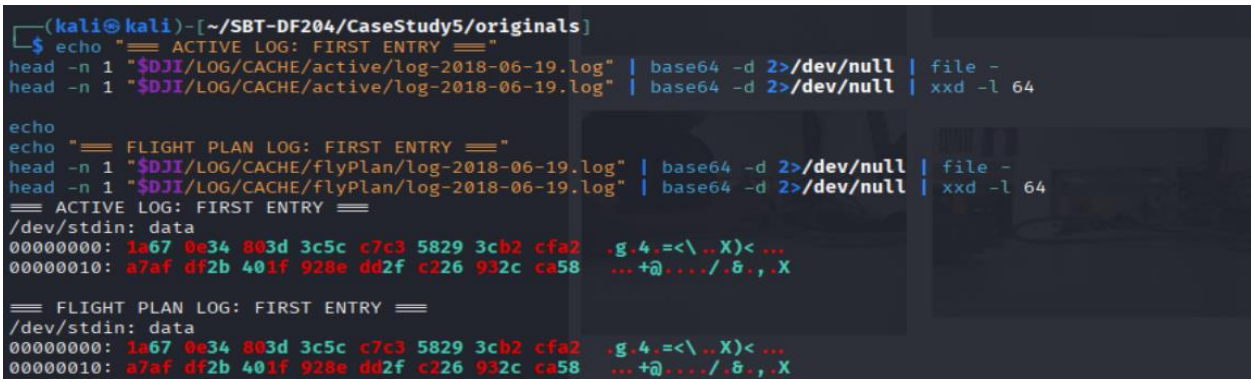
3. Correlation and evidential significance

This is where we connect these findings to the evidence already examined.

The active application and flight-plan logs did not provide independently interpretable event details during the initial examination. Consequently, they cannot presently be used to establish when the aircraft was powered on, when its motors were activated, when take-off or landing occurred, or whether a particular flight plan was executed.

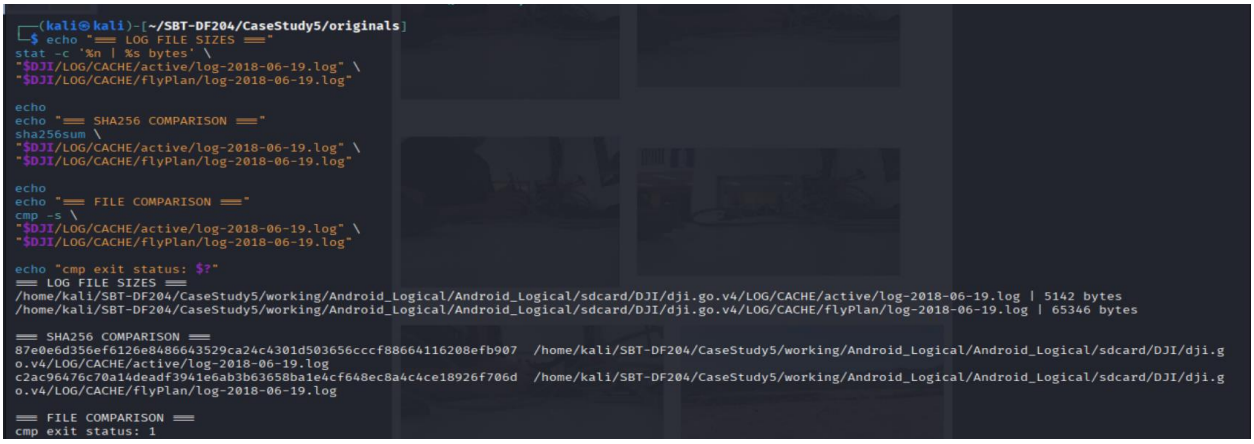


However, the separately examined DJI GO 4 notification log provides readable, timestamped warnings between 14:49:39 and 15:02:10 on 19 June 2018. The associated video metadata identifies a capture date of 14:50:39, while the MP4 contains approximately four minutes of video. These readable artefacts provide a stronger basis for the next stage of the investigation.



Comparison of DJI GO 4 Active and Flight-Plan Logs

The DJI GO 4 active application log and flight-plan log were subjected to file-size, cryptographic hash and binary comparison examinations to establish whether the two artefacts contained identical data. The active application log measured 5,142 bytes and produced a SHA-256 hash of 87e0e6d356ef6126e8486643529ca24c4301d503656cccf88664116208efb907. The flight-plan log measured 65,346 bytes and produced a SHA-256 hash of c2ac96476c70a14deadf3941e6ab3b63658ba1e4cf648ec8a4c4ce18926f706d.



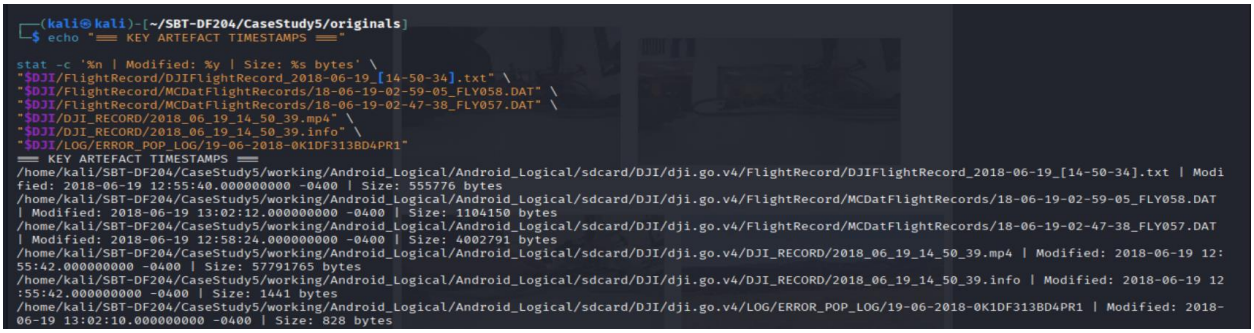
A direct comparison using the Linux cmp utility returned an exit status of 1, confirming that the complete files are not identical. This is consistent with their different file sizes and SHA-256 values. Although the preliminary decoding examination identified identical initial binary content in their first entries, this similarity does not extend to establishing that the files are duplicates. The two artefacts have therefore been retained and documented separately for subsequent examination.

Examination of Filesystem Modification Timestamps

The filesystem modification timestamps of six principal DJI GO 4 artefacts were examined using the Linux stat utility. The examination covered the TXT flight record, two DAT flight records, cached MP4 video, associated INFO metadata and DJI GO 4 error notification log. All six timestamps were displayed with a UTC−04:00 offset.

The TXT flight record showed a modification time of 12:55:40 on 19 June 2018, while the associated MP4 video and INFO metadata both showed 12:55:42. The FLY057.DAT record showed a modification time of 12:58:24, followed by the DJI error notification log at 13:02:10 and the FLY058.DAT record at 13:02:12. The matching modification

timestamps of the MP4 and INFO files, together with the two-second interval between the error log and FLY058.DAT, were recorded as observations for subsequent correlation. These timestamps represent filesystem metadata and must be distinguished from event timestamps contained in application records and video metadata. In particular, the INFO file identifies a capture date of 14:50:39, while its associated file modification timestamp is 12:55:42. The difference has not been interpreted as evidence of alteration or as proof of a particular recording sequence. Further timezone and timestamp-basis verification is required before constructing a definitive chronological timeline.



Timestamp examination findings

Artefact	Modified timestamp (displayed)	Size
DJI TXT flight record	19 Jun 2018, 12:55:40	555,776 bytes
FLY058.DAT	19 Jun 2018, 13:02:12	1,104,150 bytes
FLY057.DAT	19 Jun 2018, 12:58:24	4,002,791 bytes
MP4 video	19 Jun 2018, 12:55:42	57,791,765 bytes
INFO metadata	19 Jun 2018, 12:55:42	1,441 bytes
DJI error log	19 Jun 2018, 13:02:10	828 bytes

Observations

1. The TXT flight record was modified two seconds before the MP4 and INFO files.
2. The MP4 and INFO files have identical displayed modification timestamps.
3. FLY057.DAT has a later modification timestamp than the TXT and MP4 files.
4. FLY058.DAT has the latest modification timestamp among the six artefacts.
5. The error log modification time is two seconds before FLY058.DAT.

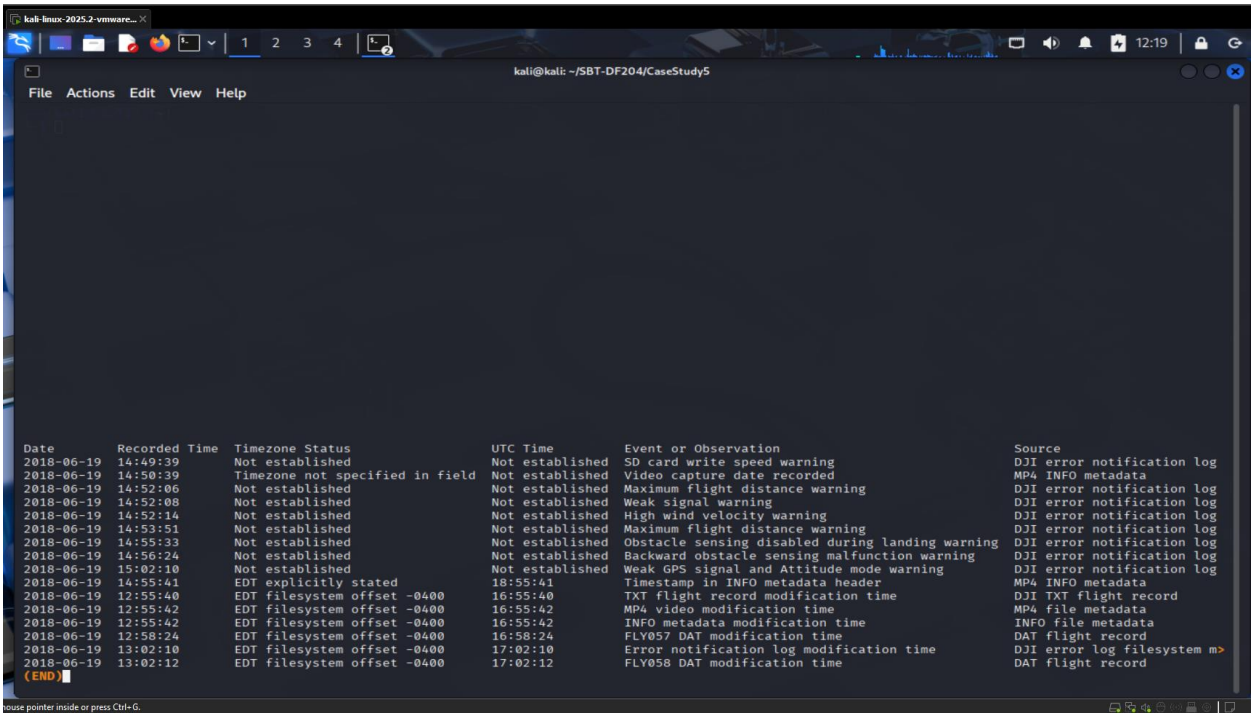
Timezone verification

The examination environment's timezone configuration was verified using the Linux date and timedatectl utilities. The system was configured to use the America/New_York timezone. At the time of examination, the system displayed Eastern Standard Time (EST), with an offset of UTC−05:00. However, the evidence timestamps relate to 19 June 2018, when the America/New_York timezone observed Eastern Daylight Time (EDT), with an offset of UTC−04:00.

This historical timezone difference explains the UTC−04:00 offset displayed by the stat utility for the examined artefacts. Accordingly, the filesystem modification timestamps were converted to Coordinated Universal Time (UTC) using the historical EDT offset. This conversion applies to the filesystem timestamps displayed by the examination environment and does not automatically establish the timezone basis of timestamps embedded in the DJI application records. Those timestamps have been retained separately pending further corroboration.

Converted filesystem timestamps

Artefact	Original time (EDT)	Converted time (UTC)
TXT flight record	12:55:40	16:55:40
MP4 video	12:55:42	16:55:42
INFO metadata	12:55:42	16:55:42
FLY057.DAT	12:58:24	16:58:24
Error notification log	13:02:10	17:02:10
FLY058.DAT	13:02:12	17:02:12



Cross-Source Correlation of DJI GO 4 Evidence

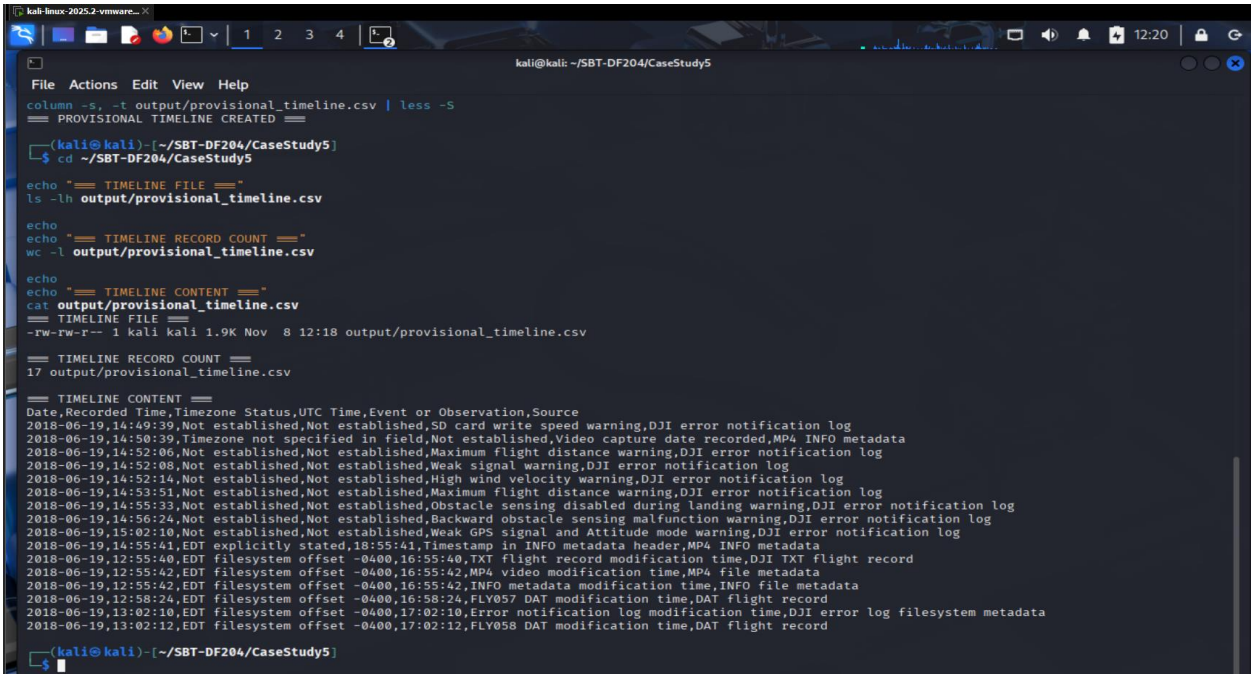
The examination identified several DJI GO 4 artefacts associated with 19 June 2018, including a flight record, two DAT records, a cached MP4 video, its associated INFO metadata, cached images and application notification logs. The filenames and recorded dates provide a basis for associating these artefacts with the same general investigation period. However, the available evidence does not establish that every artefact represents the same individual flight.

The INFO metadata identifies a video capture date of 14:50:39 and records a video duration of 234.833 seconds. Independent examination of the associated MP4 file using FFprobe identified a duration of 240.972400 seconds. The difference of approximately 6.139 seconds has been documented as an unresolved discrepancy. Both sources identify a local video resolution of 1280 × 720 pixels.

The DJI GO 4 notification log contains eight timestamped warnings between 14:49:39 and 15:02:10. These include warnings relating to SD-card write speed, maximum flight distance, weak signal, high wind velocity, obstacle sensing and GPS reception. The warnings provide additional context for the application activity recorded around the video capture period. Nevertheless, they should not be interpreted individually as conclusive proof of take-off, landing, motor activation or other physical aircraft events.

The TXT and DAT records were also examined. Their binary structures could not be reliably interpreted using the available examination methods. The two DAT records were found to be different files despite sharing an identical initial sequence. The application logs also contained encoded-looking data that could not be assigned a reliable meaning during the examination. Consequently, these records remain potentially relevant but cannot presently provide independently verified flight-event details.

The filesystem modification timestamps provide additional evidence about when the extracted artefacts were last modified according to their recorded metadata. These timestamps have been distinguished from the timestamps contained within the DJI application records and video metadata. A definitive unified UTC chronology cannot be established until the timezone basis of the application event timestamps is independently confirmed.



DJI GO 4 Evidence Inventory

Following the extraction and verification of the Android logical image, the DJI GO 4 application directory was examined to identify and catalogue the available application artefacts. A recursive file inventory was generated using the Linux find utility, recording the relative file paths and file sizes of all files within the dji.go.v4 directory. The inventory identified a total of 220 files across the application directory.

The inventory revealed several categories of potentially relevant evidence, including flight records, detailed DAT flight records, cached video recordings, associated INFO metadata, analytics files, cached images and application logs. Of particular interest were the flight record dated 19 June 2018, the two DAT files identified as FLY057 and FLY058, the cached video 2018_06_19_14_50_39.mp4, its associated INFO metadata, and the DJI GO 4 error notification log.

This inventory provided a structured overview of the available DJI GO 4 artefacts and helped identify the files selected for detailed examination and cross-source correlation. The complete inventory was saved as output/dji_evidence_inventory.txt for reference and subsequent reporting.

Following the extraction and verification of the Android logical image, the DJI GO 4 application directory was examined to identify and catalogue the available application artefacts. A recursive file inventory was generated using the Linux find utility, recording the relative file paths and file sizes of all files within the dji.go.v4 directory. The inventory identified a total of 220 files across the application directory.

The inventory revealed several categories of potentially relevant evidence, including flight records, detailed DAT flight records, cached video recordings, associated INFO metadata, analytics files, cached images and application logs. Of particular interest were the flight record dated 19 June 2018, the two DAT files identified as FLY057 and FLY058, the cached video 2018_06_19_14_50_39.mp4, its associated INFO metadata, and the DJI GO 4 error notification log.

```
┌──(kali㉿kali)-[~/SBT-DF204/CaseStudy5]
└─$ cd ~/SBT-DF204/CaseStudy5

DJI="working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4"

echo "== PRINCIPAL DJI EVIDENCE SHA-256 HASHES =="

sha256sum \
"$DJI/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt" \
"$DJI/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT" \
"$DJI/FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT" \
"$DJI/DJI_RECORD/2018_06_19_14_50_39.mp4" \
"$DJI/DJI_RECORD/2018_06_19_14_50_39.info" \
"$DJI/LOG/ERROR_POP_LOG/19-06-2018-0K1DF313BD4PR1"

echo
echo "== WORKING AND ORIGINAL ARCHIVE HASHES =="

sha256sum \
originals/Android_Logical.zip \
working/Android_Logical.zip
== PRINCIPAL DJI EVIDENCE SHA-256 HASHES ==
0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a  working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecor
d_2018-06-19_[14-50-34].txt
06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1  working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRec
ords/18-06-19-02-47-38_FLY057.DAT
fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade07a  working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRec
ords/18-06-19-02-59-05_FLY058.DAT
2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00  working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50
_39.mp4
a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a  working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50
_39.info
cb9671659ae2ff11b0fd3dd846e8702a81366479b04f3e2442300b2211bcff3e  working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/ERROR_POP_LOG/19-06-201
8-0K1DF313BD4PR1

== WORKING AND ORIGINAL ARCHIVE HASHES ==
9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d  originals/Android_Logical.zip
9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d  working/Android_Logical.zip
```

This inventory provided a structured overview of the available DJI GO 4 artefacts and helped identify the files selected for detailed examination and cross-source correlation. The complete inventory was saved as output/dji_evidence_inventory.txt for reference and subsequent reporting.

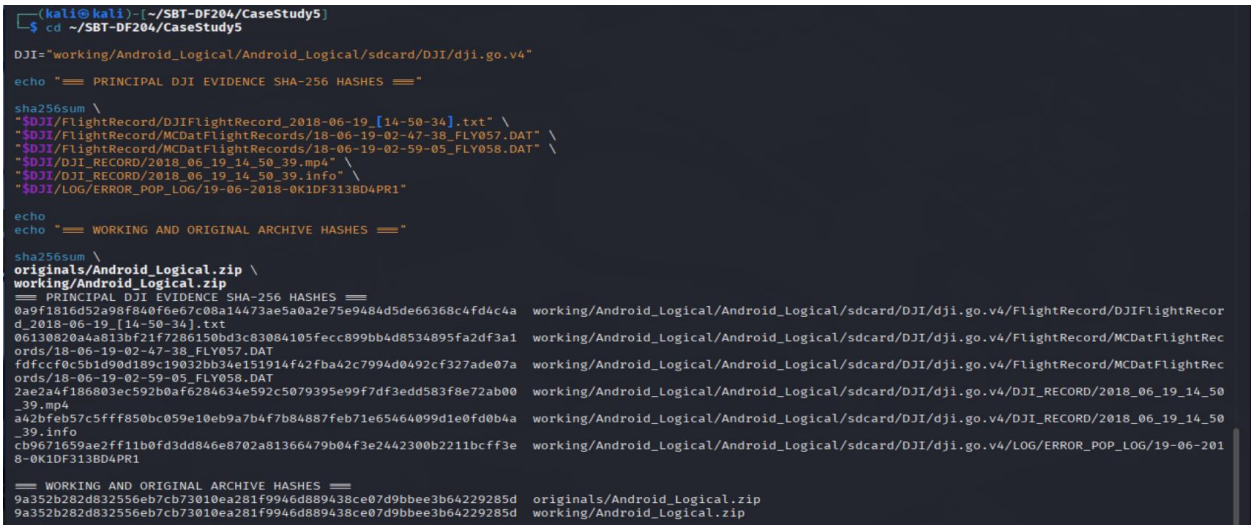
Following the inventory, cryptographic hashing was performed on the principal artefacts to establish reference values for integrity verification and to support the evidence log.

Cryptographic Hash Verification

Following the identification of the principal DJI GO 4 artefacts, SHA-256 cryptographic hashing was performed using the Linux sha256sum utility. The purpose was to establish unique reference values for the examined files and support subsequent integrity verification. The examination covered the DJI GO 4 TXT flight record, the two DAT flight records (FLY057 and FLY058), the cached MP4 video, its associated INFO metadata and the application error notification log. Individual SHA-256 values were generated and recorded for all six artefacts.

The original Android logical archive and its working copy were also hashed. Both files produced the same SHA-256 value of 9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d. This confirms that the two archive files were binary-identical at the time of examination, supporting the integrity of the working copy used for the investigation.

The individual artefact hashes were retained as reference values for the evidence register. Hash equality supports file-integrity verification but does not, by itself, establish the authenticity of the original source, the accuracy of embedded metadata or the identity of the person who operated the aircraft.



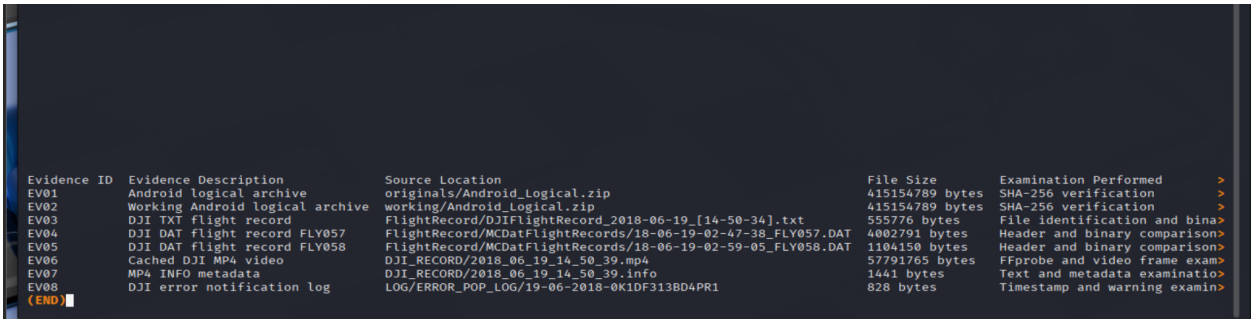
Evidence hash register

Evidence item	SHA-256
DJI TXT flight record	0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a
FLY057.DAT	06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1
FLY058.DAT	fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade07a
MP4 video	2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00
INFO metadata	a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a
DJI error notification log	cb9671659ae2ff11b0fd3dd846e8702a81366479b04f3e2442300b2211bcff3e

Preparation of the Evidence Log

Following the identification and examination of the principal DJI GO 4 artefacts, an evidence log was prepared to document the items selected for detailed analysis. The register records each evidence item's identification number, description, source location, file size and examination performed, together with its current examination status.

Eight principal evidence items were initially entered into the register, covering the original and working Android logical archives, the DJI TXT flight record, two DAT flight records, cached MP4 video, associated INFO metadata and DJI GO 4 error notification log. The register provides a structured record of the evidence examined and supports traceability between the findings presented in this report and their corresponding source files.

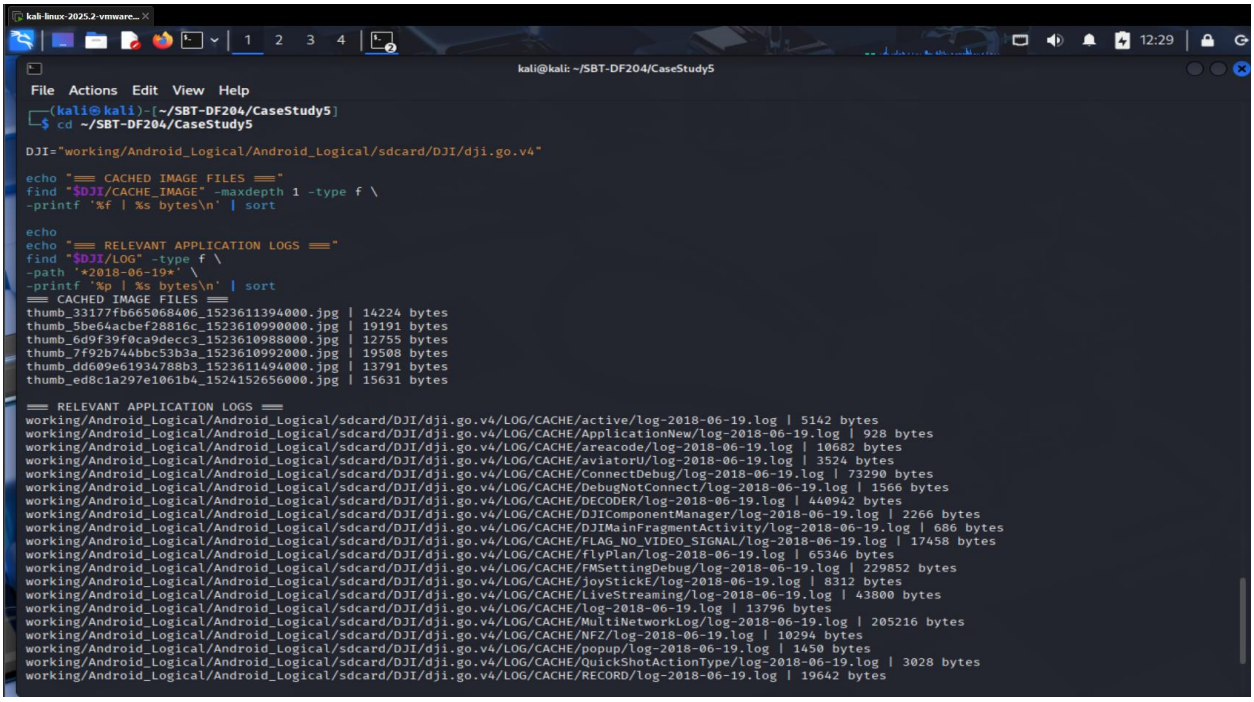


Evidence ID	Evidence Description	Source Location	File Size	Examination Performed
EV01	Android logical archive	originals/Android_Logical.zip	415154789 bytes	SHA-256 verification
EV02	Working Android logical archive	working/Android_Logical.zip	415154789 bytes	SHA-256 verification
EV03	DJI TXT flight record	FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt	555776 bytes	File identification and bina>
EV04	DJI DAT flight record FLY057	FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT	4002791 bytes	Header and binary comparison>
EV05	DJI DAT flight record FLY058	FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT	1104150 bytes	Header and binary comparison>
EV06	Cached DJI MP4 video	DJI_RECORD/2018_06_19_14_50_39.mp4	57791765 bytes	FFprobe and video frame exam>
EV07	MP4 INFO metadata	DJI_RECORD/2018_06_19_14_50_39.info	1441 bytes	Text and metadata examinatio>
EV08	DJI error notification log	LOG/ERROR_POP_LOG/19-06-2018-0K1DF313BD4PR1	828 bytes	Timestamp and warning examin>

(END)

Cached Images and Application Log Inventory

Following the examination of the principal flight records and video artefacts, the DJI GO 4 CACHE_IMAGE and LOG directories were examined to identify additional supporting evidence. The cached image directory contained six JPEG files, with individual file sizes ranging from 12,755 to 19,508 bytes. These files were retained for visual examination and potential correlation with other recovered media.



```
kali-linux-2025.2-vmware...
1 2 3 4
12:29
kali@kali: ~/SBT-DF204/CaseStudy5
File Actions Edit View Help
┌──(kali㉿kali)-[~/SBT-DF204/CaseStudy5]
└─$ cd ~/SBT-DF204/CaseStudy5

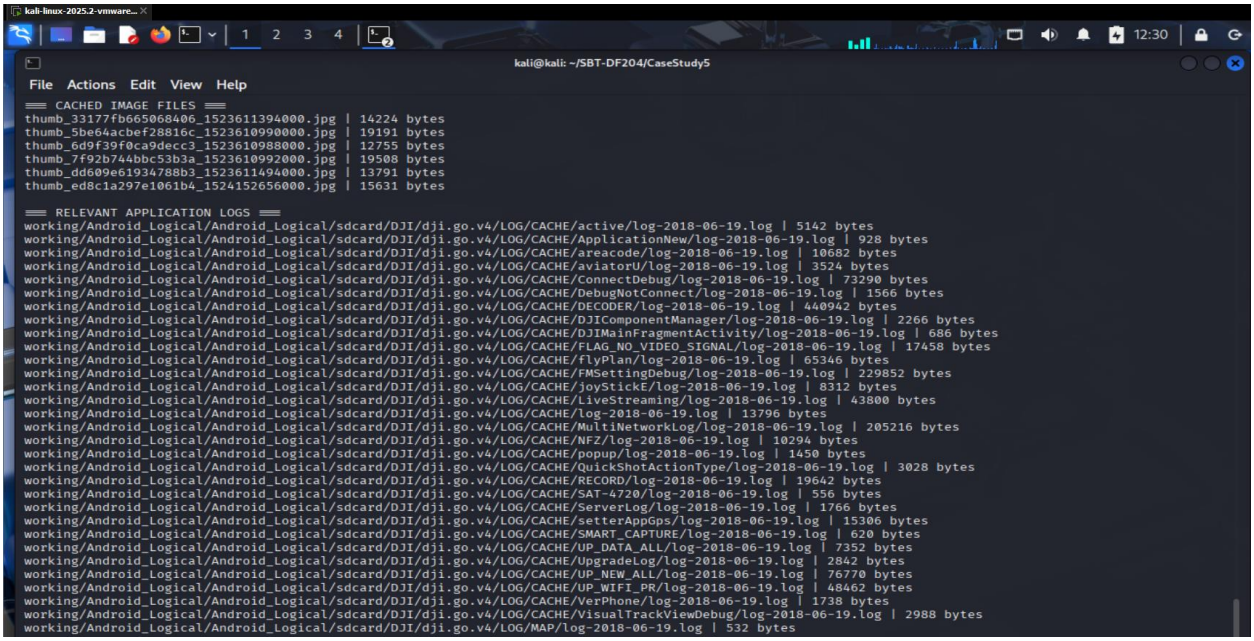
DJI="working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4"

echo "== CACHED IMAGE FILES =="
find "$DJI/CACHE_IMAGE" -maxdepth 1 -type f \
-printf '%f | %s bytes\n' | sort

echo
echo "== RELEVANT APPLICATION LOGS =="
find "$DJI/LOG" -type f \
-path '*2018-06-19*' \
-printf '%p | %s bytes\n' | sort
== CACHED IMAGE FILES ==
thumb_33177fb665068406_1523611394000.jpg | 14224 bytes
thumb_5be64acbef28816c_1523610990000.jpg | 19191 bytes
thumb_6d9f39f0ca9decc3_1523610988000.jpg | 12755 bytes
thumb_7f92b744bbc53b3a_1523610992000.jpg | 19508 bytes
thumb_dd609e61934788b3_1523611494000.jpg | 13791 bytes
thumb_ed8c1a297e1061b4_1524152656000.jpg | 15631 bytes

== RELEVANT APPLICATION LOGS ==
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/active/log-2018-06-19.log | 5142 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/ApplicationNew/log-2018-06-19.log | 928 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/areacode/log-2018-06-19.log | 10682 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/aviatorU/log-2018-06-19.log | 3524 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/ConnectDebug/log-2018-06-19.log | 73290 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/DebugNotConnect/log-2018-06-19.log | 1566 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/DECODER/log-2018-06-19.log | 440942 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/DJIComponentManager/log-2018-06-19.log | 2266 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/DJIMainFragmentActivity/log-2018-06-19.log | 686 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/FLAG_NO_VIDEO_SIGNAL/log-2018-06-19.log | 17458 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/flyPlan/log-2018-06-19.log | 65346 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/FMSettingDebug/log-2018-06-19.log | 229852 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/joyStickE/log-2018-06-19.log | 8312 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/LiveStreaming/log-2018-06-19.log | 43800 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/log-2018-06-19.log | 13796 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/MultiNetworkLog/log-2018-06-19.log | 205216 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/NFZ/log-2018-06-19.log | 10294 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/popup/log-2018-06-19.log | 1450 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/QuickShotActionType/log-2018-06-19.log | 3028 bytes
working/Android_Logical/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/RECORD/log-2018-06-19.log | 19642 bytes
```

A further examination of the application log directory identified 30 log files bearing the date 19 June 2018. These included records associated with application activity, flight planning, connection debugging, video recording, decoder operations and flight-control settings. Of particular interest were the active, flyPlan, ConnectDebug, DECODER, RECORD and FMSettingDebug logs.



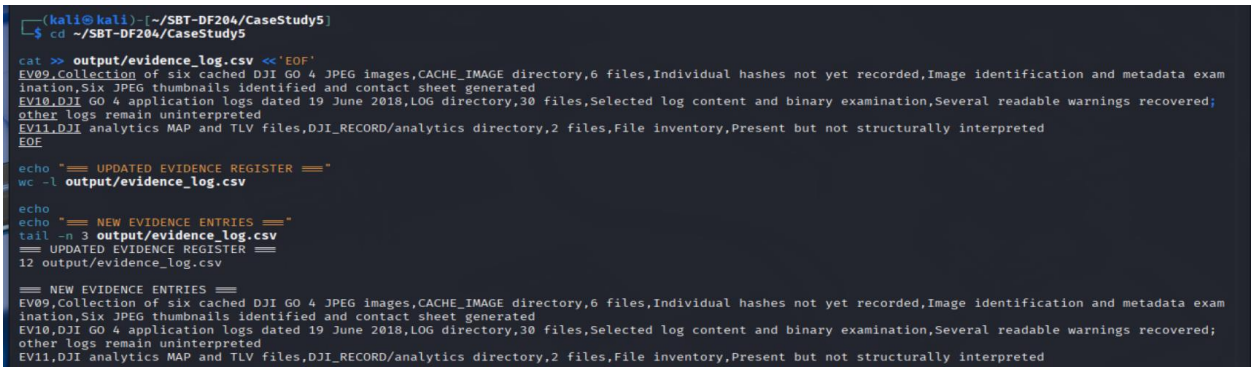
The identified files were included in the wider evidence inventory. Their presence establishes that additional application records were available for examination; however, the content and evidential significance of individual logs vary. Some contained readable information, while others contained encoded-looking binary data that could not be reliably interpreted during this examination.

Expansion of the Forensic Evidence Register

Following the identification of the cached images, application logs and analytics files, the forensic evidence register was updated to incorporate these additional artefacts. Three further evidence entries were recorded: a collection of six cached DJI GO 4 JPEG images, 30 application log files dated 19 June 2018, and two analytics files comprising MAP and TLV formats.

The cached images were documented as a collection because they were recovered from the same application directory and examined together for identification and metadata purposes. The application logs were similarly grouped to provide a consolidated record

of the available DJI GO 4 diagnostic and activity records. The MAP and TLV files were also included to ensure that potentially relevant analytics artefacts were accounted for, although their internal structures were not examined in detail.



With these additions, the evidence register contained 11 entries, covering the principal Android archive, DJI flight records, cached video, metadata, application logs, cached images and analytics files. The register was maintained as a separate CSV file to support evidence traceability and subsequent reporting.

Final verification of the evidence register

The SHA-256 hash of the DJI GO 4 INFO metadata file was recalculated during the final verification stage. The examination produced the hash:

a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a



The evidence register was also checked and confirmed to contain 13 evidence entries, identified as EV01 to EV13. These entries cover the original and working Android logical

archives, selected DJI flight records, cached video and metadata, application logs, cached images, analytics files, and the two unavailable evidence sources: the raw microSD image and EWF image.

CONCLUSION

The forensic examination of the DJI Mavic Air evidence identified several artefacts associated with the DJI GO 4 application, including flight-record files, video recordings, metadata, cached images and application logs. The examination established that the original Android logical evidence archive and its working copy had matching SHA-256 hashes, supporting the integrity of the copied archive at the time of verification.

The video recording was successfully examined using FFprobe, which confirmed its format, resolution, frame rate and duration. The associated INFO file provided additional metadata, including capture-date information, video dimensions and GPS fields. Selected DJI GO 4 application logs also contained warnings relating to SD card write speed, maximum flight distance, weak signal, high wind velocity, obstacle sensing and GPS conditions. These entries provided useful investigative leads but were not treated as conclusive proof of the aircraft's physical behaviour.

Attempts to interpret the binary flight-record files using available tools, including GNU Octave and SciPy, were unsuccessful. Their internal structures and the meaning of their recorded values therefore remain unresolved. Similarly, the raw internal microSD image and EWF forensic image were not successfully obtained, limiting the scope of the examination.

Overall, the investigation demonstrated the value of combining file-system examination, cryptographic hashing, metadata analysis, video examination and application-log review when investigating drone-related digital evidence. However, the available evidence was insufficient to establish a complete flight path, identify the aircraft operator or independently confirm every reported flight event. Further examination of the original microSD card, a verified forensic image and suitable DJI flight-data parsing tools would be necessary to develop a more complete reconstruction.